

Selected Answers

Answers to Chapter II, section 1

II.1.8 The Riemann sphere is not a vector space because of the action of the 'Point at Infinity'. To be a vector space, each point must have an additive inverse, that is, another point which summed to it yields the additive identity element 0. Since there is no such inverse for ∞ , the space is not a vector space. Note that the usual check $\alpha(x + y) = \alpha x + \alpha y$ does hold however.

The examples in 1.7 are not vector spaces for the simple reason that they are not closed under algebraic operations. That is, multiply any element in $\mathbf{X}$ by a large enough scalar, and it will not be in the space. The example in 1.6 has the same problem. If one defines multiplication by scalars in codespace in a reasonable way, namely multiply the distances by multiplying the codes and 'carrying' one will find a sequence for which there are no codes. (Multiply $x = (N - 1)(N - 1) \cdots$ by 2).

II.1.9 By definition $\{x\} \subset \mathbf{X}$ if every element in $\{x\}$ is in $\mathbf{X}$. The only element in $\{x\}$ is x which by assumption is in $\mathbf{X}$. Hence $\{x\} \subset \mathbf{X}$.

Answers to Chapter II, section 2

II.2.1 We use extensively the inequality $|x| + |y| \geq |x + y|$. It is proven by separating it into three cases: a) $x, y \geq 0$. In this case $|x| = x, |y| = y$, and since the sum of two positive numbers is positive, $|x + y| = x + y = |x| + |y|$. b) $x, y \leq 0$. In this case $|x| = -x, |y| = -y$. Since the sum of two negative numbers is negative, $|x + y| = -x - y = |x| + |y|$. c) $x \geq 0, y \leq 0$. By changing the roles of x, y this proves the other possibility as well. $|x| = x, |y| = -y, |x + y| = |x - (-y)| = ||x| - |y||$. This latter number is either equal to $|x| - |y|$ or its negative, and both are less than or equal to $|x| + |y|$ with equality only if one or the other are zero. Hence $|x + y| \leq |x| + |y|$. We proceed to verify the metric space hypotheses:

(a)

(i) $d(x, y) = |x - y| = |-(x - y)| = |y - x| = d(y, x)$.

(ii) $x - y \neq 0$ unless $x = y$. It is not infinite if x, y are not infinite. Since taking the absolute value does not alter either of these properties $0 < d(x, y) < \infty$.

(iii) $d(x, x) = |x - x| = |0| = 0$.

(iv) $d(x, y) = |x - y| = |x - z + z - y| \leq |x - z| + |z - y|$ by means of the proof above.

(b) (i), (ii), (iii) can be verified by multiplying everything in part a) by 2. (iv) $d(x, y) = 2|x - y| = 2|x - z + z - y| \leq 2(|x - z| + |z - y|) = 2|x - z| + 2|z - y|$.

(c) All properties can be done by a) substituting x^3, y^3, z^3 everywhere in the above arguments.

II.2.3 $d(x, y) = |xy|$ is not a metric for $\mathbb{R}$. Choose $x = 0$ and $y \neq 0$. Then $x \neq y$, but $d(x, y) = |0 \cdot y| = 0$, which violates property (i).

II.2.4 On the punctured plane $\mathbb{R}^2 \setminus \mathcal{O}$ define $d(x, y) = |r_1 - r_2| + \theta$. Then for two points x, y , we have

(1) $d(x, y) = |r_1 - r_2| + \theta = |r_2 - r_1| + \theta = d(y, x)$ since θ is still the smallest angle subtended by the two rays.

(2) If $x \neq y$ and $|r_1 - r_2| = 0$ then $\theta \neq 0$ so $d(x, y) > 0$ if $x \neq y$. $\theta \leq \pi$ so that for $d(x, y) = \infty$ we must have either r_1 or r_2 infinite. But $r_1 = x_1^2 + x_2^2$ and $r_2 = y_1^2 + y_2^2$ so if either is infinite so is either x or y . Hence $0 < d(x, y) < \infty$ if $x \neq y$.

$$(3) d(x, x) = |r - r| + 0 = 0 + 0 = 0.$$

$$(4) d(x, z) + d(z, y) = |r_1 - r_3| + |r_3 - r_2| + \theta_1 + \theta_2. \theta_1 + \theta_2 > \theta \text{ if the ray to } z \text{ is not between the other two, and } \theta_1 + \theta_2 = \theta \text{ otherwise, so } d(x, z) + d(z, y) \geq |r_1 - r_3| + |r_3 - r_2| + \theta \geq |r_1 - r_3 + r_3 - r_2| + \theta = |r_1 - r_2| + \theta = d(x, y) \text{ by problem 1 above.}$$

II.2.5 If x, y are not equal, then there is an x_i, y_i pair which do not agree. Hence $|x_i - y_i|/(N+1)^i \neq 0$. Since all the terms in the sum are positive or zero, there is no cancelling, and therefore $d(x, y) \neq 0$. We can switch the order in each term because of the absolute value signs, consequently $d(x, y) = d(y, x)$. The series converges since the numerators are less than $2N$ so the series is dominated by

$$2N \cdot \sum_{i=1}^{\infty} \frac{1}{(N+1)^i}$$

which converges for all $N < \infty$. Because the series converges, the proof of all the metric requirements, except the above, follow from problem on each term, and therefore for the whole series.

II.2.6 The easiest way to show that this equation for a distance measure satisfies metric properties turns out to be that we show that it really is a Euclidean metric, in the usual sense. Let x be an element of Σ the code space on N symbols. Then the series

$$\sum_{i=1}^{\infty} \frac{x_i}{(N+1)^i}$$

converges to a number in $[0, 1]$. In fact it is the $(N+1)$ -ary expansion of a real number in this range, having the property that it has no digits equal to N , because the code space symbols range from $0, \dots, N-1$. For instance, when $N=9$ we have

$$\sum_{i=1}^{\infty} \frac{x_i}{10^i} = \frac{x_1}{10} + \frac{x_2}{100} + \dots = .x_1x_2x_3\dots$$

Then the mapping $\Sigma \rightarrow [0, 1]$ given by

$$x \mapsto \tilde{x} = \sum_{i=1}^{\infty} \frac{x_i}{(N+1)^i}$$

is a metric equivalence, with constants c_1, c_2 in the definition equal to 1. Using this map, let $(x, y), (u, v)$ be two points in the space of shifts on N symbols. By the above equivalence, we have corresponding points $(\tilde{x}, \tilde{y}), (\tilde{u}, \tilde{v})$ in the square $[0, 1] \times [0, 1] \subset \mathbb{R}^2$. Since the series converge, we have

$$\tilde{x} - \tilde{u} = \sum_{i=1}^{\infty} \frac{x_i}{(N+1)^i} - \sum_{i=1}^{\infty} \frac{u_i}{(N+1)^i} = \sum_{i=1}^{\infty} \frac{(x_i - u_i)}{(N+1)^i}$$

The metric in the statement of the problem can therefore be written as

$$d((x, y), (u, v)) = \sqrt{(\tilde{x} - \tilde{u})^2 + (\tilde{y} - \tilde{v})^2}$$

which is Euclidean distance in $\mathbb{R}^2$, and hence a metric.

II.2.7 In body space, the shortest distance between two points is the length of the path through the body. The shortest distance from a point to itself is then 0. The shortest distance between two points is the length of the path which is also the shortest path in reverse order. The length of a path between two distinct points cannot be zero, and a path of infinite length inside the body must not be the shortest path between two points. Finally the shortest path between two points in the body is less than the path which takes a detour to a third point unless that point is on the shortest path already. Hence $d(x, y)$ is a metric.

II.2.8 For a function d which is not a metric, see for example exercise 2.3. For the annulus to look like a cylinder, we must have the distance around any concentric path look the same. This implies that we need the distance around all concentric

paths equal. Use the angle subtended between two points to do this. To look like height, we need a linear radial distance, use the radius. If we want to look like the surface of a cylinder with Euclidean metric (unrolled), we thus use $\sqrt{(r_1 - r_2)^2 + \theta^2}$. The proof that this is a metric follows exercise 2.4.

II.2.9 That a distance measure formed by great circle distance is a metric is true on any two-sphere. We verify the conditions as follows:

(i) The plane defined by the center C of the sphere and any two distinct points x, y on its surface cuts the sphere on the great circle path between the two points (that is its definition), so there is one and only one great circle distance between the two points. This distance is $r\theta$ where θ is the interior angle xCy . Hence $d(x, y) = 0$ if and only if $\theta = 0$ that is $x = y$.

(ii) Because we are dealing with interior angles, the angle xCy and the angle yCx are equal hence $d(x, y) = d(y, x) > 0$. The maximum value of an interior angle is π , hence $d(x, y) \leq r\pi < \infty$.

(iii) To prove the triangle inequality, we note first that if x, y, z lie on the same great circle, then if z is interior to x, y , $d(x, z) + d(z, y) = d(x, y)$, whereas if it is exterior then since an exterior angle is larger than an interior one, $d(x, z) + d(z, y) > d(x, y)$. Consequently, we deal only with z not on the great circle defined by x, y . For this case, the $\mathbb{R}^3$ distance between the points x and z is given by $\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta}$ by the law of cosines. Without loss of generality, $r_1 = r_2 = 1$ and the square root is monotonic increasing, so that this distance is proportional to $1 - \cos \theta$ and hence decreases for decreasing θ . Consider the plane triangle in $\mathbb{R}^3$ defined by x, y, z . By moving z towards the line $\overline{xy}$, we decrease the distances between x and z and between z and y (we are moving down the perpendicular from z). Since the above law of cosines formula insists that these distances can only decrease if the corresponding angles xCz and yCz decrease, the distances on the sphere are also strictly decreasing. This implies that $d(x, z) + d(y, z) > d(x, y)$ in this case. Hence the triangle inequality holds for great circle distance on any sphere. When this sphere is interpreted as $\hat{\mathbb{C}}$, the great circle between ∞ and any other point is along the projection of vertical lines in the complex plane. Hence since $1 + i$ is above the real axis, its distance to infinity is less than that of zero, which is $\pi/2$.

II.2.10 Suppose that we have

$$c_1 d_1(x, y) \leq d_2(x, y) \leq c_2 d_1(x, y)$$

where c_1, c_2 are positive real numbers. Using the first inequality, we have

$$d_1(x, y) \leq \frac{1}{c_1} d_2(x, y)$$

and the second gives

$$\frac{1}{c_2} d_2(x, y) \leq d_1(x, y)$$

Let $e_1 = \frac{1}{c_2}$ and $e_2 = \frac{1}{c_1}$, and combine the above two inequalities, we get

$$e_1 d_2(x, y) \leq d_1(x, y) \leq e_2 d_2(x, y)$$

as desired.

II.2.11 Yes in both cases. Let $d_1(x, y)$ be Manhattan metric, $d_2(x, y)$ be Euclidean, and use, for instance $c_1 = 1/2, c_2 = 2$. Since the Manhattan metric is at most $\sqrt{2}$ times the Euclidean metric (when they lie on a line at a 45° angle) these constants will do fine.

II.2.12 Two metrics are uniformly equivalent if and only if there are constants c_1, c_2 such that

$$c_1 d_1(x, y) \leq d_2(x, y) \leq c_2 d_1(x, y)$$

for every pair x, y in the space. To see that the metric $d(x, y) = |r_1 - r_2| + \theta$ is not equivalent to the Euclidean metric on the annulus, choose d_1 to be the Euclidean metric, and d_2 to be this one. Let x, y be on the real line $x < 0$ and $y > 0$. Then $d_2(x, y) > \theta = \pi$. Regardless of the value of c_2 by moving x, y towards the origin we can insure that $d_1(x, y)$ is less than any ϵ by letting $x = (-\epsilon/3, 0)$ and $y = (\epsilon/3, 0)$. Hence $c_2 d_1(x, y) < c_2 \epsilon$. Then if $\epsilon < c_2/\pi$ the inequality fails. Hence the two metrics are not equivalent.

II.2.13 Use $h(x) = x - 1$. Then $\tilde{d}_1(h(x), h(y)) = 2 \cdot |(x - 1), (y - 1)| = 2 \cdot |x - y|$. This metric is equivalent to $d_1(x, y)$ by using $c_1 = 1/4, c_2 = 1$.

II.2.14 To show that (■, Manhattan) and (■, Euclidean) are equivalent metric spaces, use $h(x) = x$. It is easiest to see the equivalence by making the substitution $x' = 0, y' = y - x = y_1 - x_1, y_2 - x_2$. Then the Euclidean distance $d(x, y) = d(\mathcal{O}, y') = \sqrt{y_1'^2 + y_2'^2} = r$. The Manhattan distance in the same notation is $|r \cos \theta + r \sin \theta|$. Maximizing the last expression with respect to θ yields $\theta = \pi/2$ or $\cos \theta = \sin \theta = \sqrt{2}/2$. So the maximum difference in size between the two metrics is $d_M = \sqrt{2}d_E$. Then

$$\frac{1}{\sqrt{2}}d_M(x, y) \leq d_E(x, y) \leq d_M(x, y).$$

II.2.15 Using $h(z) = h(z_1 + iz_2) = (z_1, z_2)$, we have $\tilde{d}(x, y) = d_E(x, y)$ on $\mathbb{R}^2$. The above problem gives the metric equivalence for this with the Manhattan metric.

II.2.16 Suppose they were equivalent. Then there are constants c_1, c_2 such that

$$c_1 d_1(x, y) \leq d_2(x, y) \leq c_2 d_1(x, y)$$

Let $x = 1$ and $y = \frac{1}{2c_2+1}$. Then $d_2(x, y) = 2c_2 d_1(x, y)$, a contradiction. Hence they are not equivalent.

II.2.18 If two metric spaces are equivalent, there is a one-to-one, onto map $h: X_1 \rightarrow X_2$ such that

$$c_1 d_1(x, y) \leq d_2(h(x), h(y)) \leq c_2 d_1(x, y)$$

The second inequality implies that if

$$d_1(x, y) < \delta = \epsilon/c_2 \text{ then } d_2(h(x), h(y)) < c_2 \delta = \epsilon.$$

Since h is invertible, we may write the first inequality as

$$d_1(h^{-1}(x), h^{-1}(y)) \leq \frac{1}{c_1} d_2(x, y)$$

for points $x, y \in X_2$. Then $d_1(h^{-1}(x), h^{-1}(y)) < \epsilon$ whenever $d_2(x, y) < \delta = c_1 \epsilon$. Hence h and its inverse are continuous, so $h(x)$ is a homeomorphism.

II.2.19 Intuitively, these metrics behave similar to base 10 decimals. In fact, if we look at Σ_{10} the symbols are 0–9, and by choosing $k = N = 10$, we get decimal expressions (with some ambiguity). To show that these metrics are topologically identical, we must demonstrate that the identity map $\iota(x) = x$ is a homeomorphism. This is equivalent in epsilon-delta format to showing that we need to show that for every $x, \epsilon > 0$ there is a δ such that for every y in Σ_N the two statements

$$d_{k_1}(x, y) < \delta \Rightarrow d_{k_2}(x, y) < \epsilon$$

and,

$$d_{k_2}(x, y) < \delta \Rightarrow d_{k_1}(x, y) < \epsilon$$

are true.

We can simplify matters by breaking the proof into cases and reducing it to a single statement to prove as follows:

Case 1: $k_1 = k_2$. In this case there is nothing to prove, since a metric is always equivalent to itself.

Case 2: $k_1 < k_2$. In this case, by comparing terms of the two series, since $k_2' > k_1'$, $d_{k_2}(x, y) \leq d_{k_1}(x, y)$ for any x, y with equality holding only for $x = y$. Consequently, we may choose $\delta = \epsilon$ in the first statement:

$$d_{k_1}(x, y) < \epsilon \Rightarrow d_{k_2}(x, y) \leq d_{k_1}(x, y) < \epsilon.$$

and we need only prove the second statement, namely

$$d_{k_2}(x, y) < \delta \Rightarrow d_{k_1}(x, y) < \epsilon$$

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Case 3: This case requires only the proof of the first statement for the same reasons as case 2. There is no loss in just assuming case 2, and reversing the subscripts for case 3.

Consequently, all we need to do is show that if $k_1 < k_2$,

$$d_{k_2}(x, y) < \delta_1 \Rightarrow d_{k_1}(x, y) < \epsilon$$

and then take $\delta = \min(\delta_1, \epsilon)$.

Looking at the analogy between decimals and Σ_N , we can get the distance $d_k(x, y)$ arbitrarily small if $x_i = y_i$ up to some big number i_0 . This would correspond in decimal arithmetic to the first i_0 decimal places being equal to zero in the distance formula. Writing it without the analogy, since the series is less than

$$C = N \sum_{i=1}^{\infty} \frac{1}{k^i}$$

then

$$\sum_{i=i_0}^{\infty} \frac{|x_i - y_i|}{k^i} < \sum_{i=i_0}^{\infty} \frac{N}{k^i} = \frac{C}{k^{i_0}}$$

To insure this is less than, say α we only need to choose i_0 large enough. This is nice, we can get a set we can describe by just using the digits that put us inside a given $B_\epsilon(x)$. What we don't know is whether it contains a ball inside it (For large enough N it is a ball, but we don't know that). On the other hand, we can look at the smallest possible distance we could get if any of the first i_0 "digits" were not zero. This would be the distance between an x and a y with *only* the i_0 digits different, namely

$$|x_{i_0} - y_{i_0}| = 1$$

and $x_i = y_i$ whenever $i \neq i_0$. This distance is

$$d_k(x, y) = \frac{1}{k^{i_0}}$$

So any ball $B_\delta(x)$ with $\delta < 1/k^{i_0}$ is guaranteed to have no elements which disagree on the first i_0 places.

We now proceed as follows: Choose i_0 such that the set

$$B = \{y : x_i = y_i \text{ if } i \leq i_0\}$$

is inside of the d_{k_1} ball $B_\epsilon(x)$. Now choose δ_0 such that the d_{k_2} ball $B_{\delta_0}(x)$. Since all elements of this d_{k_2} ball have at least i_0 "digits" of agreement with x , everything in the ball is in the d_{k_1} ball of radius ϵ . Therefore,

$$d_{k_2}(x, y) < \delta_0 \Rightarrow d_{k_1}(x, y) < \epsilon$$

which was all we decided we needed to prove.

Answers to Chapter II, section 3

II.3.1 By Definition 3.3, if $\{x_n\}$ is a Cauchy sequence in (X, d) a complete metric space, then it converges to a point $x \in X$. This implies that given an $\epsilon > 0$ there is an N such that if $n > N$ then $x_n \in B(x, \epsilon)$. Since there are infinitely many integers greater than any integer N , $x_n \in B(x, \epsilon)$ for infinitely many integers n .

II.3.2 We will use the property of $(\mathbb{R}, \text{Euclidean})$ that every set bounded above has a Least Upper Bound (LUB). This implies that a monotone increasing sequence $\{l_n\}$ which is bounded above has a limit. Since the LUB cannot be less than l_n for any n (the sequence is increasing) and there is no minimum distance to all the l_n (since it is the LUB) it follows that for any $\epsilon > 0$ there is an N such that $x_n \in B(\text{LUB}, \epsilon)$ for $n > N$. We use this property to prove completeness of $(\mathbb{R}, d)$ as follows: Let $\epsilon_i = 1/2^i$ and N_i be for each i a number such that $N_i > N_{i-1}$ and $d(x_n, x_m) < \epsilon_i$ whenever $n, m > N_i$. It follows that there is an interval of length ϵ_i such that all the x_n are in this interval if $n > N_i$. Let l_i be the greatest lower bound of the set $\{x_n : n > N_i\}$ (which exists by subtracting everything from a large number and using the LUB principle). The sequence $\{l_i\}$ is then monotone increasing, and is bounded above by the LUB of the same set of x_n which we call L_i . Then

this sequence has a limit which we will call x . We claim that x is the limit of the Cauchy sequence as well. Given $\epsilon > 0$, $B(x, \epsilon) \supset [l_i, L_i]$ for some i , hence if $n > N_i$, $x_n \in B(x, \epsilon)$ which means x is the limit of $\{x_n\}$ and the sequence converges. Hence $(\mathbb{R}, \text{Euclidean})$ is complete.

II.3.3 We show that $\mathbb{R}^2$ is complete by using the fact that $\mathbb{R}$ is complete. Let $\{(x_n, y_n)\}$ be a sequence in $\mathbb{R}^2$ which converges to (x, y) . Then $\{x_n\} \rightarrow x$, $\{y_n\} \rightarrow y$. The converse is also true in $\mathbb{R}^n$, since given $\epsilon > 0$ we can find a small rectangle $B(x, \epsilon_x) \times B(y, \epsilon_y) \subset B((x, y), \epsilon)$. In turn, we can find N_x, N_y such that $x_n \in B(x, \epsilon_x)$ if $n > N_x$ and $y_n \in B(y, \epsilon_y)$ if $n > N_y$. Choose $N = \max(N_x, N_y)$. Then for $n > N$, $(x_n, y_n) \in B((x, y), \epsilon)$ since it is in the small rectangle inside. With this in mind, let $\{(x_n, y_n)\}$ be a Cauchy sequence. Then because $\sqrt{(x_n - x_m)^2 + (y_n - y_m)^2} < \epsilon$ implies $|x_n - x_m| < \epsilon$, $|y_n - y_m| < \epsilon$, $\{x_n\}$ and $\{y_n\}$ are each Cauchy sequences in $\mathbb{R}$. By problem 2, these converge, so $\{(x_n, y_n)\}$ converges. Hence $\mathbb{R}^2$ is complete.

II.3.4 There are two ways of approaching this. We can either repeat the arguments in the answers to exercises 3.2 and 3.3 for $\blacksquare$, or we can peek ahead to the definition of a closed set. If (X, d) is a complete metric space, the only way a Cauchy sequence in a subset can fail to converge in the subset is if it converges to a limit which is not in that subset, because viewed as a Cauchy sequence in X , it has a limit. The limit of this sequence is thus a limit point of the subset, and hence if the set is closed, it is in the set. So any closed subset of a complete metric space is complete. $\blacksquare$ is a closed subset of $\mathbb{R}^2$ so it is complete by problem 3.

II.3.5 The sphere can be thought of as an embedded closed subset of $\mathbb{R}^3$. The spherical metric is equivalent to the $\mathbb{R}^3$ Euclidean metric. To see this, note that the Euclidean metric, used on the sphere gives to each pair of points x, y the length of the chord between them. This length, for a sphere of radius 1 is $\sqrt{2 - 2\cos\theta}$ (by the law of cosines). Using the constants $2/\pi$ and $1/\pi$ this is a metric equivalence to the distance measure of great circles. If $\{x_i\}$ is a Cauchy sequence on the sphere, it is in $\mathbb{R}^3$ as well by this equivalence, consequently it has some $(\mathbb{R}^3)$ limit x . Since the sphere is a closed subset of $\mathbb{R}^3$, it contains its limit points, and therefore contains x . Hence the sequence has a limit, and the sphere is complete.

II.3.6 Let $\{x_\sigma\}$ be a Cauchy sequence on the codespace Σ on N symbols. Then given $\epsilon > 0$ there is an M such that if $\sigma_1, \sigma_2 > M$ then $d(x_{\sigma_1}, x_{\sigma_2}) < \epsilon$. In particular, if we choose $\epsilon = \frac{1}{(N+1)^i}$ then since any of the first i terms must either be 0 or contribute more than this value, $x_{\sigma_1}, x_{\sigma_2}$ must agree on the first i terms. Choose x such that for each i , the first i terms of x agree with all those elements of $\{x_\sigma\}$ with $\sigma > M_i$, where M_i is the M corresponding to the above choice of ϵ . Then $x \in \Sigma$ and for each $\epsilon > 0$ there is an M such that $d(x, x_\sigma) < \epsilon$ if $\sigma > M$. In other words x is the limit of $\{x_\sigma\}$. Hence codespace is complete.

II.3.7 Let $\{f_i\}$ be a Cauchy sequence on $(C[0, 1], D)$. Then for each $s \in [0, 1]$, the sequence $\{f_i(s)\}$ is Cauchy in $\mathbb{R}$, since given ϵ and N guaranteed by the assumptions, $d(f_i(s), f_j(s))$ in the Euclidean metric is certainly less than the maximum value over s . We will anticipate a little and call the limit of each such sequence for each s $f(s)$. As defined, $f(s)$ is clearly a function on $[0, 1]$, we must show that it is continuous. Given $\epsilon > 0$ there is a $\delta > 0$ such that if $d(s_1, s_2) < \delta$ then for any i , $D(f_i(s_1), f_i(s_2)) < \epsilon/3$. This property, called equicontinuity, is true for bounded functions on a closed interval, such as we have. Then there is also an N such that if $i > N$ $d(f_i(s_1), f(s_1)) < \epsilon/3$ and $d(f_i(s_2), f(s_2)) < \epsilon/3$ since the sequence converges. Consequently, for our choice of δ ,

$$d(f(s_1), f(s_2)) < d(f_i(s_1), f(s_1)) + d(f_i(s_1), f_i(s_2)) + d(f_i(s_2), f(s_2)) < \epsilon$$

by the triangle inequality, and hence $f(s)$ is continuous. Then $f \in (C[0, 1], D)$ and the space is complete.

II.3.8 Let (X_1, d_1) and (X_2, d_2) be equivalent metric spaces. Then there is a homeomorphism $h: X_1 \rightarrow X_2$ such that $\tilde{d}(x, y) = d_2(h(x), h(y))$ is equivalent to $d_1(x, y)$ on X_1 . Let $\{x_n\}$ be a Cauchy sequence in X_2 , and let $\epsilon > 0$, and N large enough that $d_2(x_n, x_m) < \epsilon$ whenever $n, m > N$. There exists a sequence $\{y_n\}$ in X_1 such that $h(y_n) = x_n \forall n$, and by the metric equivalence, we have that

$$d_1(y_n, y_m) < c_2 \tilde{d}(y_n, y_m) = c_2 d_2(x_n, x_m) < c_2 \epsilon$$

Consequently, by specifying $\delta = c_2 \epsilon$, for any $\epsilon > 0$ there is an N such that $d_2(x_n, x_m) < \delta$ which forces $d_1(y_n, y_m) < \epsilon \forall n, m > N$, in other words $\{y_n\}$ is a Cauchy sequence in X_1 . This has a limit y because X_1 is complete. Since for all $\delta > 0$ there is an N such that $d_1(y, y_n) < \delta$ for $n > N$, then since $\tilde{d}(y, y_n) < (1/c_1)d_1(y, y_n) < (1/c_1)\delta = \epsilon$, it follows that

$\tilde{d}(y, y_n) = d_2(h(y), x_n)$, $\{x_n\} \rightarrow h(y) = x$ and the sequence converges in X_2 . Hence X_2 is complete. In the process, if two spaces are metrically equivalent, then a Cauchy sequence in one corresponds to a Cauchy sequence in the other under the metric equivalence.

II.3.9 Given two points $x = (x_1, x_2)$, $y = (y_1, y_2)$ in $(\blacksquare, \text{Manhattan})$, any path which consists of straight horizontal and vertical segments, which are monotonic in each coordinate have the same length. Since the sum of these lengths is equal to the distance between the two points, they are all shortest paths.

II.3.10 Suppose $\{x_n\} \rightarrow x$ in the space X . Then given $\epsilon/2 > 0$ there exists an N such that $d(x, x_n) < \epsilon/2$ if $n > N$. Choose two such points x_n, x_m then by the triangle inequality

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \epsilon$$

So given ϵ , if $n, m > N$, $d(x_n, x_m) < \epsilon$, in other words, the sequence is Cauchy.

II.3.11 Suppose $\{x_n\} \rightarrow x$ and $\{x_n\} \rightarrow y$. Then given $\epsilon/2 > 0$ there is an N_1 such that $d(x_n, x) < \epsilon/2$, and an N_2 such that $d(x_n, y) < \epsilon/2$. Choose $N = \max(N_1, N_2)$. Then for $n > N$

$$d(x, y) \leq d(x, x_n) + d(x_n, y) < \epsilon$$

Consequently, for any $\epsilon > 0$, $d(x, y) < \epsilon$. In other words, $d(x, y) = 0$, or $x = y$.

II.3.12 Problem is misstated. 0 is the limit regardless, in the second space the sequence does not converge because 0 is not in the space.

II.3.13 Define the sequence $\{x_n\}$ by $x_n = a \forall n$. This sequence is clearly Cauchy, $d(x_n, x_m) = 0 < \epsilon$ for any $\epsilon > 0$ and any n, m . It also converges since $d(a, x_n) = 0 < \epsilon$ for all n and any $\epsilon > 0$. The space contains no limit points since a limit point is defined as being the limit of a sequence but not a member of the sequence. Since this is the only sequence in the space, the space contains no limit points and therefore it contains all its limit points (there are none) but is not perfect since a is not a limit point.

II.3.14 $\{x_n = n\}$ does not converge in $(\mathbb{R}, \text{Euclidean})$. Suppose it did. Then by Theorem 3.1 this is a Cauchy sequence. But for any $n \neq m$ $d(x_n, x_m) \geq 1$ and thus is not less than any given $\epsilon > 0$ for any N . For $(\hat{\mathbb{C}}, \text{spherical})$, the distance to the Point at Infinity is calculated by looking at the tangent of the angle subtended by the real line and the projection through the sphere. The radius of the sphere is 1 so this tangent is $1/n$. Given $\epsilon > 0$ we may choose N such that $1/N < \epsilon$, and therefore if $n > N$ $d(\infty, x_n) = 1/n < \epsilon$. Hence the sequence converges in $(\hat{\mathbb{C}}, \text{spherical})$.

II.3.15 Let x be a limit point in X_1 . Then there is a sequence $\{x_n\} \rightarrow x$, $x \neq x_n \forall n$, in X_1 by definition of a limit point. Since h provides a metric equivalence, given $\epsilon > 0$ there is a $\delta > 0$ such that if $n > N$, $d_1(x, x_n) < \delta = (1/c_1)\epsilon$ implies that $d_2(h(x), h(x_n)) < \epsilon$. Hence since h is onto $\{h(x_n)\} \rightarrow h(x) \in X_2$. Since h is 1-1 and onto, $h(x_n) \neq h(x) \forall n$. Hence $h(x)$ is a limit point in X_2 .

II.3.16 $(\blacksquare, \text{Euclidean})$ is defined as being those points whose x and y coordinates both lie in $[0, 1]$. The points in this set are all greater than 1 in the y coordinate, hence are not in $\blacksquare$. The set, taken in $\mathbb{R}^2$ has two limit points in $\blacksquare$ (it does not converge as a sequence). These are at $(1, 1)$ and $(-1, 1)$.

II.3.17 Any point of the set S is the limit of a sequence which converges because eventually all the terms are the same. This type of sequence is always possible (see exercise 3.13), however such convergences do not qualify the point as a limit point of the set. The only such point would be 0, since this is the only Cauchy sequence converging to point not in the sequence. Since this point is not in the space $(0, 1]$, the set of limit points of S is empty and therefore a subset of S . Hence S is closed in $(0, 1]$, Euclidean).

II.3.18 Let $S = [0, 1] \subset (\mathbb{R}, \text{Euclidean})$. Let $x \in S$, and let $a = \min(x, 1 - x)$. If $a = 0$, set $a = 1$. In other words, a is the least distance to an end of the interval. Then the sequence $\{x_n = x + a/2^n\}_{n=1}^{\infty}$ lies entirely in S . Given $\epsilon > 0$ there is an N such that $\epsilon > a/2^N$. This implies that for $n > N$, $d(x, x_n) = d(x, x + a/2^n) = a/2^n < \epsilon$, in other words this sequence in $[0, 1]$ converges to x , hence x is a limit point. Since x was any point in the set, the set consists entirely of limit points. It is closed, and therefore perfect.

II.3.19 By reference to exercise 3.17, this set contains its only limit point, 0. It is not perfect since for any n there is no

Cauchy sequence (and hence no convergent sequence converging to $1/n$ which does not contain $1/n$. Hence $1/n$ is not a limit point, and the set is not perfect. It is closed, since it contains its limit point.

II.3.20 That Σ contains its limit points is true because it is the whole space and is therefore closed. We need only verify that each point in Σ is the limit of a sequence $\{y_n\}$ which does not include it. This is done as follows: Let $x = x_1x_2\ldots \in \Sigma$ and suppose we have N symbols. For each n let y_n be the point which agrees with x on all symbols except the n^{th} , and is equal to the next greater symbol at the n^{th} place unless that symbol in x is $N - 1$ in which case use $N - 2$. Then for each n we have

$$d(x, x_n) = \sum_{i=1}^{\infty} \frac{|x_i - y_{ni}|}{(N+1)^i} = \frac{|x_n - y_{nn}|}{(N+1)^n} = \frac{1}{(N+1)^n}$$

because there is only one non-zero term in the sum. This sequence is then convergent—given $\epsilon > 0$ there is an M such that for $n > M$ $d(x, y_n) = 1/(N+1)^n < \epsilon$. No term in the sequence is equal to x because it disagrees at the n^{th} term. Since x is arbitrary, every point in Σ is a limit point and Σ is a perfect set.

II.3.21 If (S, d) is complete, then given a Cauchy sequence in S , it converges to a point in S . Since every limit point is the limit of a convergent sequence in S , and hence a Cauchy sequence in S , S contains its limit points and is therefore closed. Suppose S is closed, and let $\{x_n\}$ be a Cauchy sequence in S . Then $\{x_n\}$ is a Cauchy sequence in $\mathbf{X}$ as well and there is a point $x \in \mathbf{X}$ such that $\{x_n\} \rightarrow x$, because $\mathbf{X}$ is complete. Hence x is a limit point for S . Since S is closed, $x \in S$.

Answers to Chapter II, section 4

II.4.1 In order for $\mathbf{X}$ to be closed, it must contain all its limit points. Thus let $\{x_n\}$ be a convergent sequence in $\mathbf{X}$. By definition, there is an $x \in \mathbf{X}$ such that $\{x_n\} \rightarrow x$. Hence $\mathbf{X}$ is closed. A Cauchy sequence need not converge however. Let $\mathbf{X} \subset Y$, such that Y is complete. Then if $\{x_n\} \rightarrow y \notin \mathbf{X}$, then this sequence does not converge in $\mathbf{X}$, and y is not a limit point in $\mathbf{X}$. The sequence is nevertheless Cauchy, and since it does not converge, $\mathbf{X}$ is not a complete metric space. An example is $\mathbf{X} = (0, 1)$.

II.4.2 See the answer to exercise 3.21 (Chapter II).

II.4.3 By exercise II.3.15, if x is a limit point of the set S in $\mathbf{X}_1$ then $\theta(x)$ is a limit point of the set $\theta(S) \subset \mathbf{X}_2$. Suppose there is an element in $\theta(S)$ which is a limit point, namely that there exists a sequence $\{(\theta(s))_n\} \rightarrow s \in \theta(S)$. Then given $\epsilon > 0$ there is an N such that for $n > N$,

$$d_1(\theta^{-1}(\theta(s)_n), \theta^{-1}(\theta(s))) \leq (1/c_1)d_2(\theta(s)_n, \theta(s)) < \epsilon$$

Letting $\theta^{-1}(\theta(s)_n) = s_n$ we see that $\theta(s)$ is the image of a limit point in S . Since S is closed, the image of s is in $\theta(S)$, thus $\theta(S)$ contains its limit points, and is therefore closed.

II.4.4 Given $x \in \mathbf{X}$, by definition $B(x, \epsilon) = \{y \in \mathbf{X} : d(x, y) < \epsilon\}$ and hence is a subset of $\mathbf{X}$ for any ϵ . Hence $\mathbf{X}$ is open.

II.4.6 Let S be a bounded subset of $(\mathbb{R}^N, \text{Euclidean})$. By definition, there is a point $a \in S$ and a number R , such that $d(s, a) < R \forall s \in S$. Explicitly,

$$\sqrt{\sum_{n=1}^N (s_n - a_n)^2} < R$$

The terms in the sum are all non-negative, it therefore follows that $(s_n - a_n)^2 < R^2$ or that $|s_n - a_n| < R$, for each entry. This means that S is contained in the box $[a_1 - R, a_1 + R] \times [a_2 - R, a_2 + R] \times \cdots \times [a_N - R, a_N + R]$. To provide a finite ϵ -net for S , we subdivide each interval into M intervals such that $N/M < \epsilon$. Then the product of the endpoints of each interval is a point in a finite ϵ -net of this N -cube, its intersection with S is also finite and each point of S is less than ϵ away from one of the above corners. Hence S is totally bounded. To show that $\mathbb{R}^N$ is complete, it suffices to note that $\mathbb{R}$ is complete (exercise II.3.2) and that a sequence is Cauchy or convergent if and only if it is Cauchy or convergent on each of its coordinates. Finally, since $\mathbb{R}^N$ is complete, and S is totally bounded if it is bounded, it follows that if S is closed and bounded it is compact.

II.4.7 Let (X, d) be a metric space, $f : X \rightarrow X$ be continuous, and A be a compact non-empty subset of X . It follows immediately that if $a \in A$, then $f(a) \in f(A)$ by definition, so $f(A)$ is non-empty. Let $\{b_n\}$ be a sequence of points in $f(A)$. Then each b_n is the image of at least one point in A . Let $\{a_n\}$ be any sequence in A such that $f(a_n) = b_n$ for each n . Because A is compact, there is a subsequence $\{a_{n_i}\} \subset \{a_n\}$ which converges to some point $a \in A$. The sequence $\{f(a_{n_i})\}$ is a subsequence of $\{b_n\}$, we must show that it converges to a point in $f(A)$. In fact, because f is continuous, given $\epsilon > 0$, there is a $\delta > 0$ such that if $d(a_{n_i}, a) < \delta$ then $d(f(a_{n_i}), f(a)) < \epsilon$. Since this subsequence converges in A , given $\delta > 0$ there is an N such that $i > N$ implies $d(a_{n_i}, a) < \delta$ and therefore that $d(f(a_{n_i}), f(a)) < \epsilon$. This subsequence therefore converges in $f(A)$, and letting $b_{n_i} = f(a_{n_i})$ we have found a convergent subsequence for the sequence $\{b_n\} \subset f(A)$. Hence $f(A)$ is a compact non-empty subset of X .

II.4.8 Let $S \subset (X_1, d_1)$ be open. Then it follows that by example 2.4.5, $X_1 \setminus S$ is closed. By exercise II.4.3, $h(X_1 \setminus S)$ is closed in X_2 . Let $t \in h(X_1 \setminus S)$. Then it follows that since h is invertible, $h^{-1}(t) \notin S$, hence $t \in X_2 \setminus h(S)$. Hence $h(X_1 \setminus S) = X_2 \setminus h(S)$. Hence $X_2 \setminus h(S)$ is closed which means that $h(S)$ is open in X_2 . Note that the proof depends on the invertibility of h . It does *not* hold even for arbitrary continuous functions.

II.4.10 Given $x \in (0, 1)$ we have $0 < x < 1$ so choose $\epsilon = \min(x, 1 - x)$. Then the set $B(x, \epsilon) = \{y \in X : d(x, y) < \epsilon\}$ is a subset of $(0, 1)$. Hence $(0, 1)$ is open in X . Let $x \in \{2\}$. This is a singleton set, so $x = 2$. Choose $\epsilon < 1$. Then the set $B(x, \epsilon) = \{y \in X : d(x, y) < 1\}$ contains only the point 2 and $d(x, 2) = 0 < \epsilon$. Hence $\{2\}$ is open in X . The two sets are both closed: $(0, 1) = X \setminus \{2\}$ is the complement of an open set, by 2.4.5 it is closed. By the same token $\{2\} = X \setminus (0, 1)$ so it is the complement of an open set and is closed. $\{2\}$ is compact, since given any sequence in $\{2\}$ all entries are 2, hence the sequence converges, and has a convergent subsequence (any subsequence). $(0, 1)$ is not compact, choose the sequence $\{x_n = 1/n\}$. This converges to $0 \notin (0, 1)$, so any subsequence must likewise converge to 0. Hence it has no convergent subsequence.

II.4.11 Let $S \subset (X, d)$. By definition, its boundary ∂S is the set of points s such that any open ball $B(s, \epsilon)$ contains points in S and in $X \setminus S$. But $S = X \setminus (X \setminus S)$, since it is the set of points which are in X which are not in $X \setminus S$. Consequently if any open ball $B(s, \epsilon)$ around a point s contains points in both sets it is in the boundary $\partial(X \setminus S)$, and the two boundaries are the same. X is open (exercise II.4.4) hence $\emptyset$ is closed, and contains its boundary. Since this implies the boundary of X is contained in the empty set, $\partial X = \emptyset$.

II.4.12 Let (X_1, d_1) and (X_2, d_2) be equivalent metric spaces and let $h : X_1 \rightarrow X_2$ be the metric equivalence. Let $S \subset X_1$, and let $s \in \partial S$. If $h(s) \notin \partial h(S)$ then there is an open ball $B(h(s), \epsilon)$ which is either a subset of $h(S)$ or of $X_2 \setminus h(S)$. If it is a subset of $h(S)$, then it is the image of an open set in S (see problem II.4.8), and by definition there is an open ball around s in S , which contradicts the statement $s \in \partial S$. If it is a subset of $X_2 \setminus h(S)$, then there are no points in $B(h(s), \epsilon)$ which are the image of points in S . Hence $B(h(s), \epsilon)$ is the image of an open set containing s which is contained in $X_1 \setminus S$. Then there is an open ball within this open set, which is centered at s , and contains no points in S . Again this contradicts $s \in \partial S$. Consequently, since this is true for all s in ∂S , $h(\partial S) \subset \partial h(S)$. If we do this again for the transformation h^{-1} with the boundary of $h(S)$ we get that the inverse image of the boundary of $h(S)$ is contained in ∂S . Then the two boundaries are images of each other, and $h(\partial S) = \partial h(S)$.

II.4.13 $S \subset (\mathbb{R}, \text{Euclidean})$ is the set of rational points in the real line. We want to show that the boundary of S is $\mathbb{R}$ itself. Let $x \in \mathbb{R}$, and let $\epsilon > 0$ be given. Suppose a, b are two integers such that $b = a + 1$ and $a < x < b$. Now let N be chosen such that $1/2^N < \epsilon$. Since the numbers $a + k/2^N$ for $k \leq 2^N$ subdivide the interval $[a, b]$ into 2^N equal parts of width $1/2^N < \epsilon$ it follows that any interval which is greater than ϵ in width contains at least one of them. But $B(x, \epsilon) = [x - \epsilon, x + \epsilon]$ is of width 2ϵ , hence there is a rational number in $B(x, \epsilon)$. The number $\sqrt{2}$ is not a rational number: Let $p/q = \sqrt{2}$ and suppose the fraction p/q is in lowest terms. Squaring both sides yields $p^2 = 2q^2$ implying that p is even. Hence p^2 is divisible by 4. Hence q is even, a contradiction. If c is irrational, so is $p/(cq)$ for any rational p/q . If not, by simple algebra we may write c in fractional form. Now take all the numbers in $[a, b]$ of the form $a + k/(\sqrt{2}2^N)$. Once again these are evenly spaced and any interval of width less than ϵ contains at least one of them. Hence there is a number in $B(x, \epsilon)$ which is in $\mathbb{R} \setminus S$. Hence $x \in \partial S$. Since x was arbitrary, $\partial S = \mathbb{R}$.

II.4.14 The only point in $\hat{\mathbb{C}}$ for which every ϵ -ball contains points in $\mathbb{C}$ and points in $\hat{\mathbb{C}} \setminus \mathbb{C} = \{\infty\}$ is ∞ . Hence the boundary of $\mathbb{C}$ in $\hat{\mathbb{C}}$ is $\{\infty\}$.

II.4.15 Let $S \subset X$ be closed. Suppose that there is a point in X such that for every $\epsilon > 0$ $B(x, \epsilon)$ contains points in S and points in $X \setminus S$. Define a sequence $\{x_n\}$ by $x_n \in B(x, 1/n)$, $x_n \neq x$. If this cannot be done, then $\partial S = \emptyset$ and the statement is true. Then the above sequence makes x a limit point of S and hence $x \in S$. Hence $\partial S \subset S$.

II.4.16 Suppose S is open. Let $s \in \partial S$. Then for every $\epsilon > 0$ there is a point $x \in B(s, \epsilon)$ such that $x \notin S$. Then there is no $\epsilon > 0$ such that $B(s, \epsilon) \subset S$. Since S is open $s \notin S$. Hence $\partial S \cap S = \emptyset$.

II.4.17 By definition $S^0 = S \setminus \partial S$. Since $S \cap \partial S = \emptyset$, $S \setminus \partial S = S$. Hence $S^0 = S$. Conversely, suppose $S^0 = S$. Then $\partial S = \emptyset$. Consequently for any $s \in S$ there is an $\epsilon > 0$ for which $B(s, \epsilon)$ contains no points in $X \setminus S$. Then $B(s, \epsilon) \subset S$, and S is open.

II.4.18 Since S is closed, by exercise II.4.16, $\partial S \subset S$. By definition $S^0 = S \setminus \partial S$. We have $S \setminus \partial S \cup \partial S = S$.

II.4.19 By exercise II.4.12, being the boundary of a set is invariant under metric equivalence. Thus if $s \in S^0$, then $s \in S$ and $s \notin \partial S$. Since $h(s) \in h(S)$, and $h(s) \in \partial h(S) = h(\partial S)$ implies that $s \in \partial S$ which contradicts hypothesis, $h(s) \in (h(S) \setminus \partial h(S))$ which means $h(s) \in (h(S))^0$.

II.4.20 $X \setminus \partial S$ is $S \setminus \partial S \cup (X \setminus S) \setminus \partial S$. Since $\partial S = \partial(X \setminus S)$ (by exercise II.4.11) $X \setminus \partial S = (X \setminus S)^0 \cup S^0$, which are disjoint sets, and by II.4.17, are open. Note that one or the other or both may be empty, however. For the examples, (a) the boundary of the set $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$ is the set $\partial S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, since every ϵ -ball around a point in this set contains $(x, y) \in S$ and as well (x, y) such that $x^2 + y^2 > 1$. This is a circle of unit radius and divides the plane into the open disk interior, and the plane with a circular hole of radius one in it. (b) $S = \mathbb{R}^2$ the whole space, so $\partial S = \emptyset$. The two sets $\mathbb{R}^2$ and $\emptyset$ are both open as shown in II.4.1 and II.4.4.

II.4.21 By problem II.4.20, $X \setminus \partial S$ is the union of two disjoint open sets. The union of open sets is open, hence ∂S is closed.

II.4.22 We start this problem with an elementary fact about compact sets. Let C be a compact set. Then if $A \subset C$ is a closed subset, let C_n be an open cover for A . We can extend this cover to an open cover of C , by adding ϵ -balls in $C \setminus A$ (which is open) we can form an open cover for C . There is therefore a finite subcover $\{C_1, \dots, C_N\}$. Since by our choice, all of the C_n which cover A are from the original open cover, and they are a subset of this finite subcover, there is a finite subcover for A . It follows that any closed subset of a compact set is compact.

We now turn to the problem at hand. S is a subset of a compact metric space. By problem II.4.21, ∂S is closed, and also a subset of the same compact metric space. Therefore ∂S is compact. The assertion is best proven by proving the following statement which is true for arbitrary subsets S (closed, open, otherwise). Let S be a subset of a metric space (X, d) . Then $A \subset S$ is open in the space (S, d) if and only if we can write $A = S \cap O$ for some open subset of X . If A is open in (S, d) then for each $a \in A$ and each $\epsilon > 0$

$$B_S(a, \epsilon) = \{y \in S : d(a, y) < \epsilon\} = \{y \in X : d(a, y) < \epsilon\} \cap S = B_X(a, \epsilon) \cap S$$

so $A = S \cap O$ where O is the union of such balls in X . Conversely, if $A = S \cap O$ for some open set $O \subset X$ then given $a \in A$, there is an $\epsilon > 0$ such that $B_X(a, \epsilon) \subset O$, and

$$B_S(a, \epsilon) = \{y \in S : d(a, y) < \epsilon\} = \{y \in X : d(a, y) < \epsilon\} \cap S = B_X(a, \epsilon) \cap S$$

The latter intersection is in $A = S \cap O$, so A is open.

The assertion that the interior of the land is open in Z despite the fact that it includes points in the “border” of Z , is true by the above statement. The fact that it appears to include border points comes about because these points are points in $\partial Z \subset X$, whereas when Z is itself the metric space, the border of the whole space is the empty set.

Answers to Chapter II, section 5

II.5.1 The properties of connectedness preserved by metric equivalence: (i) Connectedness. Let (X_1, d_1) and (X_2, d_2) be equivalent metric spaces and $h : X_1 \rightarrow X_2$ be the invertible function giving this equivalence. It was proven last section (problem II.4.8) that if S is an open set, then $h(S)$ is open, and (problem II.4.3) if S is a closed set, then $h(S)$ is closed. Because h^{-1} also establishes an equivalence, the same is true about closed and open sets in X_2 under h^{-1} . Let X_1 be connected. Suppose $S \subset X_2$ is both closed and open. Then $h^{-1}(S)$ is closed because S is closed, and open because S is open. Since X_1 is connected, $h^{-1}(S)$ is either X_1 or $\emptyset$. In the first case, because h is onto, $S = X_2$. In the second $S = \emptyset$. Hence X_2 is connected.

(ii) By studying the definition of metric equivalence, we see that given $\epsilon > 0$ there is a $\delta = (1/c_2)\epsilon$ such that $d_2(h(x), h(y)) < \epsilon$ whenever $d_1(x, y) < \delta$. Consequently h is a continuous function. The composition of continuous functions is continuous. Suppose X_1 is pathwise connected. Given two points $x, y \in X_2$ because h is onto, there are a pair of points $u, v \in X_1$ such that $h(u) = x, h(v) = y$. Because X_1 is pathwise connected, there exists a continuous function $f : [0, 1] \rightarrow X_1$ taking $f(0) = u, f(1) = v$. But then there is a continuous function $h(f(y))$ taking a point $y \in [0, 1]$ to X_2 such that $h(f(0)) = h(u) = x$ and $h(f(1)) = h(v) = y$. Consequently, X_2 is pathwise connected.

(iii) Suppose X_1 is disconnected. Then there exist a set $A \neq X_1$ and $A \neq \emptyset$ which is both open and closed in X_1 . As in the proof of i), since this set is open, $h(A)$ is open in X_2 , since the set is non-empty, $h(A)$ is non-empty, and since $A \neq X_1$ and h is onto, $h(A) \neq X_2$. Hence X_2 is also disconnected.

(iv) Suppose X_1 is pathwise disconnected. Then there exist points $x, y \in X_1$ such that there is no continuous function mapping $[0, 1]$ to X_1 with $f(0) = x, f(1) = y$. Suppose there were a continuous function $g : [0, 1] \rightarrow X_2$ such that $g(0) = h(x), g(1) = h(y)$. Then, because h^{-1} is continuous, the function $h^{-1}(g(w))$ taking the interval to X_1 would take $h^{-1}(g(0)) = x, h^{-1}(g(1)) = y$, a contradiction. Hence X_2 is pathwise disconnected.

(v) Suppose X_1 is simply connected, and, because h is continuous and onto, let $f_1, f_2 : [0, 1] \rightarrow X_2$ be two paths connecting points $h(x), h(y)$ in X_2 . Then there exist paths $h^{-1}(f_1([0, 1])), h^{-1}(f_2([0, 1]))$ connecting $x, y \in X_1$. There is then a continuous deformation $g(s, t)$ transforming $h^{-1}(f_1([0, 1]))$ to $h^{-1}(f_2([0, 1]))$. Then $h(g([0, 1] \times [0, 1])) \rightarrow X_2$ is a continuous function, and

$$h(g(s, 0)) = h(h^{-1}(f_1(s))) = f_1(s),$$

$$h(g(s, 1)) = h(h^{-1}(f_2(s))) = f_2(s),$$

while

$$h(g(0, t)) = h(h^{-1}(h(x))) = h(x), \text{ and}$$

$$h(g(1, t)) = h(h^{-1}(h(y))) = h(y),$$

hence X_2 is simply connected.

(vi) Finally, assume the converse, that X_1 is multiply connected. Then there exist two points $x, y \in X_1$ and two paths $f_1, f_2 : [0, 1] \rightarrow X_1$ with no continuous deformation between them. We again transfer everything to X_2 . The points $h(x), h(y)$ in X_2 have the two paths $h(f_1([0, 1])), h(f_2([0, 1]))$ connecting them. If there were a continuous deformation $g : [0, 1] \times [0, 1] \rightarrow X_2$ deforming one path to the other, then $h^{-1}(g([0, 1] \times [0, 1])) \rightarrow X_1$ would provide a deformation in X_1 . Hence there is no such deformation, and X_2 is multiply connected.

II.5.2 The metric space $(\blacksquare, \text{Euclidean})$ is simply connected: Choose x, y and two functions from $[0, 1] \rightarrow \blacksquare$ such that $f_0(0) = f_1(0) = x$ and $f_0(1) = f_1(1) = y$. Let g in the definition be given by $g(s, t) = t \cdot f_0(s) + (1 - t) \cdot f_1(s)$, where $a(x, y) = (ax, ay)$. This satisfies the four demands of the definition, we must only check that for all values of (s, t) it remains within the square. It would be maximized in, say, the x coordinate by setting $f_0(s) = f_1(s) = (1, y)$ in which case for any t the function would equal $t + 1 - t = (1, y)$. Since this is clearly in the square, there is no problem there. The other cases are identical. Hence $\blacksquare$ is simply connected.

II.5.3 To show that $X = ((0, 1) \cup \{2\}, \text{Euclidean})$ is disconnected, we examine the subset $\{2\}$. This set is closed since points are closed in metric spaces. It is also open: The ball $B_\epsilon(2) \subset \{2\}$ for $\epsilon = 1/2$ since by definition this ball is the set of all points in X which are closer than $1/2$ to the point 2 of which there is only the point itself. Consequently, for every point in the set $\{2\} \subset X$, there is an open ball around that point in the subset, hence the subset is open. Since this set is both open and closed, and not empty or equal to X , the set is disconnected.

II.5.4 Given the codespace metric we have used, for the space Σ on the N symbols $0, \dots, (N - 1)$, there is a metric equivalence to the subset of $[0, 1]$ written in base $N + 1$ of real numbers with no $N + 1$ -ary digits equal to N . The equivalence is generated by the mapping $x_1x_2 \dots \mapsto .x_1x_2 \dots$. The map has the same distance formula in both cases, and is clearly 1-1 and onto. The set of such real numbers is a totally disconnected subset of $([0, 1], \text{Euclidean})$: Given any closed subset of the real line, either that subset is a single point, or an interval (possibly $[a, \infty)$). If not, we may write the subset as the union of two closed disjoint subsets, and there is a distance between them say c . Then for any point in each subinterval, an ϵ -ball around that point of radius $c/3$ contains only other points from the same subinterval, hence each subinterval is

open as well as closed and the set is disconnected. Since the set given above contains no intervals of with numbers with the digit N it is clearly not an interval, and it clearly contains more than one point. Hence codespace is disconnected. To see that it is totally disconnected, suppose there were an interval enclosed in the equivalent subspace of $[0, 1]$. Then if $a = .x_1x_2\ldots$ and $b = .y_1y_2\ldots$ suppose they agree to $x_i \leq y_i$. Then the point $.x_1x_2\ldots(z_i + 1 = N)\ldots$ is in the interval $[a, b]$ and not in the set. Hence there are no intervals in the set, and it is totally disconnected, by our metric equivalence, codespace is totally disconnected.

II.5.5 Put the annulus in $\mathbb{R}^2$ centered at the origin, take x, y on either side of the origin, and take f_0 to be a path which goes around the upper half of the annulus, f_1 a path which goes around the lower half. Suppose the function $g(s, t)$ exists as in the definition. Then for each point $f_0(s), s \neq x, y$ we have $f_0(s)|_y > 0$ and the corresponding value for $f_1(s)$ is negative. Holding s fixed the intermediate value theorem says that there is a value of t such that $g(s, t)|_y = 0$. Since we may take the annulus as thin as we please, this says that every point must pass through either x or y as it is deformed from one path to the other. If some points pass through x and some through y there is a value of t such that $g(s, t)$ is not connected, hence is not a path. Hence the annulus is multiply connected.

II.5.6 Choose the set of nested subsets of $\mathbb{R}^2$ given by $S_i = \mathbb{R}^2 B_i$ where $B_i = \{(x, y) : -1 \leq x \leq 1, y \leq i\}$. These sets are nested, they are all connected, and the intersection is the plane minus a strip around the y -axis 2 units wide, which is disconnected.

II.5.8 The body space is multiply connected, it has a hole corresponding to the entire gastro-intestinal tract, if not others (e.g. pierced ears).

Answers to Chapter II, section 6

II.6.1 If x and y are elements of $\mathcal{H}(\mathbf{X})$ then by definition they are both non-empty and both compact. Their union is therefore non-empty. To see that it is also compact, let $\mathcal{C}$ be a covering of $x \cup y$ by open sets. Then there is a subset of $\mathcal{C}$ which covers x , say $\mathcal{C}_x$ and another $\mathcal{C}_y$ which covers y . Because x is compact, there is a finite subset of $\mathcal{C}_x$ which covers x , likewise for y . The collection of sets formed by taking all the sets in the finite subcover of x and all of those in the finite subcover y forms a subcover of $x \cup y$, and since its cardinality is the sum of two finite integers, is also finite. Hence $\mathcal{C}$ admits a finite subcover of $x \cup y$ which is therefore compact. Since $x \cup y$ is compact and non-empty, it is in $\mathcal{H}(\mathbf{X})$. To see that the intersection of two elements in $\mathcal{H}(\mathbf{X})$ need not be in $\mathcal{H}(\mathbf{X})$, take x and y disjoint. Then $x \cap y$ is empty and therefore not in $\mathcal{H}(\mathbf{X})$.

II.6.2 The elements of $\mathcal{H}(\mathbf{X})$ are compact non-empty subsets of $\mathbf{X}$, while a compact non-empty subset of $\mathbf{X}$ is only a single point in $\mathcal{H}(\mathbf{X})$. An infinite collection of compact non-empty subsets of $\mathbf{X}$ is a valid subset of $\mathcal{H}(\mathbf{X})$, while its union may or may not be a compact non-empty subset of $\mathbf{X}$.

II.6.3 Let $B, C \in \mathcal{H}(\mathbf{X})$ and $B \subset C$. Let $x \in \mathbf{X}$. Then

$$\begin{aligned} d(x, C) &= \min_{y \in C} (d(x, y)) \\ &= \min_{y \in B} (\min_{y \in C} (d(x, y))), \min_{y \in (C \setminus B)} (d(x, y)) \\ &\leq \min_{y \in B} (d(x, y)) = d(x, B) \end{aligned}$$

II.6.4 The distance from $(1, 1)$ to the closed disk of radius $\frac{1}{2}$ centered at $(\frac{1}{2}, 0)$ is given by the distance to the point on the boundary of the disk closest to $(1, 1)$, since this point is not in the disk. The closest point on the boundary is on a radius of the disk, by the triangle inequality. This means the distance is the distance from $(1, 1)$ to $(\frac{1}{2}, 0)$ minus the distance along this chord which is inside the disk, namely $\frac{1}{2}$. The whole distance is $\sqrt{1 + (1 - 1/2)^2} = \sqrt{5}/4$ so the distance $d(x, B) = (\sqrt{5} - 1)/2$.

II.6.5 In this case, the closed disk of radius $\frac{1}{2}$ looks like a square with corners at $(0, 0), (0, 1), (1/2, 1/2), (1/2, -1, 2)$ since this is the set of points equidistant from $(1/2, 0)$ at a distance $1/2$. In the Manhattan metric, the distance to any point on the right upper face of this square is $1/2 + 1/2 = 1$. Since the distance to any other face is larger, $d(x, B) = 1$.

II.6.6 The elements of the set B are less than 3 only if n is odd. The sequence $\{\frac{-n}{n^2+1}\}$ increases monotonically towards zero,

consequently it has its least value at $1/2$. The least value in B then is the value $x_1 = 3 - 1/2$ or $5/2$. The distance to the point $x = 1/2$ is then $d(x, B) = 2$.

II.6.7 It is easiest to provide a counterexample. Let $B \subset A$ be a proper subset, such that there is an $x \in A$ with $d(x, B) = 1$. (For instance two disks centered at the origin one with radius 1 one with radius 2). Then $d(A, B) \geq 1$. Since $B \subset A$ the minimum distance from any point in B to A is to the point itself, namely 0. Hence $d(B, A) = 0$. The two are not equal, so this is not a metric on $\mathcal{H}(X)$.

II.6.11 Let $A, B \in \mathcal{H}(X)$, $A \neq B$. Suppose $d(A, B) = 0$. Then for every $x \in A$, $d(x, B) = 0$, since we take the maximum over these to form $d(A, B)$. This can only occur if $\min_{y \in B}(d(x, y)) = 0$, for every point $x \in A$, which means that for each one, there is a $y \in B$ with $y = x$. In other words $A \subset B$. By the same argument, if $d(B, A)$ is also 0, then $B \subset A$. If $A \subset B$ and $B \subset A$ then $A = B$. This is not the case so one or the other distance cannot be 0. The converse of the above is that if $A \subset B$, then for each point in A ,

$$d(x, B) = \min_{y \in B}(d(x, y)) = d(x, x) = 0$$

so that the maximum over these is 0. Hence if $A \subset B$ then $d(A, B) = 0$.

II.6.12 Using the solution to problem II.6.3, for each $x \in A$ we have $d(x, C) \leq d(x, B)$. Suppose $d(A, B) < d(A, C)$. Then there is a point $x \in A$ such that $d(x, C) > d(x, B)$ for all $y \in A$. This is a contradiction, choosing $y = x$. Hence $d(A, C) \leq d(A, B)$.

II.6.14 We write out the expressions on either side of the inequality.

$$d(A, B) = \max_{x \in A}(\min_{y \in B}(d(x, y)))$$

On the other side, we have

$$d(A, C) + d(C, B) = \max_{x \in A}(\min_{z \in C}(d(x, z))) + \max_{w \in C}(\min_{y \in B}(d(w, y)))$$

Since the maximum over $w \in C$ is greater than or equal to the value if w is replaced by z , we can write

$$d(A, C) + d(C, B) \geq \max_{x \in A}(\min_{z \in C}(d(x, z)) + \min_{y \in B}(d(z, y)))$$

For any choice of $x \in A$ and $y \in B$ the sum is greater than the distance $d(x, y)$ by the triangle inequality, including the case where z has been chosen to minimize $d(x, z)$ and where y has been chosen to minimize $d(z, y)$. Hence for the choice of y the sum of the minima above is greater than $d(x, y)$ for that choice of y and hence greater than the distance $d(x, y)$ which is the minimum chosen to minimize $d(x, y)$ directly. We may then write

$$d(A, C) + d(C, B) \geq \max_{x \in A}(\min_{y \in B}(d(x, y))) = d(A, B)$$

For the other case, whether or not $d(A, B) \leq d(C, A) + d(C, B)$, we use the case $C \subset B \subset A$. Then the right hand side is identically zero, while if $A \neq B$, the left hand side is not. Hence the expression is not true in general.

II.6.16 We make extensive use of II.6.12. To this end, we have

$$h(A \cup B, C \cup D) = d(A \cup B, C \cup D) \vee d(C \cup D, A \cup B).$$

If $h(A \cup B, C \cup D) = d(A \cup B, C \cup D)$ then either $h(A \cup B, C \cup D) = d(A, C \cup D)$ or $h(A \cup B, C \cup D) = d(B, C \cup D)$. Choose the former. then by II.6.12,

$$h(A \cup B, C \cup D) \leq d(A, C) \leq h(A, C) \leq h(A, C) \vee h(B, D).$$

If we choose the latter,

$$h(A \cup B, C \cup D) \leq d(B, D) \leq h(B, D) \leq h(A, C) \vee h(B, D)$$

If $h(A \cup B, C \cup D) = d(C \cup D, A \cup B)$ then either $h(A \cup B, C \cup D) = d(C, A \cup B)$ or $h(A \cup B, C \cup D) = d(D, A \cup B)$.

Choose the former, then by II.6.12,

$$h(A \cup B, C \cup D) \leq d(C, A) \leq h(A, C) \leq h(A, C) \vee h(B, D).$$

If we choose the latter,

$$h(A \cup B, C \cup D) \leq d(D, B) \leq h(B, D) \leq h(A, C) \vee h(B, D).$$

Since this is all the possibilities, the result follows.

Answers to Chapter II, section 7

II.7.3 The reasonable assumptions we could make are that each image is a set of points characterized by spatial coordinates and color coordinates. The spatial coordinates for a photograph are in two dimensions and form a filled square, they are thus compact. If we assume that the colors are a closed and totally bounded region, say grey scales from 1–100, then the product space of pictures is compact. We assume some continuity in time, this is backed up by physical assumptions, the tree will probably not do something totally disconnected from its motion in the interval before. Finally, we assume that the tree waves at finite speed. This implies that the change between photographs in picture space will decrease as the time interval shortens. This should yield a Cauchy sequence given that the distance between two photographs is the Hausdorff metric, applied to successive pictures as sets of points with (x, y, color) coordinates, and, say a Euclidean metric on this 3-space. A looks like a photograph of the tree at 1 second.

II.7.4 In the space of non-empty compact subsets of the Sierpinski triangle, an example of an infinite set is (among others) the set of all singleton sets $\{x\}$ with $x \in \Delta$. A Cauchy sequence in this set is given by successive righthand corners of triangles with lefthand (lower) corners at the lefthand lower corner of Δ . This sequence converges to the lefthand lower corner of Δ , the convergence implies that it is Cauchy.

II.7.6 If X is complete, then so is $\mathcal{H}(X)$. Since this is the case if X is compact, it remains to show that $\mathcal{H}(X)$ is totally bounded. Let ϵ be given, and let $\{y_1, \dots, y_k\}$ be an ϵ -net for X . The singleton sets $\{y_1\}, \dots, \{y_k\}$ are elements of $\mathcal{H}(X)$ since such sets are compact in a metric space. Let $A \in \mathcal{H}(X)$. The distance $h(\{y_i\}, A)$ is given by

$$d(y_i, A) \vee d(A, y_i)$$

where by definition, the first distance is the distance from y_i to the closest point in A , and the second is given by the maximum over points in A of the shortest distance from one of those points to y_i that is the maximum distance of a point to y_i so it is clear that this set of singleton sets does not in itself form an ϵ -net of $\mathcal{H}(X)$. However, since the collection is finite, its power set is also finite (the collection of all possible sets made from the points y_i . Take that union of singletons such that the distance from each of the singletons to A is less than ϵ and call this set Y . This certainly exists, since each element of A is closer than ϵ to some y_i . Then the first element of the above formula is less than ϵ . So is the second. For every element of A the distance to one of the elements in $\{y_i\}$ is less than ϵ consequently $d(a, Y) < \epsilon \forall a \in A$. Hence $\mathcal{H}(X)$ is totally bounded and hence compact.

Answers to Chapter III, section 1

III.1.1 For all $x \in X$ we may write

$$\begin{aligned} f^m \circ f^n(x) &= \overbrace{f(f(\dots f(f^n(x)) \dots))}^{m \text{ copies}} \\ &= \underbrace{f(f(\dots f(f(f(\dots f(x) \dots)) \dots)))}_{m+n \text{ copies}} \\ &= f^{m+n}(x). \end{aligned}$$

III.1.2 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 2x$. Let $f(x_1) = f(x_2)$. Then $2x_1 = 2x_2$ which implies that $x_1 = x_2$ so f is one-to-one. Let $y \in \mathbb{R}$. Then for any y , $1/2y \in \mathbb{R}$, so that $f(1/2y) = 2(1/2y) = y$. Then f is onto. Since f is both one-to-one and onto,

it is invertible. Each iteration multiplies the result of the previous iteration by 2. Hence we have

$$f^n(x) = \underbrace{2(2 \cdots (2x) \cdots)}_{n \text{ copies}} = 2^n x.$$

Similarly,

$$f^{-n}(x) = \underbrace{\frac{1}{2}(\frac{1}{2}(\frac{1}{2} \cdots (\frac{1}{2}x) \cdots))}_{n \text{ copies}} = 2^{-n}x.$$

Thus for any integer, we have $f^n(x) = 2^n x$.

III.1.3 Let $f: [0, 1] \rightarrow [0, 1]$ with $f(x) = 1/2x$. f is one-to-one since $1/2x_1 = 1/2x_2 \Rightarrow x_1 = x_2$. But f is not onto, since $f(x) = 1 \Rightarrow x = 2 \notin [0, 1]$. Hence f is not invertible.

III.1.4 Let $f: [0, 1] \rightarrow [0, 1]$ with $f(x) = 4x(1-x)$. Taking the first derivative, we have $f'(x) = 4 - 8x$, which has a zero at $x = 1/2$ which is a maximum. The value of f there is 1. The minimum value must be at one or the other or both endpoints. The function takes the value 0 at both endpoints. Hence the function is onto. It is not one-to-one however, for example $f(1/4) = f(3/4) = 3/4$.

III.1.5 The first thing to show is that $f(x) \in C$ if $x \in C$. We demonstrate this by noting that if $x \in C$ then x is the endpoint of some closed interval of the form either $[(2^n 3^m)/(3^k), (2^n 3^m + 1)/(3^k)]$ or $[(3^n - 1)/(3^k), (3^n)/(3^k)]$. Hence f raises the power of the denominator by 1 and is still of this form. So let $f(x) \in C$. f is one-to-one since $x_1/3 = x_2/3 \Rightarrow x_1 = x_2$. It is not onto, since $1 \in C$ and $x/3 = 1 \Rightarrow x = 3 \notin C$. To find another transformation, note that if $x \in C$ then $1 - x \in C$. So $f(x) = (1 - x)/3$ is another such transformation.

III.1.6 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $f(x_1, x_2) = (2x_1, x_2^2 + x_1)$. f is not invertible because it is not one-to-one. Let $y = f(x) = f(x_1, x_2)$. Then $y = f(x_1, -x_2)$ as well, since x_2 appears in the expression for f only in squared form.

III.1.8 An affine transformation applied to an interval takes it to another interval, possibly reflecting it (transposing the endpoints in interval notation), hence

$$f([1, 2]) = a[1, 2] + b = \begin{cases} [a + b, 2a + b] & \text{if } a > 0, \text{ and} \\ [2a + b, a + b] & \text{if } a < 0. \end{cases}$$

To map into the interval we must have

$$1 \leq a + b \leq 2, \quad 1 \leq 2a + b \leq 2$$

The new length of the interval will be a , so that $0 \leq a \leq 1$, with $a = 0$ mapping the interval to a single point.

If $f, g: [1, 2] \rightarrow [1, 2]$ then $f([1, 2]) \subset [1, 2]$, hence

$$g \circ f([1, 2]) \subset g([1, 2]) \subset [1, 2]$$

$$f \circ g([1, 2]) \subset f([1, 2]) \subset [1, 2]$$

so that the compositions also map $[1, 2]$ into itself.

We now want the conditions such that

$$f \circ g(\mathbf{X}) \cup g \circ f(\mathbf{X}) = \mathbf{X}$$

Let $f(x) = ax + b$, and $g(x) = cx + d$. We make the simplifying assumption that $f \circ g(\mathbf{X})$ maps the interval to the lower end of itself, and $g \circ f(\mathbf{X})$ to the upper end, and the two overlap across the middle at at least one point. We write out the compositions explicitly:

$$f \circ g(x) = a(cx + d) + b = acx + ad + b;$$

$$g \circ f(x) = c(ax + b) + d = acx + bc + d.$$

The contraction of the interval is the same in each case, and is equal to ac . In order for the two maps to overlap, then, the length of each interval must be at least $1/2$. We therefore have $1/2 \leq |ac| \leq 1$. There are two cases: If $ac > 0$ then $f \circ g$

must map the interval to $[1, 1 + ac]$ and $g \circ f$ must map it to $[2 - ac, 2]$. We have

$$\begin{aligned} ac + ad + b &= 1 \\ 2ac + bc + d &= 2 \\ ad + b &= 1 - ac \\ bc + d &= 2 - 2ac \end{aligned}$$

If $ac < 0$ then $f \circ g$ must take $1 \mapsto 1 - ac$ and $2 \mapsto 1$. Similarly $g \circ f$ takes $2 \mapsto 2 + ac$ and $1 \mapsto 2$. This gives

$$\begin{aligned} 2ac + ad + b &= 1 \\ ac + bc + d &= 2 \\ ad + b &= 1 - 2ac \\ bc + d &= 2 - ac \end{aligned}$$

Given two values for a, c satisfying $1/2 \leq |ac| \leq 1$, these equations allow calculation of b, d such that the above condition is satisfied.

III.1.10 $I_n = f^n(I_0)$. Consequently, we have $f(I_n) = f(f^n(I_0)) = f^{n+1}(I_0) = I_{n+1}$. We are given that $I = \bigcup_{n=0}^{\infty} I_n$ is a finite connected interval, hence

$$f(\bigcup_{n=0}^{\infty} I_n) = \bigcup_{n=0}^{\infty} I_{n+1} = \bigcup_{n=1}^{\infty} I_n = I \setminus I_0$$

Then the length $\ell = \text{length}(I_0) + \text{length}(f(I))$. The length of $f(I)$ is the amount that it was contracted, hence is equal to $a\ell$. This is equal to the length of $I \setminus I_0 = \ell - b$. This gives $a\ell = \ell - b$ or $\ell = b/(1 - a)$. This yields

$$\sum_{n=1}^{\infty} b \cdot a^n = \frac{b}{(1 - a)}$$

III.1.11 Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be polynomial transformations. Then if

$$f(x) = a_0 + a_1x + \cdots + a_Nx^N; g(x) = b_0 + b_1 + \cdots + b_Mx^M$$

we have

$$f \circ g(x) = a_0 + a_1g(x) + a_2(g(x))^2 + \cdots + a_N(g(x))^N$$

If we expand $g(x)$ in the last term in the expression, the term with the highest power of x is $a_N(b_Mx^M)^N$. Since by definition, $a_N, b_M \neq 0, a_Nb_M^N \neq 0$ so this is a polynomial transformation of degree $M \cdot N$. Using this, the degree of $f^m(x)$ is N^m .

III.1.12 For $n > 1$ a polynomial function $f : \mathbb{R} \rightarrow \mathbb{R}$ of degree n is generally not invertible. A quick example is any polynomial having $a_i = 0$ for all odd i . More precisely, suppose that $f(x)$ is invertible. Then there is exactly one number b such that

$$a_0 + a_1b + \cdots + a_Nb^N = 0$$

because the function must be one-to-one. This polynomial must then factor into the two terms

$$(x - b)(c_0 + c_1x + \cdots + c_{n-1}x^{n-1}) = a_0 + \cdots + a_Nx^N$$

The polynomial $c_0 + c_1x + \cdots + c_{n-1}x^{n-1}$ either has no zeroes or it has only one at $x = b$. Assume not. Then by adding or subtracting from c_0 , we can force it to have a zero, thus creating a new function with two roots. Such a function is then not one-to-one and therefore not invertible.

III.1.13 For large enough x , $|x^N| \gg |x^{N-1}| \gg \cdots$. So for large $|x|$, $f(x)$ looks like a_Nx^N which, for $|x| > 1$ is expanding, since $|x_1^n - x_2^n| > |x_1 - x_2|$. If I is an interval $\{x : |x - a| \leq b\}$ for fixed $a, b \in \mathbb{R}$, then if $M > 0$, $\ell(f(I))/\ell(I) > M$ for $b > \beta$. Let $I = [a - b, a + b]$. Construct the polynomials and subtract.

III.1.15 Parabolae opening up or down will work. For example $f(x) = a(x - 1)^2$ where $a \leq 2$.

III.1.16 The function is continuous. $f_\lambda(0) = 0$ for all λ . It is maximized by setting the derivative equal to zero, as follows:

$$\begin{aligned}f_\lambda(x) &= 2\lambda x - \lambda x^2 = \lambda(2x - x^2) \\f'_\lambda(x) &= \lambda(2 - 2x)\end{aligned}$$

so that $f'_\lambda(x) = 0$ when $x = 1$. This is a maximum, since the second derivative is negative, and has the value $f_\lambda(1) = \lambda$. Hence if $\lambda \in [0, 2]$ then $f_\lambda(x) : [0, 2] \rightarrow [0, 2]$. The fold occurs at $(1, \lambda)$.

III.1.17 To be a fold point of $f(x)$, x must be a local maximum or minimum. If $f(x)$ is differentiable, the condition for this is $f'(x) = 0$ and $f''(x) \neq 0$. Polynomial transformations are differentiable. If $f''(x) = 0$ we have an inflection for some functions, (for example $f(x) = x^3$) so no fold exists at x .

III.1.21 To show that the Möbius transformation is invertible, we show that it is one-to-one and onto, as follows:

one-to-one:

Let $f(x_1) = f(x_2)$. Then we have

$$\frac{ax_1 + b}{cx_1 + d} = \frac{ax_2 + b}{cx_2 + d}$$

which yields

$$acx_1x_2 + bcx_2 + adx_1 + bd = acx_1x_2 + bcx_1 + adx_2 + bd$$

or, $bcx_2 + adx_1 = bcx_1 + adx_2$. That is

$$(ad - bc)x_1 = (ad - bc)x_2$$

Since the ‘determinant’ $ad - bc$ is assumed to be non-zero, $x_1 = x_2$ and the function is one-to-one.

onto:

Let $y \in \mathbb{R}$, we need to find x such that

$$y = \frac{ax + b}{cx + d}$$

We have

$$\begin{aligned}cxy + dy &= ax + b \\cyx - ax &= b - dy \\(cy - a)x &= b - dy \\x &= \frac{-dy + b}{cy - a}\end{aligned}$$

Since $ad - bc \neq 0$ we have $(-a)(-d) - bc \neq 0$ so this is another Möbius transformation, and f is onto.

III.1.22 We want to show that if f_1, f_2 are Möbius transformations then $f_1 \circ f_2$ is. Writing

$$f_1 = \frac{a_1x + b_1}{c_1x + d_1}; f_2 = \frac{a_2x + b_2}{c_2x + d_2}$$

we make the necessary substitutions:

$$\begin{aligned}\frac{a_1 \frac{a_2x + b_2}{c_2x + d_2} + b_1}{c_1 \frac{a_2x + b_2}{c_2x + d_2} + d_1} &= \frac{a_1(a_2x + b_2) + b_1(c_2x + d_2)}{c_1(a_2x + b_2) + d_1(c_2x + d_2)} \\&= \frac{(a_1a_2 + b_1c_2)x + (a_1b_2 + b_1d_2)}{(c_1a_2 + d_1c_2)x + (b_2c_1 + d_1d_2)}\end{aligned}$$

which has the form of a Möbius transformation. We need to check:

$$\begin{aligned}
 & (a_1a_2 + b_1c_2)(b_2c_1 + d_1d_2) - (a_1b_2 + b_1d_2)(c_1a_2 + d_1c_2) \\
 &= (a_1a_2b_2c_1 + a_1a_2d_1d_2 + b_1c_2b_2c_1 + b_1c_2d_1d_2) \\
 &\quad - (a_1b_2c_1a_2 + a_1b_2d_1c_2 + b_1d_2c_1a_2 + b_1d_2d_1c_2) \\
 &= (a_1a_2d_1d_2 + b_1c_1b_2c_2) - (a_1d_1b_2c_2 + a_2d_2b_1c_1) \\
 &= b_2c_2(b_1c_1 - a_1d_1) + a_2d_2(a_1d_1 - b_1c_1) \\
 &= (a_2d_2 - b_2c_2)(a_1d_1 - b_1c_1) \neq 0
 \end{aligned}$$

because f_1, f_2 are Möbius transformations by assumption.

III.1.23 The action of the transformation $f(z) = 1/z$ on the Riemann sphere is to flip it over. It takes $1 \mapsto 1$, $-1 \mapsto -1$ so there is no rotation, and $0 \mapsto \infty$, $\infty \mapsto 0$ which inverts the two poles.

III.1.24 When $f(\infty) = \infty$ then $a/c \rightarrow \infty$, since $a < \infty$ we have $c = 0$. Conversely, if $c = 0$ then $f(\infty) = \infty$ by definition. Hence if $\infty \mapsto \infty$ then the transformation looks like

$$f(x) = \frac{ax + b}{d} = \frac{a}{d}x + \frac{b}{d}$$

which is an affine transformation.

III.1.25 $f: \mathbb{R} \rightarrow \mathbb{R}$ takes $1 \mapsto 2$, $2 \mapsto 0$ and $0 \mapsto \infty$. So we have $d = 0$ and the function looks like

$$f(x) = \frac{ax + b}{cx}$$

$1 \mapsto 2$ implies $(a + b)/c = 2$. $2 \mapsto 0$ implies $2a + b = 0$ or $b = -2a$. This yields $-a/c = 2$ so that $f(\infty) = 2$. We can only determine the transformation up to a constant (that is, multiplying top and bottom by the same constant yields another identical transformation) so set $c = 1$. This gives $a = -2$, $b = 4$ so the Möbius transformation

$$f(x) = \frac{-2x + 4}{x}$$

will do it.

Answers to Chapter III, section 2

III.2.1 Let the affine transformation be denoted, as usual, by

$$f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

Since $(0, 0)$ is mapped to $(4, 5)$, we have $e = 4$, and $f = 5$. From the relation $(0, 1) \mapsto (-1, 2)$ we have

$$\begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

or,

$$\begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

which gives $b = -5$ and $d = -3$. Finally, using $(1, 0) \mapsto (3, 0)$, we have

$$\begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} a & -5 \\ c & -3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

or,

$$\begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

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which gives $a = -1$ and $c = -5$. The function then is

$$f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & -5 \\ -5 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

III.2.2 The affine transformation will be invertible if the matrix A is invertible, since if $y = Ax + t$, then $A^{-1}(y - t)$ yields x again if A^{-1} exists. Let A and A^{-1} be denoted by

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

Then we have, by definition,

$$\begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

or,

$$\begin{aligned} ea + fc &= 1 \\ eb + fd &= 0 \\ ga + hc &= 0 \\ gb + hd &= 1. \end{aligned}$$

Solving the first two for e, f ,

$$e = \frac{-fd}{b}$$

$$-fad + fbc = b$$

or

$$f = \frac{-b}{\det A}; e = \frac{d}{\det A}$$

Solving the second two for g, h ,

$$g = \frac{-hc}{a}$$

$$-hbc + had = a$$

or

$$g = \frac{-c}{\det A}; h = \frac{a}{\det A}$$

Putting these together, we get

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Consequently, A^{-1} exists if and only if $\det A \neq 0$. If it does, then if $(x_1, x_2) \mapsto (y_1, y_2)$ we have

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} y_1 - e \\ y_2 - f \end{pmatrix}$$

III.2.3 Let $f_1, f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with

$$f_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

and

$$f_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e' \\ f' \end{pmatrix}.$$

Then we have

$$\begin{aligned} f_3 = f_1 \circ f_2 &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left(\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e' \\ f' \end{pmatrix} \right) + \begin{pmatrix} e \\ f \end{pmatrix} \\ &= \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} ae' + bf' + e \\ ce' + df' + f \end{pmatrix} \end{aligned}$$

Letting

$$\begin{aligned} a'' &= aa' + bc' \\ b'' &= ab' + bd' \\ c'' &= ca' + dc' \\ d'' &= cb' + dd' \\ e'' &= ae' + bf' + e \\ f'' &= ce' + df' + f \end{aligned}$$

then,

$$f_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a'' & b'' \\ c'' & d'' \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e'' \\ f'' \end{pmatrix}$$

which is affine, as desired.

III.2.4 Let

$$\begin{aligned} A &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ B &= \begin{pmatrix} e & f \\ g & h \end{pmatrix} \end{aligned}$$

Then we have

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$$

The products subtracted to form the determinant $\det(AB)$ are given by

$$\begin{aligned} (ae + bg)(cf + dh) &= aecf + aedh + bgcf + bgdh \\ (af + bh)(ce + dg) &= aecf + afgd + bhce + bgdh \end{aligned}$$

Subtracting these yields

$$\begin{aligned} \det(AB) &= aedh + bgcf - afgd - bhce \\ &= ad(eh - fg) - bc(eh - fg) \\ &= (ad - bc)(eh - fg) \\ &= \det A \cdot \det B \end{aligned}$$

as desired.

III.2.5 The easiest way to visualize this is to break the desired transformation into two transformations, one which takes the triangle to the line between $(0, 0)$ and $(1, 1)$ and one which takes this line to the line between $(2, 2)$ and $(1, 1)$. The second transformation is an inversion and translation:

$$\begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

which gives $e = f = 2$ and

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

which gives $a + b - 2 = 1$ and $c + d - 2 = 1$, which by letting $b = c = 0$ gives $a = -1$ and $d = -1$. The first transformation takes the triangle to the line between $(0, 0)$ and $(1, 1)$. $(0, 0) \mapsto (0, 0)$ so that $e = f = 0$. We have

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

so that $b = d = 1$, and

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

so that $a = c = 1$. Multiplying the two together, we have

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

To put this in scaling ratios and rotation angles, let

$$r_1 + r_2 = \sqrt{2}$$

Then the needed rotation angles are

$$\cos \theta_1 = \sin \theta_1 = \cos \theta_2 = \sin \theta_2 = \frac{-1}{\sqrt{2}}$$

which gives $\theta_1 = \theta_2 = \frac{5\pi}{4}$.

III.2.6 We start with the hint. Let

$$w = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

Since $\begin{pmatrix} e \\ f \end{pmatrix}$ is a translation term, and translation does not affect the area of any transformed polygon, without loss of generality $e = f = 0$. As well, it does not change anything to assume that one vertex of the triangle is at the origin. A triangle is then determined by specifying two points (x_1, x_2) and (y_1, y_2) such that $(y_1, y_2) \neq a \cdot (x_1, x_2)$ (which makes them non-collinear), and neither is the origin (which makes less than three distinct vertices, hence no area). First, show that the area of the parallelogram in figure 2.6 is given by

$$\text{area} = |x_1 y_2 - x_2 y_1|$$

(this is the length of the vector in a vector (cross) product). To see this, rotate the parallelogram until (y_1, y_2) becomes $(y'_1, 0)$. Then if the new coordinates for (x_1, x_2) are given by (x'_1, x'_2) the area of the parallelogram is the base times the height, or $y'_1 x'_2$. The rotation is $-\theta$ with

$$\cos \theta = \frac{y_1}{\sqrt{y_1^2 + y_2^2}}; \quad \sin \theta = \frac{y_2}{\sqrt{y_1^2 + y_2^2}}$$

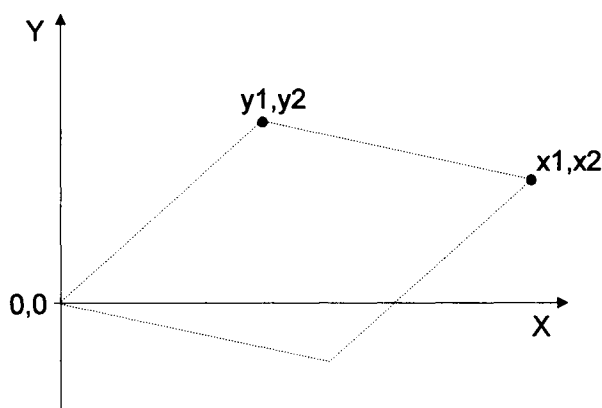


Figure 2.6. Getting the area of a triangle from a parallelogram

We have:

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \cos \theta + x_2 \sin \theta \\ x_2 \cos \theta - x_1 \sin \theta \end{pmatrix}$$

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} y_1 \cos \theta + y_2 \sin \theta \\ y_2 \cos \theta - y_1 \sin \theta \end{pmatrix}$$

so that

$$\begin{aligned} y'_1 x'_2 &= (y_1 \cos \theta + y_2 \sin \theta)(x_2 \cos \theta - x_1 \sin \theta) \\ (y_1^2 + y_2^2) y'_1 x'_2 &= y_1 x_2 y_1^2 - y_1 x_1 y_1 y_2 + y_2 x_2 y_1 x_2 - x_1 y_2 y_2^2 \\ &= y_1^2 (x_2 y_1 - x_1 y_2) + y_2^2 (x_2 y_1 - x_1 y_2) \end{aligned}$$

yielding $|y'_1 x'_2| = |x_1 y_2 - y_1 x_2|$ as desired. The area of the desired triangle is then $\frac{1}{2} |x_1 y_2 - y_2 x_1|$. Apply the transformation

$$w \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

to each of the two vectors. Then the new area is derived by

$$\begin{aligned} x'_1 &= ax_1 + bx_2; & y'_1 &= ay_1 + by_2 \\ x'_2 &= cx_1 + dx_2; & y'_2 &= cy_1 + dy_2 \end{aligned}$$

yielding

$$\begin{aligned} &\frac{1}{2} |acx_1 y_1 + adx_1 y_2 + bcx_2 y_1 + bdx_2 y_2 - acx_1 y_1 - bcx_1 y_2 - adx_2 y_1 - bcdx_2 y_2| \\ &= \frac{1}{2} |(ad - bc)x_1 y_2 - (ad - bc)x_2 y_1| \\ &= \frac{1}{2} |\det A| |x_1 y_2 - x_2 y_1| \end{aligned}$$

A polygon (and many other surfaces) can be broken into triangles on each of which this formula applies, which finishes the proof. To see that $\det A < 0$ corresponds to reflection, write

$$\det \left(\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) = -\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Note: The vector/cross/wedge product used to do the proof this way can be generalized to be applicable on other types of surfaces and hypersurfaces. The generalized version preserves the form $|\det A| \cdot (\text{"area"})$ to such surfaces and volumes.

III.2.7 Let $w(x) = Ax + t$ be an $\mathbb{R}^2$ similitude. Then w has one of the forms:

$$\begin{aligned} w \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix} \\ w \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} r \cos \theta & r \sin \theta \\ r \sin \theta & -r \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix} \end{aligned}$$

The first can be rewritten as

$$w \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = r \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix} = r R_\theta \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

By making the substitution $\phi = -\theta$ the transformation

$$\begin{aligned} R R_\theta &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \end{aligned}$$

so with this substitution, the second form for a similitude becomes

$$w \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = r R R_\phi \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

which has the proper form.

III.2.9 There are many possible ways to write a similitude $w(x) = Ax + t$ which will take the Sierpinski triangle to a subset of itself, and we give the simplest one. To take the triangle to the subtriangle with corner $(0, 0)$ we note that by taking

$$\begin{aligned} (0, 0) &\mapsto (0, 0) \\ (1, 0) &\mapsto \left(\frac{1}{2}, 0\right) \\ \left(\frac{1}{2}, 1\right) &\mapsto \left(\frac{1}{4}, \frac{1}{2}\right) \end{aligned}$$

we have completely determined w . The first transformation implies that

$$t = \begin{pmatrix} e \\ f \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

For the other two,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$$

so that $a = \frac{1}{2}$ and $c = 0$.

$$\begin{pmatrix} \frac{1}{2} & b \\ 0 & d \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} + b \\ d \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ \frac{1}{2} \end{pmatrix}$$

so that $b = 0$ and $d = \frac{1}{2}$. Then w looks like

$$w \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

III.2.10 The simplest way to show this decomposition is to multiply it out.

$$\begin{aligned} & \begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} r_3 & 0 \\ 0 & r_4 \end{pmatrix} \\ &= \begin{pmatrix} r_1 \cos \theta & -r_1 \sin \theta \\ r_2 \sin \theta & r_2 \cos \theta \end{pmatrix} \begin{pmatrix} r_3 & 0 \\ 0 & r_4 \end{pmatrix} \\ &= \begin{pmatrix} r_1 r_3 \cos \theta & -r_1 r_4 \sin \theta \\ r_2 r_3 \sin \theta & r_2 r_4 \cos \theta \end{pmatrix} \end{aligned}$$

This is a general affine transformation because the four terms $r_1 r_3$, $r_1 r_4$, $r_2 r_3$, and $r_2 r_4$ are *linearly independent*. This means that none of these can be written as a linear combination (sum with constant coefficients) of the other three, hence any arbitrary value can be written this way. It is more important to understand what is going on geometrically, which is done by looking at the action of this transformation on the coordinate axes. The transformation by r_3, r_4 rescales the coordinates, that is stretches or shrinks each coordinate and possibly reflects one or both. The rotation matrix rotates the coordinates by some angle θ , leaving the rescaled coordinates off axis. The action of the second rescaling has two functions. It further stretches or shrinks the rotated axes, but because they are no longer lined up with horizontal or vertical, it also changes the *angle* between them. This has the effect of skewing the new map of the plane. We are left with a transformation which translates, rotates, rescales, reflects and skews. This is the most general affine transformation, and requires all three parts of the decomposition to accomplish it.

III.2.12 We multiply the two matrices to get the identity matrix:

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \frac{d}{\det A} & \frac{-b}{\det A} \\ \frac{-c}{\det A} & \frac{a}{\det A} \end{pmatrix} &= \begin{pmatrix} \frac{ad-bc}{\det A} & \frac{ba-ab}{\det A} \\ \frac{cd-dc}{\det A} & \frac{-bc+ad}{\det A} \end{pmatrix} \\ &= \begin{pmatrix} \frac{\det A}{\det A} & 0 \\ 0 & \frac{\det A}{\det A} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

which proves the statement.

III.2.13 Let

$$\begin{aligned} B &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ A &= \begin{pmatrix} e & f \\ g & h \end{pmatrix} \end{aligned}$$

By exercise III.2.12 we have

$$\begin{aligned} B^{-1} &= \frac{1}{\det B} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \\ &= \begin{pmatrix} \frac{d}{\det B} & \frac{-b}{\det B} \\ \frac{-c}{\det B} & \frac{a}{\det B} \end{pmatrix} \end{aligned}$$

Then we have

$$\begin{aligned}
BAB^{-1} &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} \frac{d}{\det B} & \frac{-b}{\det B} \\ \frac{-c}{\det B} & \frac{a}{\det B} \end{pmatrix} \\
&= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \frac{ed-fc}{\det B} & \frac{af-eb}{\det B} \\ \frac{gd-hc}{\det B} & \frac{ah-gb}{\det B} \end{pmatrix} \\
&= \begin{pmatrix} \frac{aed-afc+bgd-bhc}{\det B} & \frac{a^2-aeb+bah-gb^2}{\det B} \\ \frac{ced-fc^2+gd^2-dhc}{\det B} & \frac{caf-ceb+dah-dgb}{\det B} \end{pmatrix}
\end{aligned}$$

which gives

$$\begin{aligned}
\operatorname{tr}(BAB^{-1}) &= \frac{1}{\det B} (aed - afc + bgd - bhc + caf - ceb + dah - dgb) \\
&= \frac{1}{\det B} (ade - bce + adh - bch) \\
&= \frac{\det B}{\det B} (e + h) = \operatorname{tr} A
\end{aligned}$$

The other equation is somewhat simpler:

$$\det(BAB^{-1}) = \det B \det A \det B^{-1} = \det B \det B^{-1} \det A = \det(BB^{-1}) \det A$$

Now $BB^{-1} = I$ the identity matrix, and $\det I = 1$, so

$$\det(BAB^{-1}) = \det(BB^{-1}) \det A = \det A.$$

III.2.14 We simplify the problem by noting that if q is any scalar,

$$\begin{aligned}
\frac{|A(qx)|}{|qx|} &= \left(\frac{(aqx_1 + bqx_2)^2 + (cqx_1 + dqx_2)^2}{q^2(x_1^2 + x_2^2)} \right)^{\frac{1}{2}} \\
&= \left(\frac{q^2(ax_1 + bx_2)^2 + q^2(cx_1 + dx_2)^2}{q^2(x_1^2 + x_2^2)} \right)^{\frac{1}{2}} \\
&= \frac{|Ax|}{|x|}
\end{aligned}$$

consequently, it suffices to maximize over x such that $|x| = 1$, that is, over the unit circle. For the Euclidean metric then, we let $x_1 = \cos \theta$ and $x_2 = \sin \theta$. Then recognizing that the derivative will be zero only if the derivative of the quantity inside the square root sign is zero, we maximize

$$\begin{aligned}
0 &= \frac{d}{d\theta} ((a \cos \theta + b \sin \theta)^2 + (c \cos \theta + d \sin \theta)^2) \\
&= 2(a \cos \theta + b \sin \theta)(-a \sin \theta + b \cos \theta) \\
&\quad + 2(c \cos \theta + d \sin \theta)(-c \sin \theta + d \cos \theta) \\
&= -a^2 \sin \theta \cos \theta + ab \cos^2 \theta - ab \sin^2 \theta + b^2 \sin \theta \cos \theta \\
&\quad - c^2 \sin \theta \cos \theta + cd \cos^2 \theta - cd \sin^2 \theta + d^2 \sin \theta \cos \theta \\
&= \frac{b^2 - a^2}{2} \sin 2\theta + ab \cos 2\theta + \frac{d^2 - c^2}{2} \sin 2\theta + cd \cos 2\theta \\
0 &= 2(ab + cd) \cos 2\theta + ((b^2 + d^2) - (a^2 + c^2)) \sin 2\theta, \text{ or,} \\
\tan 2\theta &= \frac{((a^2 + c^2) - (b^2 + d^2))}{2(ab + cd)}
\end{aligned}$$

This last equation has (in general) four solutions. Two are minima, two are maxima, and the norm exists. The unit circle has been transformed to an ellipse, the semi-major axis of which is the norm of A .

For the Manhattan metric, we again can restrict to the unit circle which in this metric is the set of points (x_1, x_2) such that $|x_1 + x_2| = 1$, and is a diamond centered at the origin, with vertices on the axes at $(\pm 1, 0)$ and $(0, \pm 1)$. By looking at the Euclidean case, we can see that the transformed version of this figure under A will contain two maxima and two minima. From exercise II.2.9, this transformation is a composition of a coordinate rescaling, a rotation, and another coordinate rescaling (and possibly a reflection). The result is again a parallelogram centered at the origin. The triangle inequality will now tell us that the maximum is at one pair of vertices. These are the images of the original vertices, hence one of these two images is the maximum. We have

$$A(x_1, x_2) = (ax_1 + bx_2, cx_1 + dx_2)$$

at the vertices we have $|A| = \max((a + c), (b + d))$.

Answers to Chapter III, section 3

III.3.1 (i) The most general Möbius transformation such that $f(\infty) = \infty$ is of the form $f(z) = az + b$. To show this we need to show that if $f(\infty) = \infty$ then $c = 0$. Suppose not. Then for $c \neq 0$ we have $f(\infty) = a/c \neq \infty$ because a is finite. We thus have

$$f(z) = \frac{az + b}{d} = \frac{a\bar{d}z + b\bar{d}}{|d|^2}$$

Define $a' = a\bar{d}/|d|^2$ and $b' = b\bar{d}/|d|^2$, then $f(z) = a'z + b'$ as desired.

(ii) Suppose $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a two dimensional similitude without reflection. Then we may write f as

$$f \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

Let $z = x + iy$, $b = e + if$ and $a = a_1 + ia_2$. We have $az = (a_1x - a_2y) + i(a_1y + a_2x)$ which we can write using real coordinates and matrix notation as

$$\begin{pmatrix} a_1 & -a_2 \\ a_2 & a_1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$$

Letting $\det A = r^2$ ($\det A$ is positive since we presumed no reflection) and $\theta = \cos^{-1}(a_1/r)$ we have our result. Since we divided out $\det A$ the determinant is 1 and with $a = d$ and $b = -c$, the new determinant must read $\cos^2 \theta + \sin^2 \theta = 1$, as a check that the new values match up.

III.3.2 We wish to prove that the operation $f(z) = \frac{1}{z}$ can be expressed as mapping the complex plane with infinity, to a sphere of radius 1, with equator at $|z| = 1$, rotating the sphere by π around the real axis (the axis between -1 and 1), and mapping the result back to the plane. We first establish that for at least some points this is correct.

(i) The results of this mapping on the points 0 and ∞ would be to exchange them. The map $f(z) = \frac{1}{z}$ does indeed exchange these two points.

(ii) The result on the equator is to flip it about the real line, and map it back to itself. Thus every point on the equator $|z| = 1$ is mapped to the equator. If we were to look down from above, the result of the rotation would be to exchange the halves of the equator above and below the real line, that is, each point on $|z| = 1$ should go to its complex conjugate. We have

$$f(z) = \frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

For points on the equator, the denominator of the last expression is 1 and each point does indeed move to its conjugate. We note in passing that having specified the action on these many points, as well as that we are using a Möbius transformation is in fact enough to prove the assertion, but we will perform the explicit map.

(iii) We need a coordinate system to represent points on the sphere. Since the map used to generate the Riemann sphere is to map each point to the point on the surface where a line connecting the north pole to the plane point intersects the sphere, we choose the angle ϕ between this line and the diameter of the sphere connecting the poles (interior angle). This gives us a “latitude” coordinate, for a longitude, we take the polar coordinate angle of the plane point, with respect to the positive real line. Expressing the point z in the plane as $re^{i\theta}$, this amounts to choosing θ . We need the coordinate transformation for ϕ . The line from the north pole to the point z forms the hypotenuse of a right triangle, the side adjacent is the radius of the sphere to the north pole, and the side opposite is the plane vector to z . Consequently, we have

$$r = |z| = \tan \phi$$

We may thus express the coordinate transformation as

$$z = z_1 + iz_2 = \tan \phi \cos \theta + i \tan \phi \sin \theta$$

or more simply as

$$z = \tan \phi e^{i\theta}.$$

How do these coordinates change when we rotate the sphere? As with the equator, the θ coordinate becomes the negative angle with the positive real line, that is $\theta \mapsto -\theta$. For ϕ we need some plane geometry. First, we note that if we have a diameter of a circle in the plane, and a point not on that diameter, the triangle formed by the chords between the endpoints and the third point is a right triangle. To see this, add the radius to the third point to divide the triangle into two isosceles triangles, and sum up the angles at the center of the circle. Second, we take the same planar circle, and draw the lines connecting a point on the perpendicular bisector of the diameter to the endpoints. If we now draw a line from one of the ends of the diameter to the point at which the line from the other end to the bisector intersects the circle, the triangle so formed is similar to the two triangles formed by the lines previously drawn to the bisector. Consequently, given that the tangent of the angle from the north pole to a point on the sphere is $\tan \phi$, the tangent of the angle formed by the *south* pole is $\cot \phi$.

When we rotate the Riemann sphere around the real line, we have exactly this situation, namely that the angle ϕ of the original point on the sphere is now the angle measured from the south pole. Consequently, the tangent of the new angle to the north pole is $\cot \phi$. We now look at the action of $f(z)$ on $z = \tan \phi e^{i\theta}$. We have

$$f(z) = \frac{1}{z} = \frac{1}{\tan \phi e^{i\theta}} = \cot \phi e^{i(-\theta)}$$

so the spherical coordinates have taken $\theta \mapsto -\theta$ and $\tan \phi \mapsto \cot \phi$ as desired.

III.3.3 If $f(z)$ is not a similitude then, by problem 1, $c \neq 0$. We may thus let

$$\begin{aligned} e &= \frac{a}{c} \\ f &= -\frac{ad - bc}{c^2} \\ g &= \frac{d}{c}. \end{aligned}$$

This writes $f(z)$ in the proper form as we now check:

$$\begin{aligned} e + \frac{f}{z + g} &= \frac{a}{c} - \frac{(\frac{ad-bc}{c^2})}{z + \frac{d}{c}} \\ &= \frac{az + \frac{d}{c}a}{cz + d} - \frac{(\frac{ad-bc}{c})}{cz + d} \\ &= \frac{az + b}{cz + d} \\ &= f(z) \end{aligned}$$

as desired.

III.3.6 We wish to show that a Möbius transformation $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ is always invertible. We divide the problem into its two cases: Let

$$f(z) = \frac{az + b}{cz + d}$$

(i) Suppose $c = 0$. Then the Möbius transformation can be rewritten as $f(z) = a'z + b'$. Let $w = f(z)$. Then the inverse is given by

$$z = \frac{w - b}{a}$$

(ii) Suppose $c \neq 0$. Then the transformation can be rewritten in the form

$$f(z) = e + \frac{f}{z + g}$$

Let $w = f(z)$. Then the inverse may be written

$$z = g + \frac{f}{w - e}$$

III.3.7 See answer to exercise 1.22 in Chapter III.

III.3.8 We wish to transform the real line to the unit circle in $\hat{\mathbb{C}}$. This means taking $z = z_1 + iz_2$ such that $z_2 = 0$ to $|z| = 1$. Looking at the Riemann sphere, this can be accomplished by rotating the north and south poles to the equator. Specifically, we take $\infty \mapsto i$ and $0 \mapsto -i$. Since we need one more point to define the transformation, we may do this by insisting that we rotate about the axis between 1 and -1 by taking $1 \mapsto 1$. We have

$$\infty \mapsto i \Rightarrow \frac{a}{c} = i$$

$$0 \mapsto -i \Rightarrow \frac{b}{d} = -i$$

$$1 \mapsto 1 \Rightarrow \frac{a+b}{c+d} = 1$$

From this we get

$$a = ic$$

$$b = -id$$

$$\frac{a+b}{c+d} = 1$$

$$a+b = c+d$$

$$ic - id = c + d$$

$$d = -c \left(\frac{1-i}{1+i} \right)$$

$$= -c \left(\frac{1-2i+i^2}{1-i^2} \right)$$

$$d = ic$$

Choose $a = 1$. Then we have

$$f(z) = \frac{z+i}{-iz+1} = \frac{iz-1}{z+i}$$

III.3.9 We do out the first few terms of $f^n(z)$ to see what we are dealing with:

$$\begin{aligned} f^0(z) &= z \\ f^1(z) &= \frac{1}{1+z} \\ f^2(z) &= \frac{1}{1 + \frac{1}{1+z}} \\ f^n(z) &= \frac{1}{1 + \frac{1}{1 + \frac{1}{\ddots \frac{1}{z}}}}} \end{aligned}$$

This object is known as a *continued fraction* and is usually written as

$$[a_0, a_1, \dots, a_k] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots \frac{1}{a_k}}}}$$

There is a whole body of mathematics to deal with these interesting creatures, they can be used to represent numbers, functions, traces of matrices, and other things. We need only the following relations: Let $p_0 = a_0$, $p_1 = a_1 a_0 + 1$, $p_k = a_k p_{k-1} + p_{k-2}$ and $q_0 = 1$, $q_1 = a_1$, $q_k = a_k q_{k-1} + q_{k-2}$. Then the following relation holds:

$$[a_0, a_1, \dots, a_k] = \frac{p_k}{q_k}$$

For our function, we have

$$f^n(z) = [0, 1, 1, \dots, 1, z]$$

with n one's implied. If we look at the growth of the terms p_i they represent (for terms before the n^{th} term, the terms in a *Fibonacci series*. That is, let $F_0 = 1$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$. The first few terms look like

$$1, 1, 2, 3, 5, 8, 13, \dots$$

Since $p_0 = 0$, $p_1 = 1$, $p_2 = 1$, $p_3 = 2$, and so forth, we may write $p_k = F_{k-1}$, for $k < n$. Similarly, we may write $q_k = F_k$ for $k < n$. Then $f^n(z)$ may be written, using the above relations,

$$f^n(z) = \frac{zF_{n-2} + F_{n-3}}{zF_{n-1} + F_{n-2}}$$

For the inverse transformations, we look at

$$w = \frac{1}{1+z} \Rightarrow z = -1 + \frac{1}{w}$$

Writing out the first few iterations, we have

$$\begin{aligned} f^{-1}(w) &= -1 + \frac{1}{w} \\ f^{-2}(w) &= -1 + \frac{1}{-1 + \frac{1}{w}} \end{aligned}$$

Again we have the form of a continued fraction, this time

$$f^{-n}(w) = [-1, -1, -1, \dots, -1, w]$$

Using exactly the same kind of analysis as before, we have

$$\begin{aligned} p_0 &= -1 & q_0 &= 1 \\ p_1 &= 2 & q_1 &= -1 \\ p_2 &= -3 & q_2 &= 2 \\ &\vdots & &\vdots \\ p_k &= \mp F_{k+1} & q_k &= \pm F_k \end{aligned}$$

which gives

$$f^{-n}(w) = -\frac{wF_n + F_{n-1}}{wF_{n-1} + F_{n-2}}$$

III.3.10 We want the geometrical interpretation of the map

$$f(z) = i + \frac{1}{z-i}$$

To get this we first note that by rewriting

$$f(z) = i + \frac{1}{z-i} = \frac{iz+1+1}{z-i} = \frac{iz+2}{z-1}$$

we have $f(\infty) = a/c = i$, $f(i) = \infty$. This suggests we first translate the plane to bring the point i to the origin. For this we have

$$\begin{aligned} \theta(z) &= z - i \\ \theta^{-1}(z) &= z + i \\ \theta \circ f \circ \theta^{-1}(z) &= \left(i + \frac{1}{z+i-i} \right) - i = \frac{1}{z} \end{aligned}$$

Consequently we can describe the mapping as translating the plane up by one unit, mapping the plane to the Riemann sphere, rotating the Riemann sphere about the real line, and mapping back. The net effect is a translation and an inversion.

Answers to Chapter III, section 4

III.4.2 To be correct inverse functions for $f(z)$ we must verify that $f(f^{-1}(z)) = z$. For points in the upper half plane, we have

$$\begin{aligned} &f(a(x_1, x_2) + ib(x_1, x_2)) \\ &= (a^2 - b^2 - 1) + 2iab \\ &= \frac{\sqrt{(1+x_1)^2 + x_2^2} + 1 + x_1}{2} - \frac{\sqrt{(1+x_1)^2 + x_2^2} - 1 - x_1}{2} - \frac{2}{2} \\ &\quad + 2i \sqrt{\frac{\sqrt{(1+x_1)^2 + x_2^2} + 1 + x_1}{2}} \sqrt{\frac{\sqrt{(1+x_1)^2 + x_2^2} - 1 - x_1}{2}} \\ &= x_1 + 2i \sqrt{\frac{(1+x_1)^2 + x_2^2 - (1+x_1)^2}{4}} \\ &= x_1 + ix_2 = z \end{aligned}$$

so that w_1 does indeed invert $f(z)$ there. w_2 differs from w_1 in the above derivation only in the imaginary part which replaces $2iab$ with $2i(-a)(-b)$ and hence also inverts $f(z)$. Finally, for the lower half of the plane, we notice that we are in need of a negative sign preceding the imaginary term, since the positive square root is indicated. This is provided by replacing a

by $-a$. This has no effect on the real part, because a appears there only as an a^2 term, completing the test for the inverse of $f(z)$.

III.4.3 Let $f(z) = z^2 + 1$. The critical points are those values of z such that $f'(z) = 0$, that is, the first derivative vanishes. $f'(z) = 2z = 0$ when $z = 0$ so this is the critical point of $f(z)$. The critical values of $f(z)$ are by definition the range values at which $f'(z)$ vanishes, that is $f(z)$ such that z is a critical point. This occurs at $f(0) = 1$ so 1 is a critical point for $f(z)$.

III.4.5 We determine the local approximate affine transformations for these functions by expanding about z_0 in a Taylor (Laurent) series for $f(z)$, and restricting to the first two coefficients a_0 and a_1 .

(a) $f(z) = z^2$ and $z_0 = 1$. $a_0 = f(1) = 1$. For the linear coefficient, we have

$$a_1 = \frac{f'(1)}{1!} = 2$$

Letting $w(z)$ be the linear approximation, we have

$$w(z) = 1 + 2(z - 1) = 2z - 1$$

(b) $f(z) = \frac{1}{z}$ and $z_0 = 1 + i$. We have

$$a_0 = f(z_0) = \frac{1}{1+i}$$

$$a_1 = f'(z_0) = \frac{-1}{z_0^2} = \frac{-1}{(1+i)^2}$$

Using the same notation as in (a), we have

$$\begin{aligned} w(z) &= \frac{1}{1+i} + \frac{-1}{(1+i)^2}(z - (1+i)) \\ &= \frac{1}{1+i} + \frac{-z}{(1+i)^2} + \frac{1+i}{(1+i)^2} \\ &= \frac{2}{1+i} - \frac{z}{2i} \\ &= \frac{i}{2}z + \frac{2}{1+i} \end{aligned}$$

(c) $f(z) = (z - 1)^3$ and $z_0 = 1 - i$. We have

$$a_0 = f(z_0) = ((1-i) - 1)^3 = (-i)^3 = i$$

$$a_1 = f'(z_0) = 3(z_0 - 1)^2 = 3(-i)^2 = -3$$

Putting these together gives

$$\begin{aligned} w(z) &= i + -3(z - (1-i)) \\ &= i + 3 - 3i - 3z \\ &= -3z + 3 - 2i \end{aligned}$$

Answers to Chapter III, section 5

III.5.3 Let x_f be the fixed point of $f(x) = Ax + t$. Let $\theta(x) = x - x_f$. Then $\theta^{-1}(x') = x' + x_f$, and we have

$$\begin{aligned} \theta \circ \theta^{-1}(x') &= A(x' + x_f) + t - x_f \\ &= Ax' + Ax_f + t - x_f \\ &= Ax' + x_f - x_f \\ &= Ax' = f'(x') \end{aligned}$$

as desired.

III.5.4 Using the coordinate change notation $\theta : \mathbf{X} \rightarrow \mathbf{X}$ we have $\theta(x) = 2x - 1$, which yields $\theta^{-1}(x') = (x' + 1)/2$. We get the transformed f' in the new space by

$$f'(x') = \theta \circ f \circ \theta^{-1}(x') : \mathbf{X}' \rightarrow \mathbf{X}'$$

which we get from

$$f \circ \theta^{-1}(x') = \left(\left(\frac{x' + 1}{2} \right) - 1 \right)^2 + 1 = \left(\frac{x' - 1}{2} \right)^2 + 1$$

giving us,

$$\theta \circ f \circ \theta^{-1}(x') = 2 \left(\left(\frac{x' - 1}{2} \right)^2 + 1 \right) - 1 = \frac{1}{2}(x' - 1)^2 + 1$$

III.5.5 We first find the fixed points of the Möbius transformation

$$f(z) = \frac{(z + 2)}{(4 - z)} = z$$

Multiplying through we get

$$z + 2 = 4z - z^2$$

or

$$z^2 - 3z + 2 = 0$$

which factors into

$$(z - 1)(z - 2) = 0$$

There are two fixed points, at $z = 1$ and $z = 2$.

Make the coordinate transformation which takes $1 \mapsto 0$ and $2 \mapsto \infty$. The transformation

$$\theta(z) = \frac{z - 1}{z - 2}$$

will work. We need the inverse of this:

$$\begin{aligned} w &= \frac{z - 1}{z - 2} \\ (z - 2)w &= z - 1 \\ zw - 2w &= z - 1 \\ wz - z &= 2w - 1 \\ z &= \theta^{-1}(w) = \frac{2w - 1}{w - 1} \end{aligned}$$

We make the composition $f'(w) = \theta \circ f \circ \theta^{-1}(w)$ as follows:

$$f \circ \theta^{-1}(w) = \frac{\frac{2w-1}{w-1} + 2}{4 - \frac{2w-1}{w-1}} = \frac{2w - 1 + 2w - 2}{4w - 4 - 2w + 1} = \frac{4w - 3}{2w - 3}$$

so that,

$$\theta \circ f \circ \theta^{-1}(w) = \frac{\frac{4w-3}{2w-3} - 1}{\frac{4w-3}{2w-3} - 2} = \frac{4w - 3 - 2w + 3}{4w - 3 - 4w + 6} = \frac{2}{3}w$$

This map shrinks the sphere by a factor of $2/3$. The whole process can be written by expanding the real line from 1 to 2 to the whole positive real line, then shrinking the resultant Riemann sphere, then reducing the positive real line to the line between 1 and 2 again. The net result is shrinking the sphere in a non-uniform manner.

III.5.6 Let $W(x) = Ax + t$ which we express as usual. The fixed point of this transformation is

$$x_f = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

which we rewrite as

$$x_{f1} = ax_{f1} + bx_{f2} + e; x_{f2} = cx_{f1} + dx_{f2} + f$$

We solve this as a system of linear equations:

$$0 = (a - 1)x_{f1} + bx_{f2} + e$$

$$0 = cx_{f1} + (d - 1)x_{f2} + f$$

which gives

$$0 = c(a - 1)x_{f1} + bcx_{f2} + ce$$

$$0 = c(a - 1)x_{f1} + (a - 1)(d - 1)x_{f2} + e$$

$$0 = ((a - 1)(d - 1) - bc)x_{f2} + (a - 1)f - ce$$

$$0 = (\det(A - I))x_{f2} + (a - 1)f - ce$$

yielding

$$x_{f2} = \frac{ce - (a - 1)f}{\det(A - I)}$$

A nearly identical derivation gives

$$x_{f1} = \frac{bf - (d - 1)e}{\det(A - I)}$$

Notice that these equations do indeed become invalid if $\det(A - I) = 0$, when the “fixed point” tends to infinity. To transform coordinates so that x_f is the origin, choose the transformation $\theta(x) = x - x_f$. The inverse of this, letting $w = \theta(x)$ is just $x = w + x_f$. The whole coordinate transformation is then

$$\begin{aligned} \theta \circ f \circ \theta^{-1}(w) &= (A(w + x_f) + t) - x_f \\ &= Aw + Ax_f + t - x_f \\ &= Aw + (Ax_f + t) - x_f \\ &= Aw + x_f - x_f \\ &= Aw \end{aligned}$$

So that the effect, in the transformed system is just the action of the linear transformation Ax .

III.5.7 By exercise III.2.13 we know that

$$\det D = \lambda_1 \lambda_2 = \det A$$

and

$$\operatorname{tr} D = \lambda_1 + \lambda_2 = \operatorname{tr} A$$

Then by multiplying $(\lambda - \lambda_1)(\lambda - \lambda_2)$ each λ_i is a solution to the equation

$$\lambda^2 - (\lambda_1 + \lambda_2)\lambda + (\lambda_1 \lambda_2) = 0$$

or

$$\lambda^2 - \operatorname{tr} A \lambda + \det A = 0$$

Now

$$\begin{aligned} \det \begin{pmatrix} e - \lambda & f \\ g & h - \lambda \end{pmatrix} &= (e - \lambda)(h - \lambda) - fg \\ &= eh - e\lambda - h\lambda + \lambda^2 - fg \\ &= \lambda^2 - (e + h)\lambda + eh - fg \\ &= \lambda^2 - \operatorname{tr} A \lambda + \det A \end{aligned}$$

as desired. The above equation is called the *eigenvalue equation* for A , the numbers λ_1, λ_2 are called *eigenvalues*, and have the property that there exist lines in $\mathbb{R}^2$ such that for any v on those lines

$$Av = \lambda v.$$

The lines are called the *eigenvectors* of the matrix A .

III.5.8 To make the transformation $h(z) = \frac{1}{z}$ for analyzing the behavior at the point at infinity, we first note that $h^{-1}(w) = \frac{1}{w}$ has the same functional form. Composing, we have

$$\theta \circ f \circ \theta^{-1} = \frac{1}{7(\frac{1}{w}) + 1} = \frac{w}{7 + w}$$

The point at 0 is of interest, and is now a fixed point. Locally we may expand

$$\frac{w}{7 + w} \approx a_0 + a_1 w$$

where, $a_0 = 0$ and a_1 is given by the Taylor expansion

$$a_1 = \frac{d}{dw} \left(\frac{w}{7 + w} \right) (0) = \frac{(7 + w) - w}{(7 + w)^2} \Big|_{w=0} = \frac{7}{49}$$

so that near zero, this looks like $f'(w) = (1/7)w$ which is a contraction.

III.5.9 We have

$$f_\mu(x) = x^2 - \mu; \quad g_\lambda(x) = \lambda x(1 - x)$$

and we want to reexpress f as $f_{\mu(\lambda)}$. f is a parabola opening up, centered at the y -axis with vertex at $-\mu$, g is a parabola centered at $x = 1/2$ opening down, with zeroes at 0, 1. Consequently, we need to reflect f , and stretch it so that its zeroes are one unit apart, and shift it to the right by $1/2$. First, the transformation $\theta(x) = -x$ will reflect the parabola as follows:

$$\theta \circ f \circ \theta^{-1}(x) = \theta \circ f(-x) = -f(-x)$$

which for any value of μ is

$$-f(-x) = -((-x)^2 - \mu) = -x^2 + \mu$$

a parabola pointing down with vertex μ centered on the y -axis. The transformation which will move the vertex to $x = 1/2$ is given by $\theta(x) = x - 1/2$. Combining these yields $\theta(x) = -x - 1/2$, or,

$$\theta \circ f \circ \theta^{-1}(x) = -x^2 + x - 1/4 + \mu - 1/2 = -x^2 + x + (\mu - 3/4)$$

We now need to adjust this so as to produce a multiplication by λ and remove the constant term. We use the transformation $\theta(x) = x/\lambda$. Once again composing, we now have

$$\theta(x) = \frac{-x}{\lambda} - \frac{1}{2}$$

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so that

$$\theta^{-1}(x) = -\lambda x + \frac{\lambda}{2}$$

We have

$$\begin{aligned}\theta \circ f_{\mu} \circ \theta^{-1}(x') &= \theta \circ f_{\mu}(-\lambda x' + \frac{\lambda}{2}) \\ &= \theta(\lambda^2(\frac{1}{4} - x' + x'^2) - \mu) \\ &= -\lambda(\frac{1}{4} - x' + x'^2) - \frac{\mu}{\lambda} - \frac{1}{2} \\ &= -\lambda x'^2 + \lambda x' - (\frac{\lambda}{4} + \frac{1}{2} + \frac{\mu}{\lambda})\end{aligned}$$

This is identical to the expression for $g_{\lambda}(x')$ except for the constant term, we now find $\mu(\lambda)$ by setting this term to zero and solving for μ .

$$\begin{aligned}0 &= \frac{\lambda}{4} + \frac{1}{2} + \frac{\mu}{\lambda} \\ \frac{-\mu}{\lambda} &= \frac{\lambda + 2}{4} \\ \mu &= \frac{-1}{4}\lambda^2 + \frac{-1}{2}\lambda\end{aligned}$$

We have

$$\theta(x) = \frac{1}{2} - \frac{x}{\lambda} \quad ; \quad \mu(\lambda) = \frac{-1}{4}\lambda^2 + \frac{-1}{2}\lambda$$

III.5.10 We want the behavior of $g(x) = x^2 - 1/2$ near the fixed points. We have

$$\begin{aligned}x_f &= x_f^2 - \frac{1}{2} \\ x_f^2 - x_f - \frac{1}{2} &= 0 \\ x_f &= \frac{1 \pm \sqrt{3}}{2}\end{aligned}$$

and we show the behavior around these points using two methods.

Analysis by moving the fixed points to the origin:

Let

$$\begin{aligned}\theta(x) &= x - x_{f1} = x - (\frac{1 + \sqrt{3}}{2}) \\ \theta^{-1}(x) &= x + x_{f1} = x + (\frac{1 + \sqrt{3}}{2})\end{aligned}$$

Then we have

$$\begin{aligned}g \circ \theta^{-1}(x') &= x'^2 + (1 + \sqrt{3})x' + \frac{1 + 2\sqrt{3} + 3}{4} - \frac{1}{2} \\ &= x'^2 + (1 + \sqrt{3})x' + \frac{1 + \sqrt{3}}{2} \\ g'(x') &= \theta \circ g \circ \theta^{-1}(x') = x'^2 + (1 + \sqrt{3})x'\end{aligned}$$

which has a fixed point at the origin. Very close to the origin, since $x'^2 \ll x'$, we have

$$g'(x') \approx (1 + \sqrt{3})x'$$

which is an expansion. For the other fixed point, take

$$\theta(x) = x - x_{f_2} = x - \left(\frac{1 - \sqrt{3}}{2}\right)$$

$$\theta^{-1}(x) = x + x_{f_2} = x + \left(\frac{1 - \sqrt{3}}{2}\right)$$

which gives us

$$\begin{aligned} g \circ \theta^{-1}(x') &= x'^2 + (1 - \sqrt{3})x' + \frac{1 - 2\sqrt{3} + 3}{4} - \frac{1}{2} \\ &= x'^2(1 - \sqrt{3})x' + \frac{1 - \sqrt{3}}{2} \end{aligned}$$

$$g'(x') = \theta \circ g \circ \theta^{-1}(x') = x'^2 + (1 - \sqrt{3})x'$$

Again looking at behavior near the origin, we have

$$g'(x') \approx (1 - \sqrt{3})x'$$

Since the coefficient is less than zero, there is a reflection, and since $|1 - \sqrt{3}| < 1$, it is composed with a contraction. Analysis by expanding $g(x)$ in a Taylor series:

The derivatives needed to expand $g(x)$ in Taylor series up to the linear term, at the fixed points are

$$g^{(0)}(x) = x^2 - \frac{1}{2}$$

$$g^{(1)}(x) = 2x$$

At the first fixed point, x_{f_1} we have

$$a_0 = g^{(0)}(x) \Big|_{\frac{1+\sqrt{3}}{2}} = \frac{1 + \sqrt{3}}{2}$$

$$a_1 = g^{(1)}(x) \Big|_{\frac{1+\sqrt{3}}{2}} = 1 + \sqrt{3}$$

and the expansion to the linear term looks like

$$\frac{1 + \sqrt{3}}{2} + (1 + \sqrt{3})\left(x - \frac{1 + \sqrt{3}}{2}\right) = (1 + \sqrt{3})x + \left(\frac{1 + \sqrt{3}}{2} - 2 - \sqrt{3}\right)$$

which is an expansion. For the fixed point x_{f_2} we have

$$a_0 = g^{(0)}(x) \Big|_{\frac{1-\sqrt{3}}{2}} = \frac{1 - \sqrt{3}}{2}$$

$$a_1 = g^{(1)}(x) \Big|_{\frac{1-\sqrt{3}}{2}} = 1 - \sqrt{3}$$

and the expansion to the linear term looks like

$$\frac{1 - \sqrt{3}}{2} + (1 - \sqrt{3})\left(x - \frac{1 - \sqrt{3}}{2}\right) = (1 - \sqrt{3})x + \left(\frac{1 - \sqrt{3}}{2} - 2 = \sqrt{3}\right)$$

which is a reflection and contraction as before. In general, it depends on the nature of the function which method is easier.

If it can be transformed by inspection, transforming the fixed points to the origin is preferable. In general, the Taylor series expansion method is more straight forward and simpler.

III.5.11 In exercise III.5.7 we showed that the entries of the diagonal matrix D satisfy the eigenvalue equation

$$\lambda^2 - \operatorname{tr} A \lambda + \det A = 0$$

Using the quadratic equation, the solutions to this equation are

$$\lambda_i = \frac{\operatorname{tr} A \pm \sqrt{(\operatorname{tr} A)^2 - 4 \det A}}{2}$$

and the condition stated in the problem is that the discriminant of this equation be greater than zero, in order to have two real solutions to the equation. If the discriminant is less than zero, then the solutions are complex conjugates of the form $\lambda_k = x \pm iy = re^{\pm i\theta}$. The matrix defines a transformation which contains a rotation by θ and therefore cannot be put into diagonal form, although it can be written as a rescaling and a rotation.

Turning to the problem of finding the matrix B , we separate out for a moment the normalizing factor of $1/\det B$ for the inverse of B , which will turn out not to matter, and write

$$\det B \cdot B A B^{-1} = \det B \cdot D = \det B \cdot \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

Letting

$$B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

we have

$$\det B \cdot B A B^{-1} = \begin{pmatrix} aed - afc + bgd - bhc & a^2 - aeb + bah - gb^2 \\ ced - fc^2 + gd^2 - dhc & caf - ceb + dah - dgb \end{pmatrix}$$

which yields

$$ade - afc + bdg - bch = \det B \cdot \lambda_1$$

$$adh + afc - bdg - bce = \det B \cdot \lambda_2$$

$$a^2 f - b^2 g - abe + abh = 0$$

$$d^2 g - c^2 f + cde - cdh = 0$$

The first two equations when subtracted or added, merely return to us the result of exercise III.2.13, that the trace and determinant are invariant under this kind of coordinate change. Since only these are dependent on the $\det B$ term in front, we can safely ignore it in what follows, which is why it was carried separately. Rewriting the last two equations as

$$a^2 f - ab(e - h) - b^2 g = 0$$

$$c^2 f - cd(e - h) - d^2 g = 0$$

we may solve either one, solving the last one for c , we have

$$\begin{aligned} c &= \frac{d(e - h) \pm \sqrt{d^2(e - h)^2 + 4fd^2g}}{2f} \\ &= \frac{d}{f} \left(\frac{(e - h) \pm \sqrt{(e - h)^2 + 4fg}}{2} \right) \\ &= \frac{d}{f} \left(\frac{\operatorname{tr} A \pm \sqrt{(e - h)^2 + 4fg}}{2} - h \right) \end{aligned}$$

Examining the discriminant, we have

$$\begin{aligned}(e-h)^2 + 4fg &= e^2 + h^2 - 2eh + 4fg = e^2 + h^2 + 2fg - 2\det A \\ &= e^2 + h^2 + 2\det A + 2fg - 4\det A = e^2 + h^2 + 2eh - 4\det A \\ &= (e+h)^2 - 4\det A = (\operatorname{tr} A)^2 - 4\det A\end{aligned}$$

which is the discriminant of the eigenvalue equation. Putting this in, we have

$$c = \frac{d}{f} \left(\frac{\operatorname{tr} A \pm \sqrt{(\operatorname{tr} A)^2 - 4\det A}}{2} - h \right) = \frac{d}{f} (\lambda_i - h)$$

or

$$c\left(\frac{f}{\lambda_i - h}\right) = d ; a\left(\frac{f}{\lambda_i - h}\right) = d.$$

Since $\det B \neq 0$ the columns are independent, so we must choose a different λ_i for each of the above equations. Only the ratios are important here, so we choose $a = c = 1$, and have

$$B = \left(\frac{(\lambda_1 - h)(\lambda_2 - h)}{f(\lambda_1 - \lambda_2)} \right) \begin{pmatrix} 1 & \frac{f}{\lambda_1 - h} \\ 1 & \frac{f}{\lambda_2 - h} \end{pmatrix}$$

In closing, we note that the *eigenvectors* of A are the solutions to the pair of equations

$$\begin{aligned}ex_1 + fx_2 &= \lambda x_1 \\ gx_1 + hx_2 &= \lambda x_2\end{aligned}$$

which appears to yield the two solutions

$$\frac{(\lambda - e)}{f} x_1 = x_2 ; \frac{g}{(\lambda - h)} x_1 = x_2$$

These are actually identical, since by the eigenvalue equation

$$\frac{\lambda - e}{f} = \frac{g}{\lambda - h}$$

To solve for the eigenvectors, simply let $x_1 = 1$ and use one of the corresponding values for x_2 . Hence the rows of the matrix B found above are precisely the eigenvectors of A . This is true in general, (for $N \times N$ matrices the eigenvalue equation will have N real roots if it is diagonalizable), and depending on the problem presents another way to find a useful transformation.

III.5.12 To solve this problem, we make use of the methods developed in exercise III.5.11. We have

$$f(x) = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Solving for the fixed point, we may rewrite this as

$$f(x) = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 + 1/2 \end{pmatrix} + \begin{pmatrix} 0 \\ -1/2 \end{pmatrix}$$

so there is a necessary translation by $(0, -1/2)$ to bring the fixed point to the origin. To diagonalize the matrix we have

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$$

so that $\operatorname{tr} A = 4$ and $\det A = -1$. This gives us the eigenvalue equation

$$\lambda^2 - 4\lambda - 1 = 0$$

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yielding

$$\lambda_i = 2 \pm \sqrt{5}$$

We may now write down immediately the final form of the transformation, we want B and the above translation to produce

$$f'(x') = \begin{pmatrix} 2 + \sqrt{5} & 0 \\ 0 & 2 - \sqrt{5} \end{pmatrix}$$

which is indeed a coordinate rescaling. A matrix for B is then given in exercise III.5.11 by

$$\begin{aligned} B &= \begin{pmatrix} 1 & \frac{c}{\lambda_1 - d} \\ 1 & \frac{c}{\lambda_2 - d} \end{pmatrix} \\ &= \begin{pmatrix} 1 & \frac{2}{\lambda_1 - 3} \\ 1 & \frac{2}{\lambda_2 - 3} \end{pmatrix} \end{aligned}$$

or to write out the whole expression for the coordinate transformation $\theta(x)$,

$$\theta(x) = \begin{pmatrix} 1 & \frac{2}{\sqrt{5}-1} \\ 1 & \frac{2}{\sqrt{5}+1} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix}$$

III.5.13 Let f be a transformation $f: \mathbf{X} \rightarrow \mathbf{X}$. We want to show that $\{f^n\}_{n=0}^\infty$ is a semigroup, which requires showing that $f^n \circ f^m: \mathbf{X} \rightarrow \mathbf{X}$ is in this set of transformations. But

$$f^n \circ f^m(x) = f^n(f^m(x)) = f^{n+m}(x)$$

so the set of iterates of f is indeed a semigroup.

III.5.14 Let $T: \Sigma \rightarrow \Sigma$ be a left shift transformation on code space, i.e.

$$\begin{aligned} x &= x_1 x_2 x_3 \dots \\ T(x) &= x_2 x_3 x_4 \dots \end{aligned}$$

Since each application of T shifts the sequence to the left and drops the initial term, we have

$$T^n(x) = x_{n+1} x_{n+2} x_{n+3} \dots$$

The fixed points of $T^3(x) = x_4 x_5 \dots$ must therefore obey

$$x_{f_1} x_{f_2} x_{f_3} \dots = x_{f_4} x_{f_5} x_{f_6} \dots$$

Listing out the first few equalities, we have

$$\begin{aligned} x_1 &= x_4 \\ x_2 &= x_5 \\ x_3 &= x_6 \\ x_4 &= x_7 \\ &\vdots \end{aligned}$$

which yields

$$\begin{aligned} x_1 &= x_{3n+1} \\ x_2 &= x_{3n+2} \\ x_3 &= x_{3n+3} \end{aligned}$$

In other words, any fixed point of $T^3(x)$ must repeat the first three symbols indefinitely. Explicitly, the fixed points of $T^3(x)$ are

$$\overline{000}, \overline{001}, \overline{010}, \overline{011}, \overline{100}, \overline{101}, \overline{110}, \overline{111}.$$

III.5.15 We need to show that the Möbius transformations on $\hat{\mathbb{R}}$ form a group under composition. To check closure, we let

$$w_1(x) = \frac{ax+b}{cx+d} ; w_2(x) = \frac{ex+f}{gx+h}$$

The composition is then

$$\begin{aligned} w_1 \circ w_2(x) &= \frac{a(\frac{ex+f}{gx+h}) + b}{c(\frac{ex+f}{gx+h}) + d} \\ &= \frac{aex + af + bgx + bh}{cex + cf + dgx + dh} \\ &= \frac{(ae + bg)x + (af + bh)}{(ce + dg)x + (cf + dh)} \end{aligned}$$

which has the proper form. We check

$$\begin{aligned} (ae + bg)(cf + dh) - (ce + dg)(af + bh) \\ &= aecf + adeh + bcgf + bdgh - aecf - bceh - adgf - bdgh \\ &= (ad - bc)eh - (ad - bc)gf \\ &= (ad - bc)(eh - gf) \neq 0 \end{aligned}$$

so the Möbius transformations on $\hat{\mathbb{R}}$ form a semigroup. They are invertible as follows: Let

$$w = \frac{ax+b}{cx+d}$$

We have

$$\begin{aligned} wx + wd &= ax + b \\ wx - ax &= b - wd \\ x &= \frac{-dw + b}{cw - a} = \frac{dw - b}{-cw + a} \end{aligned}$$

which has the proper form, with

$$da - (-b)(-c) = ad - bc \neq 0$$

so there is an inverse for each element, and they form a group.

III.5.16 This problem is identical in algebraic manipulations to exercise 5.12, with the replacement of $a, b, c, d, z, w \in \hat{\mathbb{C}}$ instead of in $\hat{\mathbb{R}}$.

III.5.17 Let $f(x) = Ax + t$, and $g(x) = Bx + u$ be invertible affine transformations on $\mathbb{R}^2$. Then if $w = f(x)$, we have

$$w - t = Ax ; A^{-1}(w - t) = x = A^{-1}w - A^{-1}t = f^{-1}(w).$$

Similarly, $g^{-1}(w) = B^{-1}w - B^{-1}u$.

We have

$$f \circ g(x) = A(Bx + u) + t = ABx + (Au + t)$$

which is an affine transformation. We need to check whether it is invertible both because it needs to be in our set of transformations, and to prove group structure. Let

$$\begin{aligned} w &= ABx + (Au + t) \\ w - (Au + t) &= ABx \\ A^{-1}(w - (Au + t)) &= Bx \\ B^{-1}A^{-1}(w - (Au + t)) &= x \end{aligned}$$

Rewriting the left hand side, we have

$$f^{-1}(w) = B^{-1}A^{-1}w - B^{-1}A^{-1}(Au + t) = B^{-1}A^{-1}w - (B^{-1}u + B^{-1}A^{-1}t),$$

which is affine. The inverse matrices B^{-1}, A^{-1} must exist because by assumption f, g are invertible. Hence the set of invertible affine transformations is a group under composition.

III.5.18 Suppose $f(\mathbb{A}) \subset \mathbb{A}$. This implies $f^n(\mathbb{A}) \subset \mathbb{A}$, as follows: It is certainly true for $n = 1$. Suppose for every $0 < i < k$, the statement holds. Since by hypothesis $f^{k-1}(\mathbb{A}) \subset \mathbb{A}$, and $f(A) \subset \mathbb{A}$ if $A \subset \mathbb{A}$, we have $f(f^{k-1}(\mathbb{A})) \subset \mathbb{A}$. But $f(f^{k-1}(\mathbb{A})) = f^k(\mathbb{A})$ so the claim is true by induction. To show that this implies that $\{f^n\}$ is a semigroup, recall that $f^n \circ f^m = f^{n+m}$ which is in the collection. Hence $\{f^n\}$ is a semigroup of transformations.

III.5.19 Let $w(x) = Ax + t$ where

$$A = \begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$$

and $ac \neq 0$. This transformation is invertible with $x = A^{-1}(w - t)$, as long as A^{-1} exists. This is the case since $\det A = ac \neq 0$. To show that the set of transformations of this form are a group under composition, we need to show that if B, A are of this form then BA is as well, and, if A is of this form then A^{-1} is as well.

Let

$$A = \begin{pmatrix} a & 0 \\ b & c \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} e & 0 \\ f & g \end{pmatrix}$$

Then

$$BA = \begin{pmatrix} e & 0 \\ f & g \end{pmatrix} \begin{pmatrix} a & 0 \\ b & c \end{pmatrix} = \begin{pmatrix} ae & 0 \\ af + bg & cg \end{pmatrix}$$

which is also of the proper form. We have

$$ae \cdot cg = (ac)(eg) \neq 0$$

so the transformation is in the set. Furthermore,

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} c & 0 \\ -b & a \end{pmatrix}$$

which is of the correct form. $ca = ac$, so the inverse is in the set. Hence the set of all such transformations $w(x)$ on $\mathbb{R}^2$ forms a group.

III.5.20 Given $f(z) = Az^2 + Bz + C$, we want $\theta(z) = az + b$ such that

$$\theta \circ f \circ \theta^{-1}(z') = (z')^2 + \tilde{C}$$

Since

$$\theta^{-1}(z') = \frac{1}{a}(z' - b)$$

we have

$$\begin{aligned} f \circ \theta^{-1}(z') &= \frac{A}{a^2}(z'^2 - 2bz' + b^2) + \frac{B}{a}(z' - b) + C \\ &= \frac{A}{a^2}z'^2 + (B - \frac{2bA}{a^2})z' + (\frac{Ab^2}{a^2} - \frac{Bb}{a} + C) \end{aligned}$$

Composing $\theta(z)$ with this yields

$$\theta \circ f \circ \theta^{-1}(z) = \frac{A}{a}z'^2 + (B - 2\frac{bA}{a})z' + (\frac{Ab^2}{a} - Bb + Ca + b)$$

We must have $A/a = 1$ and $B - 2(bA/a) = 0$ for the desired transformation, meaning $a = A$ and $b = B/2$. Using these to simplify the constant term we have

$$\theta(z) = Az + \frac{B}{2}; \quad \tilde{C} = CA + \frac{B}{2} - \frac{B^2}{4}$$

Answers to Chapter III, section 6

III.6.1

- (a) $\sup(-\infty, 3) = 3$; $\inf(-\infty, 3) = -\infty$.
- (b) $\sup(\mathbb{C}) = 1$; $\inf(\mathbb{C}) = 0$.
- (c) $\sup\{1, 2, 3, 4, \dots\} = +\infty$; $\inf\{1, 2, 3, 4, \dots\} = 1$.
- (d) $\sup(\mathbb{R}^+) = +\infty$; $\inf(\mathbb{R}^+) = 0$.

III.6.2 Let $f : X \rightarrow X$ be a contraction mapping on (X, d) , and let $s_0 = \inf\{s \in \mathbb{R} : s \text{ is a contractivity factor for } f\}$. Then s_0 is a contractivity factor for f .

Proof: Clearly $0 \leq s < 1$ since if s is any contractivity factor for f , then $0 \leq s < 1$ so that $0 \leq s_0 \leq s < 1$. As well, the infimum is taken over a non-empty set since there is at least some s which is a contractivity factor for f by definition. Consequently, we need only show that s_0 satisfies

$$d(f(x), f(y)) \leq s_0 \cdot d(x, y) \quad \forall x, y \in X.$$

Suppose not. Then there is a pair $x, y \in X$ such that

$$d(f(x), f(y)) > s_0 \cdot d(x, y).$$

But this says that

$$d(f(x), f(y)) - s_0 \cdot d(x, y) > 0$$

and since

$$d(f(x), f(y)) - d(x, y) < 0$$

by the intermediate value theorem, there is an s' with $1 > s' > s_0$ such that

$$d(f(x), f(y)) - s' \cdot d(x, y) = 0$$

Hence s' is less than all valid contractivity factors, and there is an interval (s_0, s') which contains no values which are valid contractivity factors for f . This is a contradiction of the assumption that s_0 is the infimum of the contractivity factors for f .

III.6.3 Let $f : X \rightarrow X$ and $g : X \rightarrow X$ be contractions with contractivity factors s, t respectively. Then the contractivity factor of $f \circ g$ is st .

Proof: For all $x, y \in X$ we have

$$d(g(x), g(y)) \leq t \cdot d(x, y) \Rightarrow s \cdot d(g(x), g(y)) \leq st \cdot d(x, y)$$

But we also have

$$d(f(g(x)), f(g(y))) \leq s \cdot d(g(x), g(y)) \leq st \cdot d(x, y)$$

which by definition makes st a contractivity factor for $f(g(x)) = f \circ g(x)$ as desired.

III.6.4 Let $w(x) = Ax + t : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with fixed point x_f and $\det(A - I) \neq 0$. Let $h(x) = x - x_f = x'$. Then $h^{-1}(x') = x' + x_f$. Represent

$$w(x) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

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and

$$x_f = \begin{pmatrix} x_{f1} \\ x_{f2} \end{pmatrix}$$

Then we have

$$\begin{aligned} w \circ h^{-1}(x') &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x'_1 + x_{f1} \\ x'_2 + x_{f2} \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix} \\ &= \begin{pmatrix} ax'_1 + ax_{f1} + bx'_2 + bx_{f2} + e \\ cx'_1 + cx_{f1} + dx'_2 + dx_{f2} + f \end{pmatrix} \\ &= \begin{pmatrix} ax'_1 + bx'_2 + x_{f1} \\ cx'_1 + dx'_2 + x_{f2} \end{pmatrix} \end{aligned}$$

because x_f is a fixed point. Then

$$\begin{aligned} h \circ w \circ h^{-1}(x') &= \begin{pmatrix} ax'_1 + bx'_2 + x_{f1} \\ cx'_1 + dx'_2 + x_{f2} \end{pmatrix} - \begin{pmatrix} x_{f1} \\ x_{f2} \end{pmatrix} \\ &= \begin{pmatrix} ax'_1 + bx'_2 \\ cx'_1 + dx'_2 \end{pmatrix} \\ &= Ax' = w'(x') \end{aligned}$$

Inverting this we have

$$h^{-1} \circ w \circ h(x) = h^{-1} \circ w'(x - x_f) = h^{-1}(A(x - x_f)) = A(x - x_f) + x_f$$

Finally, we check that $A(x - x_f) + x_f = Ax + t$. Since

$$\begin{pmatrix} x_{f1} \\ x_{f2} \end{pmatrix} = \begin{pmatrix} ax_{f1} + bx_{f2} + e \\ cx_{f1} + dx_{f2} + f \end{pmatrix}$$

we have

$$\begin{aligned} &\begin{pmatrix} ax_1 + bx_2 - (ax_{f1} + bx_{f2}) + (ax_{f1} + bx_{f2} + e) \\ cx_1 + dx_2 - (cx_{f1} + dx_{f2}) + (cx_{f1} + dx_{f2} + f) \end{pmatrix} \\ &= \begin{pmatrix} ax_1 + bx_2 + e \\ cx_1 + dx_2 + f \end{pmatrix} \\ &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix} \\ &= Ax + t \end{aligned}$$

To prove the next claim we have

$$w(x) = w^1(x) = A^1(x - x_f) + x_f$$

and

$$\begin{aligned} w^2(x) &= A(w(x) - x_f) + x_f \\ &= A((A(x - x_f) + x_f) - x_f) + x_f \\ &= A(A(x - x_f)) + x_f \\ &= A^2(x - x_f) + x_f \end{aligned}$$

Suppose that

$$w^n(x) = A^n(x - x_f) + x_f$$

Then we have

$$\begin{aligned} w^{n+1}(x) &= A((A^n(x - x_f) + x_f) - x_f) + x_f \\ &= A(A^n(x - x_f)) + x_f \\ &= A^{n+1}(x - x_f) + x_f \end{aligned}$$

so the result follows by induction on n .

We now describe the conditions on A under which the map is contractive for the Euclidean metric. We want $d(w(x), w(y)) \leq s \cdot d(x, y)$ for some $0 \leq s < 1$. This means,

$$d((A(x - x_f) + x_f), (A(y - x_f) + x_f)) \leq s \cdot d(x, y)$$

or

$$\begin{aligned} &(ax_1 - ax_{f_1} + bx_2 - bx_{f_2} + x_{f_1} - ay_1 + ax_{f_1} - by_2 + bx_{f_2} - x_{f_1})^2 \\ &\quad + (cx_1 - cx_{f_1} + dx_2 - dx_{f_2} + x_{f_2} - cy_1 + cx_{f_1} - dy_2 + dx_{f_2} - x_{f_2})^2 \\ &= (ax_1 - ay_1 + bx_2 - by_2)^2 + (cx_1 - cy_1 + dx_2 - dy_2)^2 \\ &= (A(x - y)_1)^2 + (A(x - y)_2)^2 \end{aligned}$$

Using the notation of exercise III.2.14 we need

$$|A(x - y)| \leq s \cdot |x - y|$$

or

$$\frac{|A(x - y)|}{|x - y|} \leq s \forall x, y \in \mathbf{X}$$

This is certainly true if we maximize over $|x - y|$ so we have as a condition that $|A| \leq s < 1$ for $w(x)$ to be contractive. The proof is identical with the Manhattan metric substituted and the appropriate changes made.

III.6.6 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = 1/2x + 1/2$. Then f is a contraction mapping as follows:

$$\begin{aligned} d(f(x), f(y)) &= d\left(\frac{1}{2}x + \frac{1}{2}, \frac{1}{2}y + \frac{1}{2}\right) \\ &= \left|\frac{1}{2}x + \frac{1}{2} - \left(\frac{1}{2}y + \frac{1}{2}\right)\right| \\ &= \left|\frac{1}{2}x - \frac{1}{2}y\right| \\ &= \frac{1}{2}|x - y| \\ &\leq \frac{1}{2}d(x, y) \end{aligned}$$

so that $s = \frac{1}{2}$ is a contractivity factor. To find the fixed point, we write

$$\begin{aligned} x_f &= \frac{1}{2}x_f + \frac{1}{2} \\ \frac{1}{2}x_f &= \frac{1}{2} \end{aligned}$$

or $x_f = 1$. Rewriting in the form $f(x) = A(x - x_f) + x_f$, we have

$$\begin{aligned}
 f(x) &= \frac{1}{2}(x-1) + 1 \\
 f^n(x) &= \frac{1}{2^n}(x-1) + 1 \\
 \lim_{n \rightarrow \infty} f^n(x) &= \lim_{n \rightarrow \infty} \frac{1}{2^n}(x-1) + 1 \\
 &= 1 = x_f
 \end{aligned}$$

With $x = 0$, $f^n(x) = 1 - 1/(2^n)$.

Writing the original,

$$f^2(x) = \frac{1}{2}\left(\frac{1}{2}x + \frac{1}{2}\right) + \frac{1}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{4}x$$

If

$$f^n(x) = \frac{1}{2}(f^{n-1}(x)) + \frac{1}{2} = \sum_{i=1}^n \frac{1}{2^i} + \frac{1}{2^n}x$$

then

$$\begin{aligned}
 f^{n+1} &= \frac{1}{2}\left(\sum_{i=1}^n \frac{1}{2^i} + \frac{1}{2^n}x\right) + \frac{1}{2} \\
 &= \left(\sum_{i=1}^n \frac{1}{2^{i+1}} + \frac{1}{2^{n+1}}x\right) + \frac{1}{2} \\
 &= \sum_{i=2}^{n+1} \frac{1}{2^i} + \frac{1}{2^{n+1}}x + \frac{1}{2} \\
 &= \sum_{i=1}^{n+1} \frac{1}{2^i} + \frac{1}{2^{n+1}}x
 \end{aligned}$$

so the formula for $f^n(x)$ follows by induction. In the limit,

$$x_f = \sum_{n=1}^{\infty} \frac{1}{2^n} = 1$$

independent of x .

III.6.7 Let (X, d) be compact, and $f : X \rightarrow X$ be a contraction with contractivity factor s and fixed point x_f . Then $f(X) \subset X$, $f(X) \neq X$.

Proof: (X, d) is compact so it is bounded. In particular, since the function $g(x) = d(x_f, x)$ is a continuous function from (X, d) to $\mathbb{R}$, it has a maximum M , and a point $x_M \in X$ such that $d(x_f, x_M) = M$. Then for all $x \in X$,

$$\begin{aligned}
 d(f(x), f(x_f)) &= d(f(x), x_f) \\
 &\leq d(f(x_M), x_f) \\
 &\leq s \cdot M < M
 \end{aligned}$$

hence there is no x such that $f(x) = x_M$, and $f(X) \neq X$.

III.6.8 The set of contraction mappings on (X, d) are a semigroup under composition. To see this, we only need to know that if f, g are contractions then so is $f \circ g$. By problem 3, if the contractivity factors for f, g are s, t respectively, then so is $f \circ g$ with contractivity factor st .

III.6.9 Let $w(x) = Ax + t$ where

$$A = \begin{pmatrix} \frac{1}{2} \cos 120^\circ & -\frac{1}{2} \sin 120^\circ \\ \frac{1}{2} \sin 120^\circ & \frac{1}{2} \cos 120^\circ \end{pmatrix}$$

and $t = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}$. The easiest way to see that w takes the equilateral Sierpinski triangle to a subset of itself, and is a contraction is to break the transformation apart. The equilateral Sierpinski triangle is expressed by the following IFS:

$$\begin{aligned} w_1 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ w_2 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} \frac{1}{4} \\ \frac{\sqrt{3}}{4} \end{pmatrix} \\ w_3 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \end{aligned}$$

Each transformation shrinks the triangle towards one of its vertices by a factor of 2, the vertices are $(0, 0)$, $(1/2, \sqrt{3}/2)$, and $(1, 0)$ respectively. Looking at w we may rewrite it as

$$\begin{aligned} w \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} \cos 120^\circ & -\sin 120^\circ \\ \sin 120^\circ & \cos 120^\circ \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \cos 120^\circ & -\sin 120^\circ \\ \sin 120^\circ & \cos 120^\circ \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} (R_\theta \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}) + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \end{aligned}$$

where R_θ is a rotation by $\theta = 120^\circ$. For the whole attractor A , the map therefore rotates A by 120° about its left lower vertex (O), by an angle equal to the exterior angle of an equilateral triangle, depositing it as an equilateral Sierpinski triangle on the left side of the y -axis with vertices $(-1, 0)$, $(0, 0)$, and $(-1/2, \sqrt{3}/2)$. The next action is a contraction, since it is identical to w_1 , and creates, setwise, a mirror image of $w_1(A)$ on the left side of the y -axis. Finally, there is a translation to the right by $1/2$ bringing the image to a half-sized equilateral triangle with vertices $(0, 0)$, $(1/2, 0)$, and $(1/4, \sqrt{3}/4)$. In other words, it lies exactly (setwise) on the image $w_1(A)$. So the map does indeed contract the triangle, and take it to a subset of itself. The fixed point of w can be calculated by

$$\begin{aligned} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} \cos 120^\circ & -\frac{1}{2} \sin 120^\circ \\ \frac{1}{2} \sin 120^\circ & \frac{1}{2} \cos 120^\circ \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{4} & -\frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & -\frac{1}{4} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix} \end{aligned}$$

which yields

$$\begin{aligned}\frac{5}{4}x_1 + \frac{\sqrt{3}}{4}x_2 - \frac{1}{2} &= 0 \\ \frac{\sqrt{3}}{4}x_1 - \frac{5}{4}x_2 &= 0 \\ \frac{\sqrt{3}}{5}x_1 &= x_2 \\ x_1 &= \frac{10}{31} \\ x_2 &= \frac{2\sqrt{3}}{31}\end{aligned}$$

III.6.10 Let Σ be a code space on the two symbols $\{0, 1\}$ and $f(x) = 1x = 1x_1x_2 \dots$. Then using the metric

$$d(x, y) = \sum_{n=1}^{\infty} \frac{|x_n - y_n|}{3^n}$$

we have

$$\begin{aligned}d(f(x), f(y)) &= \sum_{n=1}^{\infty} \frac{|x_n - y_n|}{3^{n+1}} + \frac{|1 - 1|}{3} \\ &= \sum_{n=1}^{\infty} \frac{|x_n - y_n|}{3^{n+1}} \\ &= \frac{1}{3} \sum_{n=1}^{\infty} \frac{|x_n - y_n|}{3^n} \\ &= \frac{1}{3} d(x, y)\end{aligned}$$

so that f is a contraction mapping with contractivity factor $1/3$ and fixed point $\bar{1}$.

III.6.11 Let (X, d) be a compact metric space, and $f : X \rightarrow X$ be a contraction mapping. Then $\{f^n(X)\}$ is a Cauchy sequence in $(\mathcal{H}(X), h)$ and $\lim_{n \rightarrow \infty} f^n(X) = \{x_f\}$ where x_f is the fixed point of $f(x)$.

Proof: We first verify that $\{f^n(X)\}$ is in fact a sequence in $\mathcal{H}(X)$. Since f is a contraction mapping, f is continuous, and the continuous image of a compact set is compact. Hence $f(X)$ is compact. By induction, $f^n(X)$ is compact. Furthermore, since $x_f \in X$ and $f^n(x_f) = x_f \in f^n(X)$, $f^n(X)$ is non-empty. Hence, $f^n(X) \in \mathcal{H}(X)$ for all n , so this is indeed a sequence in $\mathcal{H}(X)$. We next verify that the sequence is Cauchy. If so, since X is compact, X is complete, and therefore so is $\mathcal{H}(X)$. Hence if the sequence is Cauchy, it has a limit in $\mathcal{H}(X)$. Let s be a contractivity factor for $f(x)$, and let $\epsilon > 0$ be given. Since $s < 1$, we may choose N such that for any $n > N$, $s^n < \delta$ where δ is a number of choice less than 1. Let n, m be chosen $n < m$, and $n, m > N$. Since $m > n$, we have $f^m(X) \subset f^n(X)$, so that

$$h(f^m(X), f^n(X)) = \max_{x \in f^n(X)} d(x, f^m(X))$$

But this must be smaller than the diameter of $f^n(X)$ defined by

$$\text{diam}(f^n(X)) = \max_{x, y \in f^n(X)} d(x, y)$$

We have

$$\text{diam}(f^n(X)) \leq s^n \text{diam}(X)$$

since s is a contractivity factor for f . By choosing N such that $s^N \text{diam}(X) < \epsilon$, we guarantee that

$$h(f^m(\mathbf{X}), f^n(\mathbf{X})) = \max_{x \in f^n(\mathbf{X})} d(x, f^m(\mathbf{X})) \leq s^N \text{diam}(\mathbf{X}) < \epsilon$$

because all the distances considered for the maximum are less than $\text{diam}(f^n(\mathbf{X}))$. Hence the sequence is Cauchy and has a limit in $\mathcal{H}(\mathbf{X})$.

We now show that that limit is x_f . We know that $x_f \in \lim_{n \rightarrow \infty} f^n(\mathbf{X})$. We need only show that

$$\lim_{n \rightarrow \infty} \text{diam}(f^n(\mathbf{X})) = 0$$

But this limit is bounded by

$$\lim_{n \rightarrow \infty} \text{diam}(f^n(\mathbf{X})) \leq \lim_{n \rightarrow \infty} s^n \text{diam}(\mathbf{X}) = 0$$

Hence, if $y \in \lim_{n \rightarrow \infty} f^n(\mathbf{X})$ then $d(y, x_f) = 0$, but d is a metric, so this implies $y = x_f$. Hence

$$\lim_{n \rightarrow \infty} f^n(\mathbf{X}) = \{x_f\}$$

as desired.

III.6.13 We are asked to show that if $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is strictly positive definite (that is $a, b, c, d > 0$) then the transformation $f(\theta)$ defined on the first quadrant by

$$A \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = \lambda \begin{pmatrix} \cos f(\theta) \\ \sin f(\theta) \end{pmatrix}$$

is a contraction mapping, where λ is some positive number. Since we may scale λ to fit the matrix A , we first check the case where $\det A = 0$. This happens if the rows or columns of A are multiples of each other, and we have $f(\theta) = \theta_f$ a constant, so in this case the function f converges on the first application. Otherwise, by rescaling λ this will work for any strictly positive definite A so long as it works for a strictly positive definite A such that $\det A = 1$. We need to have real eigenvalues, so that $\text{tr } A = a + d \geq 2$. We have $ad - bc = 1$ or $bc = ad - 1$ and $bc > 0$ so that $ad \geq 1$. Suppose $ad = 1$. Then $d = 1/a$ and we have a discriminant

$$a^2 + 2ad + d^2 - 4 = a^2 + \frac{1}{a^2} - 2$$

The expression

$$a^2 + \frac{1}{a^2}$$

has derivative

$$\frac{d}{da} \left(a^2 + \frac{1}{a^2} \right) = 2a - \frac{2}{a^3}$$

showing zeroes when $a^4 = 1$ which since $a > 0$ occurs when $a = 1$. This is a minimum, and yields a discriminant of 0, but because the determinant is 1, this would yield $bc = 0$, which it can not be, so that we do indeed have such a λ , and one choice is greater than zero.

To show that f is a contraction mapping, we must show that

$$d(f(\theta_1), f(\theta_2)) \leq s \cdot d(\theta_1, \theta_2).$$

The sine of an angle is an increasing function in the first quadrant, so writing

$$\begin{aligned}
 \sin(f(\theta_1) - f(\theta_2)) &= \sin f(\theta_1) \cos f(\theta_2) - \cos f(\theta_1) \sin f(\theta_2) \\
 &= \left(\frac{c \cos \theta_1 + d \sin \theta_1}{\lambda}\right) \left(\frac{a \cos \theta_2 + b \sin \theta_2}{\lambda}\right) - \left(\frac{a \cos \theta_1 + b \sin \theta_1}{\lambda}\right) \left(\frac{c \cos \theta_2 + d \sin \theta_2}{\lambda}\right) \\
 &= \frac{1}{\lambda^2} (ac \cos \theta_1 \cos \theta_2 + bc \cos \theta_1 \sin \theta_2 + ad \sin \theta_1 \cos \theta_2 + bd \sin \theta_1 \sin \theta_2) \\
 &\quad - \frac{1}{\lambda^2} (ac \cos \theta_1 \cos \theta_2 + ad \cos \theta_1 \sin \theta_2 + bc \sin \theta_1 \cos \theta_2 + bd \sin \theta_1 \sin \theta_2) \\
 &= \frac{1}{\lambda^2} (ad(\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2) - bc(\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)) \\
 &= \frac{\det A}{\lambda^2} (\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2) \\
 &= \frac{\det A}{\lambda^2} \sin(\theta_1 - \theta_2)
 \end{aligned}$$

so that, to get a contraction mapping, we need

$$\frac{\sin(f(\theta_1) - f(\theta_2))}{\sin(\theta_1 - \theta_2)} = \frac{\det A}{\lambda^2} = s < 1$$

To test this, recall that assuming $\det A = 1$, we have $\operatorname{tr} A > 2$ and

$$\lambda = \frac{\operatorname{tr} A}{2} \pm \frac{\sqrt{(\operatorname{tr} A)^2 - 4}}{2}$$

indicating $\lambda > 1$. So f is a contraction, implying that $\{f^n(\theta)\} \rightarrow \theta_f$ as desired. The fixed angle is in fact the angle of the eigenvector, yielding

$$\theta_f = \arctan\left(\frac{\lambda - a}{b}\right)$$

Answers to Chapter III, section 7

III.7.1 We have $w_1(x) = 1/3x$ and $w_2(x) = 1/3x + 2/3$. We compute the contractivity factor for the IFS.

$$\begin{aligned}
 d(w_1(x), w_1(y)) &= \left|\frac{1}{3}x - \frac{1}{3}y\right| = \frac{1}{3}|x - y| = \frac{1}{3}d(x, y) \\
 s_1 &= \frac{1}{3} \\
 d(w_2(x), w_2(y)) &= \left|\left(\frac{1}{3}x + \frac{2}{3}\right) - \left(\frac{1}{3}y + \frac{2}{3}\right)\right| = \frac{1}{3}|x - y| = \frac{1}{3}d(x, y) \\
 s_2 &= \frac{1}{3} \\
 s &= \max(s_1, s_2) = \max\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{1}{3}
 \end{aligned}$$

We prove that the attractor of this IFS is the Cantor set by proving by induction that the action of $W(B)$ is to remove the middle third of each interval in B . To this end, our hypothesis is: Let $[a, b]$ be a component of B_k with $k < n$. Then

$$\left[a, a + \frac{1}{3}(b - a)\right] \cup \left[b - \frac{1}{3}(b - a), b\right] \subset B_{k+1}$$

and

$$\left(a + \frac{1}{3}(b - a), b - \frac{1}{3}(b - a)\right) \cap B_{k+1} = \emptyset$$

We establish that this is true for the first case: Let $B_0 = [0, 1]$. Then

$$B_1 = W(B_0) = w_1(B_0) \cup w_2(B_0) = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$$

Now suppose that $[a, b] \subset B_n$ is a component of B_n . Then there is a component $[a', b'] \subset B_{n-1}$ whose image is $[a, b]$ under some w_i ($i = 1, 2$). By hypothesis, the component $[a', b']$ has had its middle third removed in B_n . Then $[a, b]$ is replaced in B_{n+1} by the set

$$w_i([a' + \frac{1}{3}(b' - a'), [b' - \frac{1}{3}(b' - a')]) = [a + \frac{1}{3}(b - a), [b - \frac{1}{3}(b - a)]$$

since under w_i the length $b' - a' \mapsto b - a$ and clearly the two points $a', b' \mapsto a, b$ respectively. Hence this operation is a removal of middle thirds, and the induction goes through. Then the attractor of the IFS is the classical Cantor set.

III.7.2 We have $w_1(x) = s_1x$ and $w_2(x) = (1 - s_1)x + s_1$. Let $B_0 = [0, 1]$. Then

$$\begin{aligned} B_1 &= W(B_0) \\ &= w_1([0, 1]) \cup w_2([0, 1]) \\ &= [0, s_1] \cup [s_1, 1 - s_1 + s_1] \\ &= [0, s_1] \cup [s_1, 1] = [0, 1] = B_0 \end{aligned}$$

Hence $B_0 = [0, 1]$ is the attractor of the IFS.

III.7.3 Let $w_1(x) = \frac{1}{3}x$ and $w_2(x) = \frac{1}{2}x + \frac{1}{2}$. Let $B_0 = [0, 1]$. Then

$$\begin{aligned} B_1 &= W(B_0) \\ &= w_1([0, 1]) \cup w_2([0, 1]) \\ &= [0, \frac{1}{3}] \cup [\frac{1}{2}, 1] \end{aligned}$$

We have, at least on the first iteration, an operation which removes an interval of length $\frac{1}{6}$ from an interval, with this removed interval being just to the left of the center of the interval. We test this by applying the induction hypothesis as in problem 1. Assume that for each $k < n$, any component interval in B_k has a $\frac{1}{6}$ slice taken out to the left of the halfway mark, specifically, $[a, b]$ goes to

$$[a, a + \frac{1}{3}(b - a)] \cup [b - \frac{1}{2}(b - a), b]$$

and

$$(a + \frac{1}{3}(b - a), b - \frac{1}{2}(b - a)) \cap B_{k+1} = \emptyset$$

Then if $[a, b]$ is a component of B_n , then it is the image under one of the maps, say w_i of some interval $[a', b'] \subset B_{n-1}$. In B_n this interval has been replaced by

$$[a', a' + \frac{1}{3}(b' - a')] \cup [b' - \frac{1}{2}(b' - a'), b']$$

the image of which under w_i is

$$[a, a + \frac{1}{3}(b - a)] \cup [b - \frac{1}{2}(b - a), b]$$

which says that that B_{n+1} is derived from B_n by taking out the $\frac{1}{6}$ slice. A is a Cantor set, with uncountably many points, and no intervals. To see this last, if $[a, b]$ were a component interval of A , then since $W(A) = A$, this operation does not change $[a, b]$ but the above proof says it must lose a slice, hence $[a, b] \subset A \Rightarrow a = b$.

III.7.5 Let $w_1(x) = ax + b$ and $w_2(x) = cx + d$ as in the statement of the problem. The attractor A of the IFS is compact, hence it is bounded, and achieves its bounds. Then there is a closed interval $[a, b] \subset \mathbb{R}$ such that $a = \min\{x : x \in A\}$ and

$b = \max\{x : x \in A\}$. Suppose

$$w_1([a, b]) \cap w_2([a, b]) \neq \emptyset$$

Then since $[a, b]$ is connected, and the w_i are continuous, $w_1([a, b])$ and $w_2([a, b])$ are connected, their intersection is non-empty, hence $w_1([a, b]) \cup w_2([a, b])$ is connected. Furthermore, there are points $a_1, b_1 \in A \subset [a, b]$ whose image in one or the other map is a, b respectively. If there were any points in one of the images which were outside the interval $[a, b]$, then the image of either a or b under one of the maps would lie outside of $[a, b]$ and therefore not be in A which it is. Hence this image, $w_1([a, b]) \cup w_2([a, b])$ is mapped to itself exactly by W and is therefore the attractor A itself. Thus A is connected if $w_1([a, b]) \cap w_2([a, b]) \neq \emptyset$. Suppose $w_1([a, b]) \cap w_2([a, b]) = \emptyset$. Then it follows since $A \subset [a, b]$ that $w_1(A) \cap w_2(A) = \emptyset$, as well. Let $[a', b']$ be the longest connected component of A . Since $w_1(A) \cap w_2(A) = \emptyset$, there is an $i \in \{1, 2\}$ and an interval $[a'', b'']$ (connected component of A) such that

$$w_i([a'', b'']) = [a', b']$$

Let $s_i < 1$ be a contractivity factor for w_i . Then

$$d(a', b') \leq s_i d(a'', b'')$$

which says that either $[a'', b'']$ is longer than $[a', b']$ or the length of each is zero. The first contradicts the statement that $[a', b']$ is the longest connected component of A , consequently $a' = b'$, and the only connected components of A are points. Hence A is totally disconnected.

III.7.6 No. The reason for this is as follows: Because each map must have a contractivity factor of less than one, each map must take the proposed attractor A , that is the union of two disjoint intervals, to a subset of either one interval or the other. The image of each map will be the union of two disjoint intervals itself. Consequently, let w_1 take A into the first interval, and w_2 take A into the second. This situation must arise (possibly by renumbering the maps. It is possible to generate a single connected interval of either of these two images with w_3 , constructing one of the intervals of A . It is not possible to cover the gap in the other, since the image of w_3 must lie entirely in one interval or the other. (Note that the attractor could be constructed from four maps).

III.7.7 Let $\{\mathbb{R}^2 : w_1, w_2\}$ be an IFS with w_i given by

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned}$$

Let the attractor of the IFS be A . Then $(x, y) \in A$ implies that $x = y$, and $x, y \in [0, 1]$.

Proof: Let $B = \{(x, y) \in \mathbb{R}^2 : x = y, x, y \in [0, 1]\}$ Then

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2}x + \frac{1}{2} \\ \frac{1}{2}y + \frac{1}{2} \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2}x \\ \frac{1}{2}y \end{pmatrix} \end{aligned}$$

Since by supposition, $x, y \leq 1$, $1/2x, 1/2y \leq 1/2$ so adding $1/2$ to each remains in $[0, 1]$. Furthermore, if $x = y$ then $1/2x = 1/2y$ and $1/2x + 1/2 = 1/2y + 1/2$ so the image of B is contained in B . The image of B under w_1 is $\{(x, y) \in B : x, y \in [1/2, 1]\}$ exactly. This is because $0 \mapsto 1/2$ in each coordinate, and $1 \mapsto 1$ in each coordinate. The image of B under w_2 is $\{(x, y) \in B : x, y \in [0, 1/2]\}$ exactly as well. Hence the image $W(B) = B$, and since the attractor is unique, $B = A$.

III.7.8 We show that the attractor is the point $\{1\}$, together with the images $w_2^n(0)$. That these points are in the attractor of the IFS follows from the fact that the two fixed points $0, 1$ must be in the attractor, and therefore all forward images of them must be as well. Let

$$A = \{x \in \mathbb{R} : x = 1 \text{ or } x = w_2^n(0); n = 0, \dots, \infty\}$$

Let $x \in A$. Then $w_1(x) = 0 = w_2^0(0)$, if $x = 1$ then $w_2(x) = 1$, and otherwise there is a k such that $w_2^k(0) = x$ so that $w_2(x) = w_2^{k+1}(0)$. Hence $W(A) \subset A$. Furthermore if $x = 0$ then there is an element of A (for instance 0) such that $x = w_1(0)$, if $x = 1$ then $x = w_2(1)$, and otherwise if $x = w_2^k(0)$ then $x = w_2(w_2^{k-1}(0))$. Consequently W takes A onto itself as well. Therefore $W(A) = A$. A is a countable sequence of increasing points by definition. We want the geometric series for x_n . Writing $w_2(x) = 2/3(x - 1) + 1$ we have $w_2^n(x) = (2/3)^n(x - 1) + 1$. Substituting $x = 0$, we have $x_n = 1 - (2/3)^n$. To put this in the form of a geometric series,

$$\begin{aligned} x_n - x_{n-1} &= 1 - \left(\frac{2}{3}\right)^n - 1 + \left(\frac{2}{3}\right)^{n-1} \\ &= \left(1 - \frac{2}{3}\right)\left(\frac{2}{3}\right)^{n-1} \\ &= \frac{1}{2} \cdot \left(\frac{2}{3}\right)^n \end{aligned}$$

We may thus write the terms x_n as the partial sums of the geometric series

$$\frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$$

Answers to Chapter III, section 8

There are no answers to section 8, all the exercises are programming experiments.

Answers to Chapter III, section 9

III.9.2 We assume that $w_0(B) = C$ for any $B \in \mathcal{H}(X)$. We have

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0.75 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0.25 \\ 0 \end{pmatrix}$$

To show that $\{\mathbb{R}^2 : w_0, w_1\}$ we need only establish the contractivity factor for w_1 . We may rewrite w_1 using multiplication by a scalar as

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= 0.75 \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0.25 \\ 0 \end{pmatrix} \\ d(w_1(x), w_1(y)) &= \sqrt{(.75x_1 + .25 - .75y_1 - .25)^2 + (.75x_2 - .75y_2)^2} \\ &= \sqrt{(.75^2((x_1 - y_1)^2 + (x_2 - y_2)^2))} \\ &= .75\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \\ &= .75d(x, y) \end{aligned}$$

so the contractivity factor for w_1 is 0.75, the map is contractive and this is a hyperbolic IFS with condensation. Let $A_0 = C$. We calculate $w_1^n(A_0)$ by using the $A(x - x_f) + x_f$ form of the transformation. First the fixed point.

$$\begin{aligned} x_{f_1} &= .75x_{f_1} + .25 \Rightarrow .25x_{f_1} = .25 \Rightarrow x_{f_1} = 1 \\ x_{f_2} &= .75x_{f_2} \Rightarrow x_{f_2} = 0 \end{aligned}$$

Rewriting w_1^n as

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = (0.75)^n \begin{pmatrix} x - 1 \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

we see that $w_1^n(A_0)$ is C shrunk by $(0.75)^n$ and moved to $1 - (0.75)^n$. In other words, as the figure shows, the trees get smaller, closer together, and closer to 1. By example 1,

$$A = \bigcup_{n=0}^{\infty} W^n(A_0)$$

with

$$\begin{aligned} A_1 &= C \cup w_1(C) \\ A_2 &= C \cup w_1(C \cup w_1(C)) = C \cup w_1(C) \cup w_1^2(C) \\ &\vdots \end{aligned}$$

Suppose that for $k \leq n$ we have

$$A_k = \bigcup_{i=0}^k w_1^i(C)$$

then

$$\begin{aligned} A_{n+1} &= C \cup w_1(A_n) \\ &= C \cup w_1\left(\bigcup_{i=0}^n w_1^i(C)\right) \\ &= C \cup \bigcup_{i=1}^{n+1} w_1^i(C) \\ &= \bigcup_{i=0}^{n+1} w_1^i(C) \end{aligned}$$

Counting the number of transformations of C for A_n produces $n + 1$ trees as desired. Let C be the trunk of the tree. That is $w_0(B) = C$ for any $B \in \mathcal{H}(X)$. We define w_1, w_2 so as to form the two branches off the trunk, as follows:

$$\begin{aligned} w_1\begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ w_2\begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} r \cos \theta & r \sin \theta \\ -r \sin \theta & r \cos \theta \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

In other words, w_1 rotates the trunk by θ , shrinks it by r and moves its base to $(0, 1)$. w_2 rotates the trunk by $-\theta$, shrinks it by r and moves its base to $(0, 1)$. Successive applications of either of these transformations rotates the trunk by $\pm n\theta$, shrinks it by r^n , and moves the base to the top of the last piece. The figure becomes disconnected if the translation moves the piece above the top of the last trunk, it becomes not simply connected if r is close to 1 and θ is large.

III.9.8 Without the 'cls' instruction we get effectively the union of W applied to the initial set. By example 1, this is the attractor of an IFS with condensation, the condensation set being the initial set chosen.

Answers to Chapter III, section 10

III.10.10 We make reference to the general form of the affine transformation expressed in terms of the fixed point derived in exercise III.6.4. That is, for a contractive affine map, we have

$$\begin{aligned} f(x) &= Ax + t = A(x - x_f) + x_f \\ f^n(x) &= A^n(x - x_f) + x_f \end{aligned}$$

which we regard as the *canonical* form of a contraction mapping. We have, in this case

$$w_i(z) = s_i + (1 - s_i)a_i = s_i(z - a_i) + a_i$$

Regarding multiplication by a complex number as an $\mathbb{R}^2$ transformation, namely

$$\begin{aligned} s_i z &= s_{i1} z_1 + i s_{i1} z_2 + i s_{i2} z_1 - s_{i2} z_2 \\ &= (s_{i1} z_1 - s_{i2} z_2) + i(s_{i1} z_2 + s_{i2} z_1) \\ &\sim \begin{pmatrix} s_{i1} & -s_{i2} \\ s_{i2} & s_{i1} \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \end{aligned}$$

this transformation is equivalent to the $\mathbb{R}^2$ transformation

$$w_i \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} s_{i1} & -s_{i2} \\ s_{i2} & s_{i1} \end{pmatrix} \begin{pmatrix} x - a_{i1} \\ y - a_{i2} \end{pmatrix} + \begin{pmatrix} a_{i1} \\ a_{i2} \end{pmatrix}$$

from which we can read off the fixed point as a_i . Note that s_i is a rotation and scaling, making the whole operation a similitude.

Answers to Chapter III, section 11

III.11.3 In order that the attractor of an IFS be two lines joined at the origin, and variable with a parameter p , we may pick three maps as follows: Let $w_1(x, y)$ shrink the angle towards the origin with no rotation. This creates the angle again, with smaller clock hands. To get the hands back to the same length, create two maps which map the whole attractor to the tip of each hand, that is to a line at the same angle as each hand, with a fixed point at the end of the hand. Consequently, choose w_1 to be

$$w_1(x, y) = \begin{pmatrix} .9 & 0 \\ 0 & .9 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

which shrinks the clock face slightly. To collapse the attractor to a line of slope θ , we need

$$w(x, y) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

such that

$$\frac{cx + dy}{ax + by} = \tan \theta \quad \forall x, y$$

This is most easily accomplished by starting with the relations

$$\frac{c}{a} = \frac{d}{b} = \tan \theta$$

for which one solution is the matrix

$$A = \begin{pmatrix} \cos \theta & \cos \theta \\ \sin \theta & \sin \theta \end{pmatrix}$$

This matrix has $\det A = 0$, and generates the proper slope. Now suppose we want the length of the new line to be less than 1. The clock hands can be any set of lengths, we will take the minute hand to be length 1, and the hour hand to be length $1/2$. Choosing the minute hand itself, we have

$$A \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = \begin{pmatrix} \cos^2 \theta + \sin \theta \cos \theta \\ \sin^2 \theta + \sin \theta \cos \theta \end{pmatrix}$$

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so that the right hand side is of length less than two. Add to this the possibility of the whole other hand, which must be less than one, and we have to divide this matrix by 3 to assure that it fits within the given hand. So let

$$A = \begin{pmatrix} \frac{1}{3} \cos \theta & \frac{1}{3} \cos \theta \\ \frac{1}{3} \sin \theta & \frac{1}{3} \sin \theta \end{pmatrix}$$

Now we arrange to put the fixed point at the end of the hand.

$$\begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = A \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

or

$$\begin{aligned} e &= \cos \theta - \frac{\cos^2 \theta}{3} - \frac{\sin \theta \cos \theta}{3} \\ f &= \sin \theta - \frac{\sin^2 \theta}{3} - \frac{\sin \theta \cos \theta}{3} \end{aligned}$$

We now let $\theta_h = \pi/2 - (\pi/6)p$ be the hour angle, which increases clockwise from $\pi/2$ so that it reaches 2π when $p = 12$. We let $\theta_m = \pi/2 - (2/\pi)p$ which travels the same direction but reaches 2π every hour. Having arranged that

$$w_m(x, y) = A|_{\theta=\theta_m} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_m \\ f_m \end{pmatrix}$$

will be the map to the minute hand. The map to the hour hand is similar, but we must shorten the hand by a factor of 2. So,

$$w_h(x, y) = A_h \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_h \\ f_h \end{pmatrix}$$

where $A_h = \frac{1}{2}A$ and

$$\begin{aligned} e &= \cos \theta_h - \frac{\cos^2 \theta}{6} - \frac{\sin \theta_h \cos \theta_h}{6} \\ f &= \sin \theta_h - \frac{\sin^2 \theta}{6} - \frac{\sin \theta_h \cos \theta_h}{6} \end{aligned}$$

Answers to Chapter IV, section 1

IV.1.1 The addresses are $a = 21211\overline{22}$, $b = 33\overline{3}$, and $c = 122312\overline{22}$. the height, and half the width, and then, depending on the map, moves it to the appropriate corner of the square. To shrink the square by one half on each side, we define the map

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$$

and assign the symbol 0 to an iteration of this map. To get the quarter labelled 1, we must do the same shrink, and move the lower left corner over by $1/2$ in the x direction, which gives the map

$$w_2 \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$$

An iteration by w_2 corresponds to an address symbol of 1. The third map must move the lower left corner to $(1/2, 1/2)$, consequently we define

$$w_3 \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

An iteration by w_3 corresponds to the address symbol 2. Finally, the last map moves the lower left corner up by $1/2$, giving

$$w_3 \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix}$$

and an iteration by this map gets the address label 3.

IV.1.4 The easiest way to see this problem is to look for a subset of the IFS maps which have as their attractor one of the sides of the Sierpinski triangle. Consider the IFS $\{\mathbb{R}^2; w_1, w_2\}$ giving rise only to addresses which contain no 3's. If a point is on the x -axis, neither of the two IFS

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \end{aligned}$$

Either of these two will return a point on the x -axis to the x -axis, and their attractor is the line between $1\bar{1}$ and $2\bar{2}$. On the other hand, any point with a y component not equal to zero will on a single iteration have a new y value equal to half the original, which is still non-zero. Hence this edge of the Sierpinski triangle consists of all points whose addresses contain only 1 and 2.

We may use identical arguments to verify that the points along the left edge of the triangle have addresses containing only 1's and 3's. The final edge contains all addresses containing only 2's and 3's.

IV.1.6 One scheme is to use the Cartesian product of the two codespaces. Let the IFS with condensation be $\{X; w_0, w_1, \dots, w_N\}$ and the IFS whose attractor A is used for the condensation set $\{X; v_1, v_2, \dots, v_M\}$. The attractor B for the IFS with condensation is then the union of the successive iterates

$$B = \bigcup_{n=0}^{\infty} W^n(A)$$

Any point on B may be described by giving the sequence of iterates done on A to get to the particular image on B of A , combined with the address on the image, which corresponds to an address on the attractor A . Consequently, we have two code space strings, possibly on two code spaces, which give the whole address. Any address can occur.

IV.1.8 This is part of the proof of Theorem 1 of section 4.5. Let $\{X; w_1, \dots, w_N\}$ be the IFS in question, let its contractivity be s , and let its attractor be A . Consider the images of A for two points which are near by in code space. That is let some small $\epsilon > 0$ be given, and let $x = x_1 x_2 x_3 \dots$ and $y = y_1 y_2 y_3 \dots$ be two code space points $d(x, y) < \epsilon$. The points in A having these addresses are contained in the images

$$\begin{aligned} \tilde{x} &\in w_{x_1} \circ w_{x_2} \circ \dots \circ w_{x_n}(A) \\ \tilde{y} &\in w_{y_1} \circ w_{y_2} \circ \dots \circ w_{y_n}(A) \end{aligned}$$

Now, for ϵ small, we may find a number n such that

$$\frac{1}{(N+1)^n} < \epsilon$$

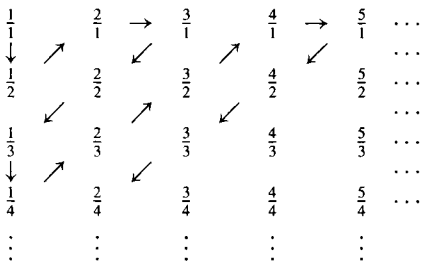
which means that $x_i = y_i$ for all $i < n$. Then both points $\tilde{x}, \tilde{y}$ are contained in

$$w_{x_1} \circ w_{x_2} \circ \dots \circ w_{x_n}(A)$$

since the contractivity is s this image must be smaller than $s^n \cdot \text{diam}(A)$. Hence the closer the addresses in code space, the closer the points having those addresses are on the attractor.

IV.1.12 The scheme which is used in Figure IV.99 is the following: Write the numbers with the same denominators in horizontal lists beginning with the denominator 1, and putting each successive denominator under the list before. Starting at the upper right hand corner, go down if we have just arrived at the left edge, go right if we have just arrived at the top edge,

go up and to the right if we are leaving the left edge, or have just gone right and up, go down and to the left if we are leaving the top edge or have just gone down and left. This winds us through the rationals as shown below:



There are two things to note: The scheme works regardless of the entries as long as every rational is in some horizontal list, with the lists ordered in a countable manner, and, all redundant terms are not in lowest terms. Since any horizontal list becomes ‘shorter’ but still infinite if such terms are removed, we may use the same scheme if we agree to write down only those numbers in lowest terms in each horizontal list. To get the negative rationals, put the negative list corresponding to each positive one in below it. That is, the second row is numbers with denominator -1 the fourth with -2 and so forth. This method of converting the rationals to a linear list indexed by the counting numbers was invented by Cantor.

IV.1.13 The triangles in the Sierpinski triangle can be labelled by size: There is 1 triangle of size 1, 3 of size $\frac{1}{3}$ and in general 3^n triangles of size $\frac{1}{3^n}$ since on each iteration of $W(A)$ we generate 3 times as many triangles of $\frac{1}{3}$ the size. Suppose we label each of the triangles with size $\frac{1}{3^n}$ with the labels

$$3^n + 1, 3^n + 2, \dots, 3^n + 3^n$$

Since this last term is $2 \cdot 3^n < 3^{n+1}$, each is a unique label, there is a label for each triangle, and the collection of labels is a proper subset of the positive integers. The number of triangles is therefore countable.

Answers to Chapter IV, section 2

IV.2.1 The IFS $\{\mathbb{R}; \frac{1}{2}x, \frac{1}{2}x + \frac{1}{2}\}$ is just touching. Let A be the attractor, and let $\mathcal{O}$ be the set

$$\mathcal{O} = (0, \frac{1}{2}) \cap A \cup (\frac{1}{2}, 1) \cap A.$$

Then we have

$$w_1(\mathcal{O}) = (0, \frac{1}{4}) \cup (\frac{1}{4}, \frac{1}{2})$$

$$w_2(\mathcal{O}) = (\frac{1}{2}, \frac{3}{4}) \cup (\frac{3}{4}, 1)$$

which are disjoint. By construction

$$\bigcup_{i=1}^n w_i(\mathcal{O}) \subset \mathcal{O}$$

Furthermore, $\mathcal{O}$ is open because it is the finite union of open intervals. The IFS is not totally disconnected because it has the attractor

$$A = [0, 1] = [0, \frac{1}{2}] \cup [\frac{1}{2}, 1] = w_1(A) \cup w_2(A)$$

and $w_1(1) = w_2(0)$.

IV.2.2 The attractor of the IFS $\{\mathbb{R}; \frac{1}{2}x, \frac{3}{4}x + \frac{1}{4}\}$ is $[0, 1]$ as follows:

$$w_1([0, 1]) = [0, \frac{1}{2}]$$

$$w_2([0, 1]) = [\frac{1}{4}, 1]$$

$$w_1([0, 1]) \cup w_2([0, 1]) = [0, 1]$$

Since there is a unique attractor with the property that $W(A) = A$ this is it. Furthermore, there is an open set $(\frac{1}{4}, \frac{1}{2})$ interior to the overlap between the images, that is

$$(\frac{1}{4}, \frac{1}{2}) \subset w_1(A) \cap w_2(A) = [\frac{1}{4}, \frac{1}{2}]$$

so the IFS is overlapping.

IV.2.3 The first thing to show is that the attractor of this IFS is $[0, 1]$, as follows: For any affine transformation $f(x) = ax + b$ on $\mathbb{R}$, an interval transforms to either

$$[c, d] \mapsto \begin{cases} [ac + b, ad + b] & \text{if } a > 0 \\ [ad + b, ac + b] & \text{if } a < 0 \end{cases}$$

For each w_n here, $a = \frac{1}{10} > 0$, so we have,

$$w_n([0, 1]) = [\frac{n-1}{10}, \frac{n}{10}]$$

Thus the union of these is

$$W([0, 1]) = \bigcup_{n=1}^{10} w_n([0, 1]) = [0, \frac{1}{10}] \cup \dots \cup [\frac{9}{10}, 1] = [0, 1]$$

Since the attractor A is unique, this is it. The IFS is clearly not totally disconnected because $w_n([0, 1]) \cap w_{n+1}([0, 1]) = \{\frac{n}{10}\}$. However, let

$$\mathcal{O} = \bigcup_{n=1}^{10} (\frac{n-1}{10}, \frac{n}{10}) \subset A$$

Then

$$w_n(\mathcal{O}) \subset w_n([0, 1]) = (\frac{n-1}{10}, \frac{n}{10}) \subset \mathcal{O}$$

and for any $i \neq j$

$$w_i(\mathcal{O}) \cap w_j(\mathcal{O}) \subset w_i([0, 1]) \cap w_j([0, 1]) = (\frac{i-1}{10}, \frac{i}{10}) \cap (\frac{j-1}{10}, \frac{j}{10}) = \emptyset$$

So the attractor is just-touching.

IV.2.4 Our IFS is $\{[0, 1]; w_1 = \frac{1}{3}x, w_2(x) = \frac{1}{3}x + \frac{2}{3}\}$. The attractor A is a subset of $[0, 1]$, and $w_1([0, 1]) = [0, \frac{1}{3}]$ and $w_2([0, 1]) = [\frac{2}{3}, 1]$ so that $w_1([0, 1]) \cap w_2([0, 1]) = \emptyset$. Then

$$w_1(A) \cap w_2(A) \subset w_1([0, 1]) \cap w_2([0, 1]) = \emptyset$$

so the IFS is totally disconnected.

IV.2.6 We have $w_1(x) = \frac{1}{2}$ and $w_2(x) = \frac{1}{2}$. The attractor is clearly $A = \{\frac{1}{2}\}$. The images of the two maps intersect, so the IFS is not totally disconnected. Furthermore, since the entire attractor is in the intersection there is clearly no open set $\mathcal{O}$ satisfying the conditions for a just-touching IFS. Consequently the IFS is overlapping, by default.

IV.2.8 The smallest number with no non-zero entries in its decimal expansion is $\overline{.1}$. Any other choice, with non-zero entries, must have at least one digit which is greater than 1. Reducing this digit to 1 produces a smaller choice. Hence reducing all of them to 1 produces the smallest.

IV.2.10 Let $\{X; w_1, w_2, \dots, w_N\}$ be our IFS with attractor A . We first prove $I \iff II$. Let x be a periodic point. Then by definition there exists a finite sequence of transformations $w_{\sigma_1}, \dots, w_{\sigma_n}$ such that

$$w_{\sigma_1} \circ \dots \circ w_{\sigma_n}(x) = x$$

If we define $f(x)$ to be this transformation, we have

$$f(x) = f^n(x) = x = (w_{\sigma_1} \circ \dots \circ w_{\sigma_n}) \circ \dots \circ (w_{\sigma_1} \circ \dots \circ w_{\sigma_n})(x)$$

consequently x has the address $\overline{\sigma_1 \dots \sigma_n}$, which is a periodic address. Now suppose that x has the periodic address $\overline{\sigma_1 \dots \sigma_n}$. Then x is given by

$$\begin{aligned} \phi(\overline{\sigma_1 \dots \sigma_n}) &= x = \phi(\sigma_1 \dots \sigma_n \overline{\sigma_1 \dots \sigma_n}) \\ &= w_{\sigma_1} \circ \dots \circ w_{\sigma_n} \phi(\overline{\sigma_1 \dots \sigma_n}) = w_{\sigma_1} \circ \dots \circ w_{\sigma_n}(x) \end{aligned}$$

so x is periodic. We next prove $I \iff III$. The semigroup of transformations generated by $\{w_1, \dots, w_N\}$ includes all finite compositions of the maps for the IFS. In particular since $w_{\sigma_1} \circ \dots \circ w_{\sigma_n}$ is a finite composition of these maps, it is an element of the semigroup. It's fixed point is x . Now suppose that x is a fixed point of an element of the semigroup of transformations generated by these maps. Then by definition of the semigroup, there is an element of the form $w_{\sigma_1} \circ \dots \circ w_{\sigma_n}$ with the property that

$$w_{\sigma_1} \circ \dots \circ w_{\sigma_n}(x) = x$$

Consequently x is periodic.

IV.2.11 We wish to show that any periodic point of the IFS $\{[0, 1] : \frac{1}{2}x, \frac{1}{2}x + \frac{1}{2}\}$ is of the form $p/(2^n - 1)$. This is certainly the case for the two fixed points 0, 1. Choose any point x in $[0, 1]$. Then we claim that after any sequence of k operations, we have

$$w_{\sigma_1} \circ \dots \circ w_{\sigma_k}(x) = \frac{1}{2^k}x + C$$

for some number C . This is true since each iteration multiplies x by $\frac{1}{2}$ and multiplies any constant term by the same factor, and possibly adds another part to the constant term. If x is k periodic then we have

$$x = \frac{1}{2^k}x + C \quad \text{or} \quad x = \frac{C2^k}{(2^k - 1)}$$

Now C is composed by taking the constant term from the last iteration and multiplying it by $\frac{1}{2}$ and either adding a half or adding 0. Thus the binary expansion of C contains at each digit a 1 if $\frac{1}{2}$ was added on that iteration, and a 0 if it was not. Hence C has a binary expansion of length k , which, multiplied by 2^k gives any possible number from 0 to $2^k - 1$, as desired.

IV.2.14 The IFS $\{\mathbb{R}; w_1(x) = 0, w_2(x) = \frac{1}{2}x + \frac{1}{2}\}$ is an IFS with condensation, the condensation set is $\{0\}$ and consequently the attractor A is given by

$$A = \bigcup_{n=0}^{\infty} w_2^n(0) \cup \{1\}$$

Any point $p \in A$ is periodic since there is some k for which $p = w_2^k(0)$ unless $p = 1$ which is a fixed point and therefore periodic. But this means that

$$p = w_2^k(w_1(p)) = w_2^k(p) \circ w_1(p)$$

so p is $k + 1$ periodic. a totally disconnected IFS is invertible, it follows that the cardinality of A is the same as that of its related code space. But code space on more than one symbol is uncountable by Theorem 1.1 in Chapter IV. Hence the attractor is too.

IV.2.16 Any uncountable attractor which is overlapping has uncountably many points with multiple addresses. This is because it contains an open set of points with multiple addresses which maps to an open set in code space by ϕ^{-1} . Any open set with non-empty interior in code space on more than one symbol is uncountable.

IV.2.17 One example of such an IFS is any overlapping IFS of two maps on the interval $[0, 1]$ such that the end points of the overlapping region $w_1(A) \cap w_2(A)$ are 3-periodic. This causes some of the sets of 4 symbols in *any* given position to map to the same point on the attractor, allowing uncountably many addresses for some points.

IV.2.18 The set of interval end points in $\mathcal{C}$ is equal to exactly twice the number of intervals used in the derivation. Following the proof of the countable number of triangles in the Sierpinski triangle, (exercise 1.13 in Chapter IV), we have at each subdivision, 2^n intervals added, and we number them $3^n + 1, 3^n + 2, \dots, 3^n + 2^n < 3^{n+1}$. Since this numbering scheme produces an integer in order for each interval without repeating, the intervals have been put into correspondence with the positive integers. Then their end points can too, since we may give all the left-hand end points the negative integer corresponding to the right hand end point labelled as above. Since the integers are countable so are the endpoints.

Answers to Chapter IV, section 3

IV.3.10 We first show that the point $12\overline{12}$ is a periodic point of period 2.

$$\begin{aligned} f(12\overline{12}) &= 2\overline{12} = 21\overline{21} \\ f^2(12\overline{12}) &= f(21\overline{21}) = 1\overline{21} = 12\overline{12} \end{aligned}$$

so this is a periodic point of period two, and the other element of the cycle is $21\overline{21}$. Next we show that it is repulsive. Let $x_f = 12\overline{12}$, and $y \in \Sigma$. Then

$$\begin{aligned} d(f^2(x_f), f^2(y)) &= d(x_f, f^2(y)) \\ &= \sum_{i=3}^{\infty} \frac{|x_i - y_i|}{3^{(i-2)}} = 3 \cdot 3 \cdot \sum_{i=3}^{\infty} \frac{|x_i - y_i|}{3^i} \\ &= 9 \cdot d(x_f, y) - (3|1 - y_1| + |2 - y_2|) \end{aligned}$$

If we choose $\frac{1}{9} > \epsilon > 0$ then for $y \in B_{\epsilon}(x_f)$ we have $y_1 = 1$ and $y_2 = 2$, so for any such ϵ

$$d(x_f, f^2(y)) \geq 9 \cdot d(x_f, y)$$

and the point is repulsive, the cycle is therefore repulsive too.

IV.3.11 The dynamical system $\{[0, 1]; 2x(1 - x)\}$ has an attractive fixed point at $x_f = \frac{1}{2}$. $f(0) = 0$ so that 0 is also a fixed point of the system. To examine the behavior near 0, we first notice that if $x \neq 0$ then as $x \rightarrow 0$ we have $(1 - x) \rightarrow 1$. Consequently, as we approach 0 the function approaches $f(x) \rightarrow 2x$. Since for any $x > 0$ we have $d(2x, 0) = 2d(x, 0) > d(x, 0)$ it is easy to find an ϵ such that the function is repulsive on an ϵ -ball around 0. For example, let $\epsilon = .1$. Then $d(0, f(x)) \geq 1.8d(0, x)$ for any $x \in B_{\epsilon}(0)$.

IV.3.14 We want the behavior of the dynamical system $\{\mathbb{R} \cup \{\infty\}; f(x) = \lambda x\}$ in four cases:

- (i) $\lambda = 0$ gives us $f(x) = 0$ which makes 0 the only fixed point, and attractive.
- (ii) $0 < |\lambda| < 1$. ∞ is a repulsive fixed point since $|f(x)| < |x|$, likewise, 0 is an attractive fixed point.
- (iii) $\lambda = -1$. 0 is a neutral fixed point, as is ∞ . Every other point is period 2, and neutral.
- (iv) $1 < \lambda < \infty$. Since $|f(x)| > x$ for any $x \neq 0, \infty$, 0 is a repulsive fixed point, and ∞ an attractive one.

IV.3.17 We have

$$f(x) = \begin{cases} x + 1 & \text{if } x \leq 0 \\ -x + 1 & \text{if } x \geq 0 \end{cases}$$

If $x = 0$ then $f(x) = 1$ and $f^2(x) = 0$ so that $\{0, 1\}$ is a cycle of period 2. For any other x split x into its integer part and fractional part, that is

$$x = n + \tilde{x} \quad |\tilde{x}| < 1, n \in \mathbb{Z}$$

Then if $x < 0$, $f^n(x) = \tilde{x}$ which is between -1 and zero, and $f^{n+1}(x) = 1 + \tilde{x}$ which is between 0 and 1. Applying f to this number gives $f(1 + \tilde{x}) = -\tilde{x}$, which is again greater than zero, and

$$f^2(1 + \tilde{x}) = f(-\tilde{x}) = 1 + \tilde{x}$$

which is a cycle of period 2. Consequently for any $x < 0$, x is eventually periodic of period 2. For $x > 0$, we have

$$f(x) = f(n + \tilde{x}) = -(n - 1) - \tilde{x}$$

so that redefining $n := -(n - 1)$ and $\tilde{x} := -\tilde{x}$ gives us the case just examined, and these points are eventually periodic of period 2 as well.

IV.3.18 If $\{X; w_1, \dots, w_N\}$ is a hyperbolic IFS, we may assume a contractivity $s < 1$. By Theorem 7.1 in Chapter III, the dynamical system $\{\mathcal{H}(X); W\}$ possesses a unique fixed point $A \in \mathcal{H}(X)$. Also by Theorem 7.1, for any $B, C \in \mathcal{H}(X)$ we have $h(W(B), W(C)) \geq s \cdot h(B, C)$. Consequently,

$$d(W(A), W(B)) = d(A, W(B)) \geq s \cdot d(A, B)$$

and the fixed point A is globally attractive (globally attractive means ϵ may be chosen any size).

IV.3.20 We need a small lemma to proceed. If a space is compact, then for any $\epsilon > 0$, there is an ϵ -net $\{y_i\}_{i=1}^{n(\epsilon)}$ of points such that every point in the space is less than ϵ from some element of the net. We need to produce a sequence of such nets.

Lemma: Given $\epsilon_1 > \epsilon_2$ and an ϵ_1 -net $\{y_i\}_{i=1}^{n(\epsilon_1)}$ there is a minimal ϵ_2 -net $\{z_i\}_{i=1}^{n(\epsilon_2)}$ such that $\{y_i\} \subset \{z_i\}$.

Proof: There is always some ϵ_2 -net which has the ϵ_1 -net as a subset, since given any ϵ_2 -net $\{z_i\}_{i=1}^{n(\epsilon_2)}$, the finite collection of points

$$\{y_i\}_{i=1}^{n(\epsilon_1)} \cup \{z_i\}_{i=1}^{n(\epsilon_2)}$$

is an ϵ_2 -net containing the ϵ_1 -net. Now taking the collection of all ϵ_2 -nets containing this particular ϵ_1 -net, there is an $n(\epsilon_2) \geq n(\epsilon_1)$ which is the minimum size of any of the nets in the collection. Let $\{z_i\}_{i=1}^{n(\epsilon_2)}$ be one of these.

Now suppose that our code space has the metric

$$d(x, y) = \sum_{i=1}^{\infty} \frac{|x_i - y_i|}{k^i}$$

with some $k > 1$, perhaps $k = N + 1$. Let $\epsilon_i = \frac{1}{k^i}$, and, with the help of the above lemma, construct a *nested* sequence of ϵ_i -nets,

$$\{y_{1i}\}_{i=1}^{n(\epsilon_1)} \subset \{y_{2i}\}_{i=1}^{n(\epsilon_2)} \subset \{y_{3i}\}_{i=1}^{n(\epsilon_3)} \subset \dots$$

We build a sequence of points $\{x_j\}_{j=1}^{\infty}$ by letting

$$x_j = y_{11} \dots y_{1j+1} y_{21} \dots y_{2j+1} \dots y_{n(\epsilon_j)1} \dots y_{n(\epsilon_j)j+1}$$

By construction, x_j visits each ϵ_j ball around each element of the ϵ_j net by repeated applications of the shift transformation T . But code space is compact, and therefore the sequence $\{x_j\}_{j=1}^{\infty}$ must have a convergent subsequence, which converges to a point x which visits every neighborhood in code space. Hence x has a dense orbit under the shift transformation.

IV.3.21 Let A, B be two open sets in code space. Choose an $\epsilon > 0$ and points $\sigma_A \in A$, and $\sigma_B \in B$, such that $B_\epsilon(\sigma_A) \subset A$. Then if the metric for Σ is

$$d(x, y) = \sum_{i=1}^{\infty} \frac{|x_i - y_i|}{(N + 1)^i}$$

there is an M such that for any $\sigma \in \Sigma$ such that

$$\sigma_1 = \sigma_{A_1}, \sigma_2 = \sigma_{A_2}, \dots, \sigma_M = \sigma_{A_M}$$

we have $\sigma \in B_\epsilon(\sigma_A) \subset A$. Then we only need to choose

$$\sigma = \sigma_{A_1} \cdots \sigma_{A_M} \sigma_B$$

which is in A , but $T^M(\sigma) = \sigma_B \in B$. Since by choosing the sequence of σ 's which agree on $M, M+1, M+2, \dots$ first symbols with σ_A , followed by σ_B , as above, we get that for all $m > M$ there is an element of A such that $T^m(\sigma) \in B$, T is mixing on code space.

Answers to Chapter IV, section 4

All the problems are examples or figure exercises.

Answers to Chapter IV, section 5

IV.5.1 Let $\{X_1; f_1\}$ and $\{X_2; f_2\}$ be equivalent dynamical systems with the equivalence provided by the homeomorphism $\theta: X_1 \rightarrow X_2$. Let $\{x_1, x_2, \dots, x_p\}$ be a cycle of period p in X_1 . That is $f_1^p(x_i) = x_i$, and

$$f_1(x_i) = \begin{cases} x_{i+1} & \text{if } i \neq p \\ x_1 & \text{if } i = p \end{cases}$$

Then

$$f_2(\theta(x_i)) = \theta \circ f_1 \circ \theta^{-1} \circ \theta(x_i) = \theta(f_1(x_i))$$

and

$$f_2^p(\theta(x_i)) = \theta \circ f_1 \circ \theta^{-1} \circ \theta \circ \cdots \circ f_1 \circ \theta^{-1} \circ \theta(x_i) = \theta(x_i)$$

so that $\{\theta(x_1), \dots, \theta(x_p)\}$ is a cycle of period p . To show that the statement holds in reverse, define the homeomorphism $\theta_2: X_2 \rightarrow X_1$ by $\theta_2 = \theta^{-1}$, and repeat the above argument.

The second statement, regarding attractive fixed points, is highly dependent on the restricted definition of the term given in the text, and, since it is not preserved by homeomorphism must be dependent on the metric. Choose two copies of the real line, $(\mathbb{R}, d_1)$ and $(\mathbb{R}, d_2)$ where $d_1(x, y)$ is the usual metric, and $d_2(x, y)$ is given by

$$d_2(x, y) = \begin{cases} |3x - 3y| & \text{if } x \leq 0 \\ |x - y| & \text{if } x \geq 0 \end{cases}$$

The identity map is then a homeomorphism between these spaces. Now choose the function $f(x) = -\frac{1}{2}x$ for both systems. In $(\mathbb{R}, d_1)$, 0 is an attractive fixed point. For $(\mathbb{R}, d_2)$, no matter what the size of the ϵ ball is around 0, it will include points on the positive side that, because the ball is shrunken on the negative side (for example, $B_1(0) = (-1/3, 1)$), fail to land back in the ball on one iteration. Hence this is not an attractive fixed point, even though the two systems are equivalent.

IV.5.2 An eventually periodic orbit is has the orbit $\{x_1, x_2, \dots, x_n, x_{n+1}, \dots, x_{n+p} = x_{n+1}\}$, for some n and some period p . As in exercise IV.5.1, we write

$$f_1(x_i) = \begin{cases} x_{i+1} & \text{if } i \neq n+p \\ x_{n+1} & \text{if } i = n+p \end{cases}$$

The argument is then identical to that in IV.5.1.

IV.5.3 We modify the example in IV.5.1. Let $(\mathbb{R}, d_1)$ and $(\mathbb{R}, d_2)$ be as in the solution to problem IV.5.1. Let the homeomorphism $\theta: \mathbb{R} \rightarrow \mathbb{R}$ be $\theta(x) = x$. Then if $f_1(x) = -2x$, under our homeomorphism we have $f_2(x) = -2x$. However, in $(\mathbb{R}, d_1)$ 0 is a repulsive fixed point, but given any $\epsilon > 0$ a point $x < 0$ such that $x \in B_\epsilon(0)$ moves under 1 iteration to a point which is closer to 0 under the d_2 metric than it started.

IV.5.4 θ is a homeomorphism, which means that θ and θ^{-1} are continuous. This implies that $\theta^{-1}(\mathcal{O})$ is open if $\mathcal{O}$ is open and $(\theta^{-1})^{-1}(\mathcal{O}) = \theta(\mathcal{O})$ is open if $\mathcal{O}$ is open. Consequently, if there is an open neighborhood of x_f in $\mathbf{X}_1$, then $\theta(\mathcal{O})$ is open in $\mathbf{X}_2$ and $\theta(x_f) \in \theta(\mathcal{O})$ implies that there is a neighborhood $\mathcal{C}$ of $\theta(x_f)$ in $\mathbf{X}_2$ such that $\mathcal{C} \subset \theta(\mathcal{O})$. Suppose that for any $y \in \mathcal{O}$ we have

$$\lim_{n \rightarrow \infty} f_1^n(y) = x_f.$$

Then since θ is continuous

$$\lim_{n \rightarrow \infty} f_2^n(\theta(y)) = \lim_{n \rightarrow \infty} \theta(f_1^n(y)) = \theta(x_f)$$

for all $\theta(y) \in \theta(\mathcal{O})$, in particular for all elements of $\mathcal{C}$. Then $x_{2f} = \theta(x_f)$ is an attractive fixed point of f_2 . *Note that this is the standard definition of an attractive fixed point. We will use the more restrictive version given in the text, however.*

IV.5.5 The solution above to IV.5.4 shows that this definition of attractive fixed points is preserved under equivalence and conjugacy. (and in fact does not require that the spaces be metric spaces). All that is needed is to give a similar definition of a repulsive fixed point. For invertible maps, this is done by saying that x_f is a repulsive fixed point if x_f is an attractive fixed point of the map f^{-1} , that is, there is a neighborhood $\mathcal{O}$ of x_f such that for every $y \in \mathcal{O}$, $\lim_{n \rightarrow \infty} f^{-n}(y) = x_f$. For non-invertible maps, this is extended by saying that there is a neighborhood $\mathcal{O}$ of x_f such that for every $y \in \mathcal{O}$ there is an n such that $f^n(y) \notin \mathcal{O}$. This is likewise preserved under equivalence and conjugacy, a moment's reflection says that if this is true for $\mathcal{O}$, then it is true for any neighborhood $\mathcal{O}_1 \subset \mathcal{O}$.

IV.5.7 We want to show that if θ , a homeomorphism establishing the equivalence of $\{\mathbb{R}; f\}$ and $\{\mathbb{R}; g\}$ is a diffeomorphism, then the definition of an attractive fixed point given in the text is preserved under the equivalence. Let x_f be an attractive fixed point of $\{\mathbb{R}; f\}$. The definition here establishes that a fixed point is attractive if and only if it is a contraction on some ϵ -ball around x_f . We let $\theta(x_f) = x_g$ be the corresponding fixed point of $\{\mathbb{R}; g\}$, and we will show that g is also locally a contraction. Choose $\epsilon > 0$ such that f is a contraction on $B_\epsilon(x_f)$. Then for $y \in B_\epsilon(x_f)$, there is an s , $0 \leq s < 1$ such that $|f(x_f) - f(y)| \leq s|x_f - y|$. Since $\theta'(x_f) \neq 0$, there is an ϵ small enough (the derivative is continuous by assumption) such that for any $y \in B_\epsilon(x_f)$, the following expression is well-defined (i.e. none of the fractions are equal to zero in the denominator) for $y \neq x_f$:

$$\left| \frac{\theta(f(x_f)) - \theta(f(y))}{f(x_f) - f(y)} \right| \left| \frac{x_f - y}{\theta(x_f) - \theta(y)} \right|$$

In the limit as $y \rightarrow x_f$, the two fractions on the right approach, respectively, $|\theta'(x_f)| \frac{1}{\theta'(x_f)} = 1$. But since f is a contraction close enough to x_f , we have

$$\left| \frac{x_f - y}{f(x_f) - f(y)} \right| \geq \frac{1}{s} > 1$$

which implies that

$$\left| \frac{\theta(f(x_f)) - \theta(f(y))}{\theta(x_f) - \theta(y)} \right| \leq s$$

Then for $\epsilon > 0$ close enough to $\theta(x_f)$, we have

$$|\theta(f(x_f)) - \theta(f(y))| \leq s|\theta(x_f) - \theta(y)|$$

so that $|g(x_g) - g(z)| \leq s|x_g - z|$ for z close to x_g , and the fixed point is attractive because it is a local contraction.

IV.5.8 Assume $\{\mathbb{R}; f\}$ is a dynamical system such that f is differentiable for all $x \in \mathbb{R}$. A web diagram for a given initial condition x_0 starts at the point $y = x_0$, and successive points are then taken from $(x_n, x_n) \mapsto (x_n, x_{n+1}) \mapsto (x_{n+1}, x_{n+1})$ where $x_{n+1} = f(x_n)$, and the points of the form (x_n, x_{n+1}) are on the graph of the function $f(x)$. It follows that if a point is fixed, $x_n = x_{n+1} = f(x_n)$ so that this is a point of intersection of the graph of $f(x)$ with the line (x, x) or $y = x$.

To prove the second assertion, we note that for our definition of attractive fixed points we have an $\epsilon > 0$ such that for any

$y \in B_\epsilon(x_f)$, we have a constant $0 \leq s < 1$ such that $|f(y) - f(x_f)| \leq s|y - x_f|$. Then for $y \neq x_f$ we may write

$$\frac{|f(y) - f(x_f)|}{|y - x_f|} = \left| \frac{f(y) - f(x_f)}{y - x_f} \right| \leq s$$

Since this is true for all $y \in B_\epsilon(x_f)$, it is certainly true as $y \rightarrow x_f$, yielding

$$\lim_{y \rightarrow x_f} \left| \frac{f(y) - f(x_f)}{y - x_f} \right| = |f'(x_f)| \leq s < 1$$

To go the other way, if the derivative is continuous in a neighborhood of x_f , and $|f'(x_f)| < 1$, it follows that for some $\delta > 0$ there is a neighborhood $B_\delta(x_f)$ such that for any $y \in B_\delta(x_f)$, $|f'(y)| < 1$. We may then find an $0 < \epsilon < \delta$ and a number $s < 1$ such that $|f'(y)| < s$ for all $y \in B_\epsilon$. Then the map is a local contraction since

$$\lim_{y \rightarrow y_0} \left| \frac{f(y) - f(y_0)}{y - y_0} \right| \leq s$$

implies that close enough to y_0 , we have

$$|f(y) - f(y_0)| \leq s|y - y_0|$$

in particular, $|f(y) - f(x_f)| \leq s|y - x_f|$, as desired.

Finally, to generalize to cycles, let $x_1, x_2, \dots, x_p$ be a cycle of period p , and suppose that the cycle is attractive. By definition, this implies that x_i is an attractive fixed point of the dynamical system $\{\mathbb{R}; f^p\}$, for some x_i in the cycle. By the result above, we have $|(f^p)'(x_i)| < 1$ for this point. This says that

$$|(f^p)'(x_i)| = |f'(x_i) \cdots f'(x_p) f'(x_1) \cdots f'(x_{i-1})| < 1$$

Notice if this condition holds at x_i it holds at any x_j in the cycle since the expression is the same product. If this product is less than 1, it follows that for at least one of the elements in the product $|f'(x_j)| < 1$, but it can also be the case that some of them do not satisfy this condition, only their product must.

IV.5.9 Since we would like our IFS to have an associated shift dynamical system equivalent to

$$f(x) = \begin{cases} 1 - 2x & x \in [0, \frac{1}{2}] \\ 2x - 1 & x \in [\frac{1}{2}, 1] \end{cases}$$

we look at an IFS which is the inverse of $f(x)$ on the regions in question. Consider the IFS

$$\{[0, 1]; \frac{1}{2}x + \frac{1}{2}, -\frac{1}{2}x + \frac{1}{2}\}$$

The attractor of this system is $[0, 1]$ as follows:

$$\begin{aligned} W([0, 1]) &= \frac{1}{2}([0, 1]) + \frac{1}{2} \cup -\frac{1}{2}([0, 1]) + \frac{1}{2} \\ &= [0, \frac{1}{2}] + \frac{1}{2} \cup [-\frac{1}{2}, 0] + \frac{1}{2} \\ &= [\frac{1}{2}, 1] \cup [0, \frac{1}{2}] = [0, 1] \end{aligned}$$

That this IFS is just-touching results from the intersection above containing the point $\frac{1}{2}$, and the fact that $\mathcal{O} = (0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$ suffices for the open set. The points of multiple addresses are the images of this point. Defining the associated shift dynamical system, we have

$$S(a) = \begin{cases} w_1^{-1}(a) & \text{if } a \in (\frac{1}{2}, 1] \\ w_2^{-1}(a) & \text{if } a \in [0, \frac{1}{2}) \end{cases}$$

The point of intersection yields

$$w_1^{-1}\left(\frac{1}{2}\right) = 2\left(\frac{1}{2} - \frac{1}{2}\right) = 0$$

$$w_2^{-1}\left(\frac{1}{2}\right) = -2\left(\frac{1}{2} - \frac{1}{2}\right) = 0$$

so we define

$$S\left(\frac{1}{2}\right) = w_1^{-1}\left(\frac{1}{2}\right) = w_2^{-1}\left(\frac{1}{2}\right) = 0$$

and there is no conflict. This explicitly gives $S(a)$ as

$$S(a) = \begin{cases} 2a - 1 & a \in [\frac{1}{2}, 1] \\ 1 - 2a & a \in [0, \frac{1}{2}] \end{cases}$$

which is clearly equivalent to $f(x)$ with the homeomorphism being the identity on $[0, 1]$.

IV.5.10 When $p = 0$ we have

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{3}{8} \\ 0 \end{pmatrix}$$

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{3}{4} \\ 0 \end{pmatrix}$$

If we let $y = 0$, then

$$w_1(x, 0) = \left(\frac{1}{4}x, 0\right)$$

$$w_2(x, 0) = \left(\frac{1}{4}x + \frac{3}{8}, 0\right)$$

$$w_1(x, 0) = \left(\frac{1}{4}x + \frac{1}{4}, 0\right)$$

so that for any point on the x -axis, $w_i(x, y)$ is also on the x -axis. We now consider the image of the set $[0, 1]$. This maps to three subintervals, each of length $\frac{1}{4}$ and arranged with their left endpoints at 0 , $\frac{3}{8}$ and $\frac{1}{4}$, respectively. The set formed by repeated iteration removes two subintervals from each interval formed by the previous iteration, and the attractor must be $\lim_{n \rightarrow \infty} W^n([0, 1], 0)$. Thus for $p = 0$, the attractor is a Cantor set.

For $p = 1$ we have

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$$

which is a just-touching Sierpinski triangle.

We next prove that for fixed $\sigma \in \Sigma$, the map $\phi(p) \rightarrow A(p)$ is continuous in p , for $p \in [0, 1]$. We may write the three transformations in matrix notation as $w_i(x) = Ax + t_i$, where

$$A = \begin{pmatrix} \frac{1+p}{4} & 0 \\ 0 & \frac{1+p}{4} \end{pmatrix}$$

$$t_i = \begin{cases} (0, 0) & i = 1 \\ (\frac{3+p}{8}, \frac{p}{2}) & i = 2 \\ (\frac{3-p}{4}, 0) & i = 3 \end{cases}$$

so that we have

$$w_{\sigma_1} \circ w_{\sigma_2}(x) = A^2x + At_{\sigma_2} + t_{\sigma_1}$$

in general, we have

$$\begin{aligned} w_{\sigma_1} \circ w_{\sigma_2} \circ \cdots \circ w_{\sigma_n}(x) \\ = A^n x + A^{n-1}t_{\sigma_n} + A^{n-2}t_{\sigma_{n-1}} + \cdots + At_{\sigma_2} + t_{\sigma_1} \end{aligned}$$

so that

$$\phi(p) = \lim_{n \rightarrow \infty} A^n x + \sum_{i=0}^n A^i t_{\sigma_{i+1}} = \sum_{i=0}^{\infty} A^i t_{\sigma_{i+1}}$$

Writing this out for fixed σ , and letting

$$t_{\sigma_i} = \begin{pmatrix} e_{\sigma_i} \\ f_{\sigma_i} \end{pmatrix}$$

we have

$$\phi_{\sigma}(p) = \begin{pmatrix} \sum_{i=0}^{\infty} (\frac{1+p}{4})^i e_{\sigma_{i+1}} \\ \sum_{i=0}^{\infty} (\frac{1+p}{4})^i f_{\sigma_{i+1}} \end{pmatrix}$$

which are continuous functions of p with σ fixed (they are, for fixed σ , differentiable in p , being a form of Taylor series expansion).

That the set $J(x) = \{y \in A(0) : \theta(y) = x\}$ where $\theta(x) = \phi(1)(\phi(0)^{-1}(x))$ is the set of y whose paths meet at x is true by construction.

Answers to Chapter IV, section 6

IV.6.3 The three points in question have addresses $1\bar{3} = 3\bar{1}$, $1\bar{2} = 2\bar{1}$, and $2\bar{3} = 3\bar{2}$. Defining $S(a) = w_1^{-1}(a)$ forces the point a to become $1\bar{2}$, which is eventually periodic since $T(1\bar{2}) = \bar{2}$, a fixed point. Similarly, if we define $S(a) = w_2^{-1}(a)$ we force the point a to become $2\bar{1}$ which since $T(2\bar{1}) = \bar{1}$ is also eventually periodic. The points b, c are analyzed similarly, being forced to either of the two addresses, which after one iteration land on the fixed points $\bar{2}$ or $\bar{3}$.

IV.6.4 If we create an IFS for the equilateral Sierpinski triangle in which we rotate the subtriangle corresponding to $w_2(\Delta)$ by 240° and that corresponding to $w_3(\Delta)$ by 120° we get a just-touching Sierpinski triangle in which the multiple address points correspond to equating the addresses

$$1\bar{3} = 3\bar{3} \quad 3\bar{1} = 2\bar{1} \quad 1\bar{2} = 2\bar{2}$$

Since these points, regardless of which map is used in the equation $S(a) = w_i^{-1}(a)$ map to the points $\bar{3}$, $\bar{1}$, and $\bar{2}$, on one application of the map, $S(a)$ may be chosen without conflict. Specifically, the IFS in question is

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} \cos 240^\circ & -\frac{1}{2} \sin 240^\circ \\ \frac{1}{2} \sin 240^\circ & \frac{1}{2} \cos 240^\circ \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \\ w_3 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} \cos 120^\circ & -\frac{1}{2} \sin 120^\circ \\ \frac{1}{2} \sin 120^\circ & \frac{1}{2} \cos 120^\circ \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{4} \\ \frac{\sqrt{3}}{4} \end{pmatrix} \end{aligned}$$

IV.6.6 The IFS $\{\Sigma; t_1, \dots, t_N\}$ where $t_n = n\sigma$ has by definition an associated shift dynamical system defined by $S(\sigma) = t_{\sigma_1}^{-1}(\sigma)$ where $\sigma = \sigma_1\sigma_2\cdots$. Then $S(\sigma) = \sigma_2\sigma_3\cdots$, which is exactly $T(\sigma)$. To see whether two such dynamical systems can be equivalent for different values of N , we use exercise IV.5.1, namely if two dynamical systems are equivalent, then a cycle of period p maps to a cycle of period p . Given $N_1 > N_2$, we have for either that any of the cycles of period 1 are given by $\bar{n}$ where $0 \leq n \leq N_i$. These are distinct, and if θ were the homeomorphism establishing the equivalence, would map to distinct fixed points. But this system for Σ_{N_2} has N_2 such points, which is less than N_1 . Thus

$$\theta : \Sigma_{N_1} \rightarrow \Sigma_{N_2}$$

either does not map fixed points to fixed points or is not one-to-one, in other words there is no such equivalence.

IV.6.14 First we prove, as claimed in the text, that

$$\tilde{A} = \{(\phi(\sigma), \sigma) : \sigma \in \Sigma\}$$

is the attractor of the lifted IFS. To do this we need to confirm hyperbolicity, in order to assert the existence of a unique fixed point in $\mathcal{H}(\mathbf{X} \times \Sigma)$. Let the metric on $\mathbf{X}$ be denoted by d_X and that on Σ be denoted d_Σ . To make things simple, we use a Manhattan metric on the product space, namely,

$$d((x_1, \sigma_1), (x_2, \sigma_2)) = d_X(x_1, x_2) + d_\Sigma(\sigma_1, \sigma_2)$$

We have, by definition,

$$\tilde{w}_i((x, \sigma)) = (w_i(x), i\sigma)$$

and we must show that this is a contraction. w_i are the maps of the original IFS, and we may assume that that system has a contractivity of s . Then

$$\begin{aligned} d(\tilde{w}_i(x_1, \sigma_1), \tilde{w}_i(x_2, \sigma_2)) &= d_X(w_i(x_1), w_i(x_2)) + \frac{1}{N+1} \sum_{j=1}^{\infty} \frac{|\sigma_{1j} - \sigma_{2j}|}{(N+1)^j} \\ &\leq s d_X(x_1, x_2) + \frac{1}{N+1} d_\Sigma(\sigma_1, \sigma_2) \end{aligned}$$

Let $m = \max(s, \frac{1}{N+1})$. Then $0 < m < 1$ and

$$d(\tilde{w}_i(x_1, \sigma_1), \tilde{w}_i(x_2, \sigma_2)) \leq m d((x_1, \sigma_1), (x_2, \sigma_2))$$

so each map $\tilde{w}_i$ is contractive, and the IFS is hyperbolic. Consequently since $\mathbf{X}$ and Σ are complete, and therefore so is $\mathbf{X} \times \Sigma$, we have the attractor $\tilde{A}$ is the unique fixed point in $\mathcal{H}(\mathbf{X} \times \Sigma)$ of the map

$$\tilde{W}(B) = \bigcup_{i=1}^N \tilde{w}_i(B)$$

for B a compact, non-empty subset of $\mathbf{X} \times \Sigma$. We now verify that the attractor $\tilde{A}$ is as given in the text. For any element of the set $\tilde{A} = \{(\phi(\sigma), \sigma) : \sigma \in \Sigma\}$ and some $\tilde{w}_i$ we have

$$\tilde{w}_i(\tilde{A}) = \{(w_i(\phi(\sigma)), i\sigma) : \sigma \in \Sigma\}$$

but $i\sigma = \phi^{-1}(w_i(\phi(\sigma)))$ so the image $\tilde{W}(\tilde{A})$ is in $\tilde{A}$. Let $a \in \tilde{A}$. Then a is given by

$$a = (w_{\sigma_1}(T(\sigma)), \sigma_1 T(\sigma))$$

so a is in the image of an element of $\tilde{A}$ under $\tilde{w}_{\sigma_1}$. Hence $\tilde{W}(\tilde{A}) = \tilde{A}$ and this is the unique attractor of the lifted IFS. Finally, to complete Lemma 1, we have

$$w_1(\tilde{A}) = \{a \in \tilde{A} : a = (x, 1\sigma)\}$$

$$w_2(\tilde{A}) = \{a \in \tilde{A} : a = (x, 2\sigma)\}$$

These sets are points with mutually exclusive sets of second coordinates, so their intersection is empty. Hence $\tilde{A}$ is totally disconnected.

IV.6.15 Let $\{x_n\}$ be any orbit of the associated shift dynamical system $\{A; S\}$. Then by definition, $S(x_n) = x_{n+1} = w_i^{-1}(x_n)$ for some w_i . Thus, regardless of the rule for assigning $S(a)$ on overlapping regions, there is a rule of assignment which assigns either w_1^{-1} or w_2^{-1} to each application of S to the orbit. Let τ be the address such that if $S(x_n) = w_i^{-1}(x_{n+1})$ then $\tau_n = i$. Let

$$\sigma_n = \tau_n \tau_{n+1} \dots$$

Then $\phi(\sigma_n) = x_n$, $\sigma_{n+1} = T(\sigma_n)$, so that the orbit $\{(\phi(\sigma_n), \sigma_n)\}$ of the shift dynamical system $\{\tilde{A}; \tilde{S}\}$ has the desired property.

IV.6.16 By definition, the lift is $(\sigma_1, \sigma_2) \mapsto (T\sigma_1, \sigma_1\sigma_2)$ which looks like the shift automorphism: it takes the symbol off the first element and puts it on the second. The suitably defined inverse would then be to take a symbol off the second element and put it back on the first. This defines a system on $\Sigma \times \Sigma$, as the space of shifts, which is identical to the shift automorphism.

Answers to Chapter IV, section 7

IV.7.2 $\tilde{x}_0 = 21\bar{2}$ consequently the true orbit of $\tilde{x}_0$ is given by

$$\begin{aligned} \tilde{x}_0 &= 21\bar{2} & T(\tilde{x}_0) &= \bar{1}\bar{2} & T^2(\tilde{x}_0) &= \bar{2} \\ T^n(\tilde{x}_0) &= \bar{2} & \text{for } n &\geq 2 \end{aligned}$$

By contrast, we have $\tilde{x}_n = \bar{1}\bar{2}$ after the first iteration, so that $d(\tilde{x}_n, T^n(\tilde{x}_0)) = 1/3$ after the second iteration, under the code space metric for two symbols. We wish to find a shadowing orbit. Taking $\theta = 1/3$ and $s = 1/3$ we need

$$d(\tilde{x}_n, x_n) \leq \frac{s\theta}{1-s} = \frac{1}{6}$$

Any real orbit which has the symbol 2 in it will eventually be separated from $\bar{1}\bar{2}$ by at least $1/3$ so the only possible choice for a shadowing orbit is $\bar{1}$. Computing the distance yields

$$d(\bar{1}\bar{2}, \bar{1}) = \frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{3^n} = \frac{1}{1-\frac{1}{3}} - 1 = \frac{1}{6}$$

which satisfies the shadowing theorem.

Answers to Chapter IV, section 8

IV.8.1 It suffices to show that the positive rationals are dense in the positive reals, since the function $\theta(x) = -x$ will show that the negative rationals are then dense in the negative reals. To this end let $r \in \mathbb{R}$, and let its decimal expansion be $.r_1 r_2 r_3 \dots$. Then the sequence of rationals

$$.r_1 \bar{0}, .r_1 r_2 \bar{0}, .r_1 r_2 r_3 \bar{0}, \dots$$

converges to r . Hence every number $r \in \mathbb{R}$ is a limit point of the rationals, and since the closure of a set contains its limit points, the closure of the rationals is the reals. Hence the rationals are dense in $\mathbb{R}$.

IV.8.2 The sequence in question is an enumeration of the rationals in $[0, 1]$, i.e. $\mathbb{Q} \cap [0, 1]$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, $\mathbb{Q} \cap [0, 1]$ is dense in $[0, 1]$.

IV.8.3 The function $f(x) = 2x \bmod 1$ can be viewed as follows: On the circle of radius $1/2\pi$, that is, of circumference 1, define $x(\theta)$ to be the distance along the circumference in the counterclockwise direction from the positive x -axis. Then taking the usual equation of angles $\theta = \theta + 2\pi = \theta \bmod 2\pi$, the function $\theta \mapsto 2\theta$ gives us our function on $[0, 1]$. If θ is not a rational number times 2π , it cannot be return to itself under this map, since $2^n\theta = \theta$ implies that $2^n\theta = \theta + m\pi$ for some integer m . Hence any irrational number times 2π for θ results in a dense orbit on the circle.

IV.8.4 The dynamical system $\{[0, \infty); f(x) = 2x\}$ is sensitive to initial conditions as follows: Choose $\delta = 1$, and for any $\epsilon > 0$ let $y \neq x$, and $y \in B_\epsilon(x)$. Then since $d(f(x), f(y)) = 2d(x, y)$, we have $d(f^n(x), f^n(y)) = 2^n d(x, y)$. Choose n , given y such that $2^n d(x, y) > 1$. In particular, for any $n > N$ where $2^N > 1/d(f^n(x), f^n(y)) > \delta$.

The dynamical system $\{[0, \infty); f(x) = .5x\}$ is not sensitive. In this case, we have $d(f(x), f(y)) = 0.5d(x, y)$ so that for any $\epsilon > 0$ and any δ , there is an N such that for any $y \in B_\epsilon(x)$ $d(f^n(x), f^n(y)) = (0.5)^n d(x, y)$ is less than δ .

IV.8.5 Let Σ be the code space on two symbols, and let T be the shift operator. Then $\{\Sigma; T\}$ is sensitive to initial conditions: Choose $\delta < \frac{1}{3}$. Then if $x, y \in \Sigma$ and $x \neq y$, there is an n such that $x_n \neq y_n$. Then since

$$T^n(x) = T^n(x_1x_2\dots) = x_nx_{n+1}\dots$$

$$T^n(y) = T^n(y_1y_2\dots) = y_ny_{n+1}\dots$$

it follows that

$$d(T^n(x), T^n(y)) \geq \frac{|x_n - y_n|}{3} > \delta$$

Hence whatever ϵ is chosen, we may find a y and an n satisfying sensitivity to initial conditions.

$\{\Sigma; T\}$ is transitive as well: Let $\mathcal{U}, \mathcal{V} \subset \Sigma$ be open sets. Then we can choose $y \in \mathcal{V}$ and $\delta > 0$ such that $B_\delta(y) \subset \mathcal{V}$, and $x \in \mathcal{U}$ and $\epsilon > 0$ such that $B_\epsilon(x) \subset \mathcal{U}$. Choose n such that $\frac{1}{3^n} < \epsilon$. Then any $v \in \Sigma$ such that

$$v = x_1x_2\dots x_n v_1v_2\dots \in B_\epsilon(x) \subset \mathcal{U}$$

Similarly, choose M such that for any $m > M$,

$$u = y_1y_2\dots y_m u_1u_2\dots \in B_\delta(y) \subset \mathcal{V}$$

Then any $\sigma \in \Sigma$ of the form

$$\sigma = y_1y_2\dots y_mx_1x_2\dots x_n\sigma_1\sigma_2\dots$$

is in $\mathcal{V}$. $T^n(\sigma) = x_1x_2\dots x_n\sigma_1\sigma_2\dots$, which is in $\mathcal{U}$. Hence $T^n(\mathcal{V}) \cap \mathcal{U} \neq \emptyset$, as desired.

IV.8.6 Suppose that $\{X; f\}$ and $\{Y; g\}$ are equivalent under the homeomorphism $\theta : X \rightarrow Y$. Let $\mathcal{U}, \mathcal{V} \subset Y$ be open sets. Then because θ is continuous, $\theta^{-1}(\mathcal{U}), \theta^{-1}(\mathcal{V})$ are open sets in X . Since $\{X; f\}$ is transitive, there is an n such that

$$\theta^{-1}(\mathcal{U}) \cap f^n \circ \theta^{-1}(\mathcal{V}) \neq \emptyset$$

Let x be a member of this non-empty intersection. Then $\theta(x) \in \mathcal{U}$ and $x \in f^n \circ \theta^{-1}(\mathcal{V})$ implies

$$\theta \circ \theta^{-1}(\mathcal{U}) \cap \theta \circ f^n \circ \theta^{-1}(\mathcal{V}) = \mathcal{U} \cap g^n(\mathcal{V}) \neq \emptyset$$

so that $\{Y; g\}$ is also transitive. Reversing the roles of θ and θ^{-1} , which we may do because θ^{-1} is also continuous, yields the converse.

Answers to Chapter V, section 1

V.1.3 Let (X, d) be a metric space, $A \subset X$ with $A = \{a, b, c\}$. We wish to calculate the fractal dimension of A . Let $r = \min_{x, y \in A} (d(x, y))$. Then for any $\epsilon < r/2$, we have $\mathcal{N}(A, \epsilon) = 3$. Hence the fractal dimension is

$$D(A) = \lim_{\epsilon \rightarrow 0} \frac{\ln 3}{\ln(1/\epsilon)} = 0.$$

V.1.6 Using the Box Counting Theorem to cover the classical Cantor set $\mathcal{C}$ with boxes of size $1/3^n$, we will get

$$D = \lim_{n \rightarrow \infty} \left(\frac{\ln \mathcal{N}_n(\mathcal{C})}{\ln 3^n} \right).$$

We need to calculate $\mathcal{N}_n(\mathcal{C})$. It takes $1 = 2^0$ box when $n = 0$, subdividing and deleting the middle third gives $2 = 2^1$ boxes when $n = 1$. On each subdivision and deletion of a set of middle thirds, we divide each interval covered by a box of size $1/3^n$ in thirds and retain two of these thirds, so that it will take 2^{n+1} boxes of size $1/3^{n+1}$ to cover the set at the next step. Hence $\mathcal{N}_n(\mathcal{C}) = 2^n$, and we have

$$D(\mathcal{C}) = \lim_{n \rightarrow \infty} \frac{\ln(2^n)}{\ln 3^n} = \lim_{n \rightarrow \infty} \frac{n \ln 2}{n \ln 3} = \frac{\ln 2}{\ln 3}.$$

V.1.7 Use boxes of side length $1/2^n$. For each quartering of the figure, and its subdivisions, we divide each box into 4 parts, and one part contains none of the fractal subset points. Consequently, on each subdivision, the number of boxes increases threefold. There is one box of side length $1/2^0$ hence

$$D(\text{Figure}) = \lim_{n \rightarrow \infty} \frac{\ln 3^n}{\ln 2^n} = \frac{\ln 3}{\ln 2}.$$

V.1.8 One way to choose the boxes is to take side lengths $1/2^n$, with the first subdivision running horizontally from just below the bottom of the caption, the vertical running vertically from the lefthand 'tip' of the bottom piece. Do it with a ruler to assure yourself that this divides the figure into three congruent pieces, and leaves the fourth of the four squares empty. Following the lead of the previous problem, the dimension is $\frac{\ln 3}{\ln 2}$.

V.1.9 We wish to show that using a Cantor set constructed by removing middle thirds on the interval $[0, 3]$ has the same fractal dimension as the usual Cantor set. We start with 2^0 intervals of length $1/3^{0-1}$. The rest of the procedure is the same, except that we now have

$$D(\text{Big Cantor}) = \lim_{n \rightarrow \infty} \frac{\ln(2^n)}{\ln(3^{n-1})} = \lim_{n \rightarrow \infty} \frac{n \ln 2}{(n-1) \ln 3} = \frac{\ln 2}{\ln 3} \lim_{n \rightarrow \infty} \frac{n}{n-1} = \frac{\ln 2}{\ln 3}.$$

V.1.10 Let $A \subset \mathbb{R}^2$ be a compact non-empty subset. A has fractal dimension D_1 in the Euclidean metric, and D_2 in the Manhattan metric. The metrics are equivalent, consequently there is a transformation $\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which forms the equivalence. By Theorem 1.2 in Chapter V then, we have $D_1(A) = D_2(\theta(A))$ or $D_1 = D_2$.

V.1.11 Let $B = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$, and define two IFS

$$B \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}, B \begin{pmatrix} x \\ y \end{pmatrix}, B \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

with attractor A_1 , and

$$B \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix}, B \begin{pmatrix} x \\ y \end{pmatrix}, B \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

with attractor A_2 .

These are two Sierpinski triangles, one with right angle at the origin, and hypotenuse between $(1, 0)$ and $(0, 1)$, the other with right angle at $(1, 1)$ and hypotenuse between the origin and $(0, 2)$. We rotate one into the other, expanding by a factor of two, to generate an equivalence between the two systems. This requires an invertible affine map. One which works is

$$\theta \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The two attractors are metrically equivalent, and therefore have the same fractal dimension.

Answers to Chapter V, section 2

V.2.3 Let B be given by

$$B = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$$

Then we can use the following IFS to generate ■:

$$w_1 = Bx$$

$$w_2 = Bx + \begin{pmatrix} 0 \\ 1/2 \end{pmatrix}$$

$$w_3 = Bx + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}$$

$$w_4 = Bx + \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

Each map has a contractivity factor of $1/2$, so the fractal dimension by Theorem 2.3 in Chapter V is given by

$$1/2^D + 1/2^D + 1/2^D + 1/2^D = 1$$

This yields $1/2^D = 1/4$ or $D = 2$ as desired.

V.2.4 We get the fractal dimension of the Cantor set from the IFS

$$\{[0, 1]; w_1(x) = \frac{1}{3}x; w_2(x) = \frac{1}{3}x + \frac{2}{3}\}.$$

Each map has contractivity $1/3$ so we have

$$1/3^D + 1/3^D = 1 \text{ or } 1/3^D = 1/2.$$

Then $-D \ln 3 = -\ln 2$ or $D = \frac{\ln 2}{\ln 3}$. to the equation $3(1/2^D) + (2/5)^D = 1$. Solved numerically, this gives $D \approx 1.874$.

V.2.5 The attractor contains 3 images of itself scaled by $1/2$, with two of them rotated by $\pi/2$ and shifted. Assuming that the attractor is centered at the origin, and extends two units in either x -direction and unit in either y direction, the IFS is given by

$$w_1 = B \begin{pmatrix} x \\ y \end{pmatrix}$$

$$w_2 = B \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 3/2 \\ -3/4 \end{pmatrix}$$

$$w_3 = B \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -3/2 \\ 3/4 \end{pmatrix}$$

where B is given by

$$\begin{pmatrix} 0 & -1/2 \\ 1/2 & 0 \end{pmatrix}$$

The scaling factors are all $1/2$, leading to the equation $3/2^D = 1$, or $D = \frac{\ln 3}{\ln 2}$.

V.2.8 This attractor has 6 maps, 3 at $1/2$ scale, 3 at $1/4$ scale. It is overlapping, we need an upper bound which we get from Theorem 3.1 in Chapter V. We have,

$$3 \cdot (1/2^D) + 3 \cdot (1/4^D) = 1$$

Rewriting this as

$$3 \cdot (1/2)^{2D} + 3 \cdot (1/2)^D - 1 = 0$$

yields $1/2^D = \pm(\sqrt{21}/6) - 1/2$ or

$$D = \left| \frac{\ln(\sqrt{21}/6 - 1/2)}{\ln 2} \right|$$

V.2.9 Let $w_i : \mathbb{R}^7 \rightarrow \mathbb{R}^7$ for $i = 1, 2$, be an IFS with scaling factors s_1, s_2 respectively.

That the new maps are similitudes follows from the general fact that the composition of similitudes is a similitude. To see this write the two similitudes as

$$r_1 R_1 R_{\theta_1} \circ r_2 R_2 R_{\theta_2}(x)$$

Since reflections are diagonal, they commute with the other matrices, the scalars commute as well. The composition of rotations is a rotation, the composition of reflections is a reflection, multiplying scalars together is a scalar. We are left with

$$r_1 r_2 R_1 R_2 R_{\theta_1} R_{\theta_2}$$

which is a similitude.

The new contraction factors are

$$s_{v_1} = s_1^2 s_{v_2} = s_1 s_2 s_{v_3} = s_2 s_1 s_{v_4} = s_2^2$$

Using the original system in Theorem 3.1 in Chapter V, we have

$$|s_1|^D + |s_2|^D = 1$$

The new system yields

$$|s_1^2|^D + 2|s_1 s_2|^D + |s_2^2|^D = (|s_1|^D + |s_2|^D)^2 = 1$$

The calculation of dimension remains unchanged, since the first equation is a solution of the second.

Answers to Chapter V, section 3

The exercises in Section 3 are all experimental in nature.

Answers to Chapter V, section 4

V.4.1 The fact that $\mathcal{M}(A, p, \epsilon)$ is a non-increasing function of ϵ is proved by looking at the sums over which the infimum is taken. Suppose $\epsilon_1 < \epsilon_2$ and that $\mathcal{M}(A, p, \epsilon_1) < \mathcal{M}(A, p, \epsilon_2)$. Since any sequence $\{A_i\}$ which qualifies for ϵ_1 is a sequence of sets with diameters less than ϵ_2 , it follows that by taking the same sequence in the computation of $\mathcal{M}(A, p, \epsilon_2)$, the infimum must be less than or equal to the sum generated by this sequence. But this is a contradiction.

V.4.2 Let $A = \{a, b, c, d, e, f, g\}$ be a subset of $(\mathbb{R}^2, \text{Euclidean})$. Let $2r$ be the minimum distance between points in A . By choosing $\epsilon < r$, we must, for any ϵ cover the set with at least 7 sets $A_1, \dots, A_7$. We can also cover A exactly with 7 sets of any epsilon of this size. Consequently, the infimum is generated for any p by the limit of the finite sum as $\epsilon \rightarrow 0$ which is 0 for any $p > 0$. When $p = 0$ the sum is exactly 7.

V.4.3 Let A be a bounded countably infinite subset of $\mathbb{R}^2$, under the Euclidean metric. Let $\rho : \mathbb{Z}^+ \rightarrow A$ be a rule of assignment which takes each positive integer i to an element $a_i \in A$, where ρ is 1-1 and onto. Such a function exists by definition of countability. Then the sequence of subsets $A_i = \{a_i\}$ of A is in $\mathcal{A}$, since their union is the set A . But the diameter of any singleton set in $\mathbb{R}^2$ is zero, so that for any $p > 0$

$$\sum_{i=1}^{\infty} (\text{diam}(\{a_i\}))^p = 0$$

Hence for any $p > 0$, $\mathcal{M}(A, p, \epsilon) = 0$ for any $\epsilon > 0$, consequently, $\mathcal{M}(A, p) = 0$. On the other hand, if $p = 0$, then for any non-empty subset of A , $(\text{diam}(A_i))^p = 1$. For each $n > 1$, there is an $\epsilon > 0$ such that there are more than n disjoint subsets of A , hence $\mathcal{M}(A, 0) = \infty$.

V.4.4 For $p = 0$, given $\epsilon < \frac{1}{3^n}$, there are at least 2^n subsets in any sequence of subsets of diameter less than ϵ which cover C . To see this, the endpoints of 2^n disjoint intervals of diameter $\frac{1}{3^n}$ are elements of C , and can not lie in the same subset. Hence as $\epsilon \rightarrow 0$, $\mathcal{M}(C, 0, \epsilon) \rightarrow \infty$. Since $\mathcal{M}(C, 0)$ is the supremum of these numbers, it is also infinite. For $p = 1$, given $\epsilon > 0$, we may choose the smallest n such that $\frac{1}{3^n} < \epsilon$, and cover C with exactly 2^n subsets with diameter $< \epsilon$, by taking the intersection of C with subintervals of $[0, 1]$. The infimum for any $\epsilon > 0$ is the value of the sum as $n \rightarrow \infty$, which is

$$\lim_{n \rightarrow \infty} \frac{2^n}{3^n} = 0$$

Since $\mathcal{M}(C, p, \epsilon) = 0$ for any $\epsilon > 0$ we have $\mathcal{M}(C, 1) = 0$.

V.4.5 The limit scheme forces a minimization of the number of sets by the diameter of sets. Suppose we attempt to find $\mathcal{M}(A, p)$ such that this number is less than infinity, but non-zero (this is not always possible). For the Sierpinski triangle, we may proceed by using a sequence of ϵ values that are the diameters of the subtriangles. That this is sufficient is guaranteed by problem 1. For any p then, the values calculated for exactly the sequence of subtriangles leads to the infimum desired. We know that calculating on the subtriangles, we can minimally cover Δ with 3^n triangles of diameter $d \cdot 2^{-n}$ where d is the diameter of Δ . Then we have, for each n , p a value of $d \cdot 3^n \cdot 2^{-np}$ as a candidate for the infimum. Notice that as $n \rightarrow \infty$ for p small, this is an increasing sequence, consequently our infimum at fixed n , p is the value at n . For p large, the sequence converges to 0, so that we have one infimum for all ϵ values. These p values correspond to infinite or zero values for $\mathcal{M}(A, p, \epsilon)$. The value at which we will get a finite nonzero number is when, independent of n , we have $2^{np} = 3^n$. This value is $p = \log_2 3 = (\ln 3)/(\ln 2)$. At this value, the terms in the sequence are identically d , the diameter of the Sierpinski triangle.

V.4.6 As the following proof will indicate (exercise V.4.7), to get strict inequality one must have a situation where it takes more sets which are box shaped to cover A than if another, better shape is chosen.

V.4.7 We have by definition,

$$D(A) = \lim_{\epsilon \rightarrow 0} \frac{\mathcal{N}_\epsilon}{\frac{1}{\epsilon}}$$

for the fractal dimension, where the diameter of each box is ϵ times some fixed constant. Fixing $\epsilon > 0$ and rearranging this formula, we have

$$\begin{aligned} D(A) \ln \left(\frac{1}{\epsilon} \right) &= \ln \mathcal{N}_\epsilon \\ \frac{1}{\epsilon^{D(A)}} &= \mathcal{N}_\epsilon \\ 1 &= \mathcal{N}_\epsilon \epsilon^{D(A)} = \sum_{i=1}^{\mathcal{N}} \epsilon^{D(A)} \end{aligned}$$

Since for A_i chosen such that $A_i = B_i \cap A$, that is the intersection of each box with the set A , we have $\text{diam}(A_i) \leq B_i$, it follows that for this choice of ϵ , $\mathcal{M}(A, D(A), \epsilon) < \infty$, since it must be less than or equal to this choice of A_i . Since $D(A)$ converges, this must then be the case for every choice of ϵ , and we have that

$$\lim_{\epsilon \rightarrow 0} \sum_{i=1}^{\infty} (\text{diam}(A_i))^p \leq 1$$

for $p = D(A)$. If this sum is zero, then $D_H(A) \leq D(A)$, if it is not, then $D_H(A) = D(A)$. Notice that if $\mathcal{M}(A, D(A)) = 0$, we cannot automatically conclude that these two are unequal, if it is non-zero then they are equal.

V.4.10 Let $A_1, A_2 \in (\mathbb{R}^n, \text{Euclidean})$, be two metrically equivalent sets, and let $\theta : A_1 \rightarrow A_2$ be a metric equivalence. Then for any $B \subset A_1$, we have $\text{diam}(\theta(B)) \leq C_1 \text{diam}(B)$ and $\text{diam}(B) \leq C_2 \text{diam}(\theta(B))$ for some constants C_1, C_2 . Consequently, if $\mathcal{M}(A_1, p) = \infty$, then for any choice of subsets, and $\epsilon > 0$, we must have $\sum_{i=1}^{\infty} (\text{diam}(B_i))^p \rightarrow \infty$. Thus for any choice of subsets on A_2 this sum for the images of B_i is equal to $\sum_{i=1}^{\infty} (C_2 \text{diam}(B_i))^p$ which is therefore also infinite.

Hence $\mathcal{M}(A_2, p) = \infty$. Conversely, if for some p , we have $\mathcal{M}(A_1, p) = 0$, then the infimum over ϵ values for all choices of subsets of A_1 is zero. Consequently, for any $\delta > 0$, we may find a set of subsets satisfying $\text{diam}(C_i) < \epsilon$ for which $\mathcal{M}(A, p, \epsilon) < \delta$ for every epsilon by choosing a sequence of subsets whose sum is less than δ/C_1 and using the metric equivalence to generate the sequence in A_2 . Consequently $\mathcal{M}(A_2, p) = 0$ as well. Obviously by using θ^{-1} , and reversing the role of the constants, we can do the same, given a value on A_2 . Since the two sets agree on infinite and zero values for $\mathcal{M}(A, p)$, then the point at which the two values change are the same. Hence $D_H(A_1) = D_H(A_2)$.

V.4.11 Proving that a metric equivalence generates the same Hausdorff-Besicovitch dimension, is identical to exercise V.4.10. The second statement, that the Hausdorff D -dimensional measure $\mathcal{M}(A, D_H(A))$ is not preserved may be shown by example: Looking at example V.4.8, we may take a Sierpinski triangle with vertices at $(0, 0)$, $(2, 0)$, and $(0, 2)$. Then for each step n in the example, we cover the triangle with 3^n closed disks of diameter $2\sqrt{2} \cdot \frac{1}{2^n}$, and the example proceeds as in the text, except that

$$\mathcal{M}(\Delta, D_H(\Delta)) = (2\sqrt{2})^{\ln 3 / \ln 2}$$

Consequently we may take the original Sierpinski triangle, with the metric $d_2(x, y) = 2 \cdot d(x, y)$, or twice the Euclidean metric, and obtain a different value for the Hausdorff p -dimensional measure.

V.4.12 They should be (inches)^{1.391}.

Answers to Chapter VI, section 1

VI.1.2 The Taylor expansion for $\sin(x)$ near 0 is given by

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{(2n+1)}}{(2n+1)!} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \dots$$

giving a linear approximation of $\ell(x) = x$. Let $\epsilon > 0$. Then a linear coordinate transformation from $[0, \epsilon] \times [0, \epsilon]$ to $[0, 1] \times [0, 1]$ is given by $\theta((x, y)) = (\frac{x}{\epsilon}, \frac{y}{\epsilon})$. With the change of coordinates we still get $\ell'(x') = x'$, and the graph of this function, L is $(x, x) \in [0, 1] \times [0, 1]$. Transforming the function $y = f(x) = \sin(x)$, gives us

$$y' = f'(x') = \theta \circ f \circ \theta^{-1}(x') = \frac{\sin(\epsilon x)}{\epsilon}$$

so the graph G is the set $(x, \frac{\sin(\epsilon x)}{\epsilon}) \in [0, 1] \times [0, 1]$.

The Manhattan metric distance between a point $g = (x, \frac{\sin(\epsilon x)}{\epsilon})$ on G and a point $l = (y, y)$ on L is given by

$$d(g, l) = |x - y| + \left| \frac{\sin(\epsilon x)}{\epsilon} - y \right|$$

The shape of an ϵ ball in Manhattan metric is a square with vertical and horizontal corners. So to get the point on L closest to a given point $g \in G$, we look for the intersection of the smallest isosceles right triangle with right angle at g intersecting L , which since the slope of L is 1, intersects from (x, x) to $(\frac{\sin(\epsilon x)}{\epsilon}, \frac{\sin(\epsilon x)}{\epsilon})$ at the same Manhattan distance. Hence for any $g \in G$ the Manhattan distance to L is

$$d(g, L) = x - \frac{\sin(\epsilon x)}{\epsilon}$$

The derivative of $\frac{\sin(\epsilon x)}{\epsilon}$ is decreasing, and this function is always less than x in this domain, hence the distance from G to L is given by the maximum, reached when $x = 1$, or

$$d(G, L) = 1 - \frac{\sin(\epsilon)}{\epsilon}$$

Going the other way, we base an isosceles right triangle with the right angle at (y, y) and look for the smallest such triangle intersecting G . Once again, since the slope of G is decreasing, this will touch when we take the first coordinates equal,

hence

$$d(\ell, G) = y - \frac{\sin(\epsilon y)}{\epsilon}$$

and this is maximized at $(1, 1)$. Hence

$$d(L, G) = 1 - \frac{\sin(\epsilon)}{\epsilon} = d(L, G) = h(L, G)$$

Noting that this is indeed decreasing as $\epsilon \rightarrow 0$ since

$$\lim_{\epsilon \rightarrow 0} \frac{\sin(\epsilon)}{\epsilon} = 1$$

we calculate

$$h(L, G) \approx .01 \Rightarrow .99\epsilon = \sin(\epsilon)$$

or ϵ less than approximately .1.

Answers to Chapter VI, section 2

VI.2.2 Let

$$f(x) = F_{i-1} + \frac{x - x_{i-1}}{x_i - x_{i-1}}(F_i - F_{i-1})$$

The first step is to show that if $(x, y) \in G$ then so is $w_i(x, y)$. Let $(x, y) \in G$, that is $(x, y) = (x, f(x))$. Then (x, y) goes to

$$\begin{pmatrix} a_i x \\ c_i x \end{pmatrix} + \begin{pmatrix} e_i \\ f_i \end{pmatrix}$$

or

$$\begin{pmatrix} \left(\frac{x_i - x_{i-1}}{x_N - x_0} \right) x + \frac{x_N x_{i-1} - x_0 x_i}{x_N - x_0} \\ \left(\frac{F_i - F_{i-1}}{x_N - x_0} \right) x + \frac{x_N F_{i-1} - x_0 F_i}{x_N - x_0} \end{pmatrix}$$

To compute $f(x)$ for the transformed x we do

$$\begin{aligned} & \frac{x_i - x_{i-1}}{x_N - x_0} x + \frac{x_N x_{i-1} - x_0 x_i}{x_N - x_0} - x_i \\ &= \frac{x_i - x_{i-1}}{x_N - x_0} x + \frac{x_N x_{i-1} - x_0 x_i}{x_N - x_0} - \frac{x_{i-1} x_N - x_{i-1} x_0}{x_N - x_0} \\ &= \frac{x_i - x_{i-1}}{x_N - x_0} + \frac{x_{i-1} x_0 - x_0 x_i}{x_N - x_0} \\ &= (x - x_0) \left(\frac{x_i - x_{i-1}}{x_N - x_0} \right) \end{aligned}$$

Dividing by $x_i - x_{i-1}$ yields

$$\frac{x - x_0}{x_N - x_0}$$

and multiplying by $(F_i - F_{i-1})$ and adding F_{i-1} yields

$$\begin{aligned} & \frac{(F_i - F_{i-1})(x - x_0)}{x_N - x_0} + \frac{F_{i-1} x_N - F_{i-1} x_0}{x_N - x_0} \\ &= \left(\frac{F_i - F_{i-1}}{x_N - x_0} \right) x + \frac{F_{i-1} x_N - F_{i-1} x_0}{x_N - x_0} = c_i x + f_i \end{aligned}$$

hence

$$\bigcup_{i=1}^N w_i(G) \subset G$$

To show equality it suffices to show that the endpoints of the graph map to the endpoints of the i^{th} interval under w_i hence showing that the map is onto. We substitute x_0, x_N for x above to get

$$\begin{aligned} a_i x_N + e_i &= \left(\frac{x_i - x_{i-1}}{x_N - x_0} \right) x_N + \frac{x_N x_{i-1} - x_0 x_i}{x_N - x_0} = x_i \\ a_i x_0 + e_i &= \left(\frac{x_i - x_{i-1}}{x_N - x_0} \right) x_0 + \frac{x_N x_{i-1} - x_0 x_i}{x_N - x_0} = x_{i-1} \end{aligned}$$

so that the union of the w_i is onto. Hence

$$\bigcup_{i=1}^N w_i(G) = G$$

as desired.

VI.2.5 For this exercise we are asked to find an IFS whose attractor is a parabola passing through the points $(0, 0)$, $(1, 1)$, and $(2, 4)$. There is only one such parabola of the form $y = Ax^2 + Bx + C$, which we will find, and its equation is $f(x) = x^2$. The IFS is of the form

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} a_1 & 0 \\ c_1 & d_1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_1 \\ f_1 \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} a_2 & 0 \\ c_2 & d_2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_2 \\ f_2 \end{pmatrix} \end{aligned}$$

which we can insure passes through the interpolation points by using the equations for the various terms given in the text. To insure that this IFS has the parabola $y = f(x)$ as its attractor, we substitute a point from this graph into the equation to yield

$$\begin{aligned} w_1 \begin{pmatrix} x \\ f(x) \end{pmatrix} &= \begin{pmatrix} a_1 x + e_1 \\ c_1 x + d_1 f(x) + f_1 \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ f(x) \end{pmatrix} &= \begin{pmatrix} a_2 x + e_2 \\ c_2 x + d_2 f(x) + f_2 \end{pmatrix} \end{aligned}$$

so that in both cases,

$$(a_i x + e_i)^2 = c_i x + d_i x^2 + f_i$$

We calculate

$$\begin{aligned} a_1 &= \frac{1-0}{2-0} = \frac{1}{2} & e_1 &= \frac{0-0}{2-0} = 0 \\ a_2 &= \frac{2-1}{2-0} = \frac{1}{2} & e_2 &= \frac{2-0}{2-0} = 1 \\ c_1 &= \frac{1-0}{2-0} - d_1 \frac{4-0}{2-0} = \frac{1}{2} - 2d_1 & f_1 &= \frac{0-0}{2-0} - d_1 \frac{0-0}{2-0} = 0 \\ c_2 &= \frac{4-1}{2-0} - d_2 \frac{4-0}{2-0} = \frac{3}{2} - 2d_2 & f_2 &= \frac{2-0}{2-0} - d_2 \frac{0-0}{2-0} = 1 \end{aligned}$$

yielding

$$\begin{aligned} \left(\frac{1}{2}x\right)^2 &= \left(\frac{1}{2} - 2d_1\right)x + d_1x^2 \\ \left(\frac{1}{2}x + 1\right)^2 &= (1 - 2d_2)x + d_2x^2 + 1 \end{aligned}$$

These are solved by equation of coefficients:

$$\begin{aligned}\frac{1}{4} &= d_1 \\ \frac{1}{2} - 2d_1 &= 0 \\ d_1 &= \frac{1}{4}\end{aligned}$$

and

$$\begin{aligned}\frac{1}{4}x^2 + x + 1 &= \frac{3}{2}x - 2d_2x + d_2x^2 + 1 \\ \frac{1}{4} &= d_2 \\ 1 &= \frac{3}{2} - 2d_2\end{aligned}$$

This leaves $c_1 = 0$, and $c_2 = 1$. We have

$$\begin{aligned}w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 1 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}\end{aligned}$$

VI.2.10 By exercise II.2.11, the Euclidean and Manhattan metrics on $\mathbb{R}^2$ are equivalent, so it suffices to show that the metric in Theorem 2.1 in Chapter VI is equivalent to the Manhattan metric on $\mathbb{R}^2$. We do this by looking at the three possible cases:

If $\theta = 1$ the two metrics, d from Theorem 2.1 in Chapter VI and d_M the Manhattan metric, are identical and therefore equivalent.

If $\theta > 1$, then since $\theta|y_1 - y_2| \geq |y_1 - y_2|$ we have $d((x_1, y_1), (x_2, y_2)) \geq d_M((x_1, y_1), (x_2, y_2))$, and since $\frac{1}{\theta}|x_1 - x_2| \leq |x_1 - x_2|$ we have

$$\frac{1}{\theta}d((x_1, y_1), (x_2, y_2)) \leq d_M((x_1, y_1), (x_2, y_2)) \leq d((x_1, y_1), (x_2, y_2))$$

and the metrics are equivalent.

Finally, if $\theta < 1$ then we have, by the same argument

$$\theta d_M((x_1, y_1), (x_2, y_2)) \leq d((x_1, y_1), (x_2, y_2)) \leq d_M((x_1, y_1), (x_2, y_2))$$

so they are equivalent in this case as well.

VI.2.13 If we look at *any* IFS of the form $\{\mathbb{R}^2; w_1(x, y)\}$, then if this IFS is hyperbolic, by definition w_1 is a contraction mapping, and the attractor of the IFS is a single point. But for $N = 1$, the curve we are attempting to interpolate has two interpolation points. Hence the attempt will always fail. Another way to look at it is that by exercise VI.2.5, these interpolations can always be arranged to be quadratic, a quadratic interpolation always requires at least 3 values.

VI.2.14 By definition, we have

$$(Tf)(x) = c_n \ell_n^{-1}(x) + d_n f(\ell_n^{-1}(x)) + f_n$$

where $\ell_n(x) = a_n x + e_n$, on each subinterval $[x_{n-1}, x_n]$. We need to show that for $f \in \mathcal{F}$, Tf is an interpolation function. It is shown in the text that $Tf \in \mathcal{F}$ and hence does the proper job at the endpoints. What is necessary is to show that Tf takes each function f to a function which agrees with f at the internal interpolation points. By definition, ℓ_n^{-1} takes x_{n-1} to x_0 and x_n to x_N . Because $f \in \mathcal{F}$, f takes x_0 to F_0 , and x_N to F_N , and we have

$$\begin{aligned}(Tf)(x_{n-1}) &= c_n x_0 + d_n F_0 + f_n \\ (Tf)(x_n) &= c_n x_N + d_n F_N + f_n\end{aligned}$$

into which we substitute the definitions of c_n , d_n , and f_n , to get

$$\begin{aligned}
 (Tf)(x_{n-1}) &= \frac{1}{x_N - x_0} (F_n x_0 - F_{n-1} x_0 - d_n F_N x_0 + d_n F_0 x_0 \\
 &= quad + d_n F_0 x_N - d_n F_0 x_0 + x_N F_{n-1} - x_0 F_n - d_n x_N F_0 + d_n x_0 F_N) \\
 &= F_{n-1} \\
 (Tf)(x_n) &= \frac{1}{x_N - x_0} (F_n x_N - F_{n-1} x_N - d_n F_N x_N + d_n F_0 x_N \\
 &\quad + d_n F_N x_N - d_n F_N x_0 + x_N F_{n-1} - x_0 F_n - d_n x_N F_0 + d_n x_0 F_N) \\
 &= F_n
 \end{aligned}$$

as desired. Since Tf interpolates the data for any $f \in \mathcal{F}$ if f is a fixed point of the transformation, then by definition, the graph of f is the attractor of the prescribed IFS, and f is an interpolation function to begin with.

VI.2.15 We wish to construct an interpolation using quadratic scaling, in which the $w_n(x, y)$ take the form

$$w_n(x, y) = ((a_n x + e_n), (c_n x + d_n y + g_n y^2 + f_n))$$

As before, we need to take the data pairs (x_0, F_0) to (x_{n-1}, F_{n-1}) and (x_N, F_N) to (x_n, F_n) . Since the transformation in x , is the same, we may define a_n, e_n as before, namely

$$a_n = \frac{x_n - x_{n-1}}{x_N - x_0} \quad \text{and} \quad e_n = \frac{x_{n-1} x_N - x_n x_0}{x_N - x_0}$$

This will insure that the first coordinates of the data pairs are mapped properly. For the second coordinates, we must satisfy the equations

$$\begin{aligned}
 F_{n-1} &= c_n x_0 + d_n F_0 + g_n F_0^2 + f_n \\
 F_n &= c_n x_N + d_n F_N + g_n F_N^2 + f_n
 \end{aligned}$$

Notice that c_n and f_n must coincide with the linear case when g_n is zero. This gives

$$c_n = \frac{F_n - F_{n-1}}{x_N - x_0} - d_n \frac{F_N - F_0}{x_N - x_0} - g_n X_c$$

where we need to fill in for X_g , and

$$f_n = \frac{x_N F_{n-1} - x_0 F_n}{x_N - x_0} - d_n \frac{x_N F_0 - x_0 F_N}{x_N - x_0} - g_n X_f$$

There are no quadratic terms in the equations to be satisfied, consequently, these must cancel at x_0, x_N , giving us

$$\begin{aligned}
 g_n F_0^2 &= g_n X_c x_0 + g_n X_f \\
 g_n F_N^2 &= g_n X_c x_N + g_n X_f
 \end{aligned}$$

Dividing out g_n , and subtracting these yields

$$\begin{aligned}
 F_N^2 - F_0^2 &= (x_N - x_0) X_c \\
 X_c &= \frac{F_N^2 - F_0^2}{x_N - x_0}
 \end{aligned}$$

Putting this back into the first equation after dividing out g_n , yields

$$\begin{aligned}
 X_f &= F_0^2 - \frac{F_N^2 - F_0^2}{x_N - x_0} x_0 \\
 &= \frac{x_N F_0^2 - x_0 F_0^2 - F_N^2 x_0 + F_0^2 x_0}{x_N - x_0} \\
 &= \frac{x_N F_0^2 - x_0 F_N^2}{x_N - x_0}
 \end{aligned}$$

so that we have

$$\begin{aligned}
 c_n &= \frac{F_n - F_{n-1}}{x_N - x_0} - d_n \frac{F_N - F_0}{x_N - x_0} - g_n \frac{F_N^2 - F_0^2}{x_N - x_0} \\
 f_n &= \frac{x_N F_{n-1} - x_0 F_n}{x_N - x_0} - d_n \frac{x_N F_0 - x_0 F_N}{x_N - x_0} - g_n \frac{x_N F_0^2 - x_0 F_N^2}{x_N - x_0}
 \end{aligned}$$

to complete the coefficients for the quadratic case.

VI.2.16 Consider the integral

$$I = \int_{x_0}^{x_N} f(x) dx$$

which here is well defined, since $f(x)$ is continuous and of bounded variation. We assume that f is the fixed point of the fractal interpolation transformation, i.e. $Tf = f$, where $(\mathcal{F}, d)$ is the metric space of functions which match the endpoint data, and are continuous. Then we have

$$\begin{aligned}
 I &= \int_{x_0}^{x_N} (Tf)(x) dx = \sum_{n=1}^N \int_{x_{n-1}}^{x_n} (Tf)(x) dx \\
 &= \sum_{n=1}^N \int_{x_0}^{x_N} (c_n x + d_n f(x) + f_n) d(a_n x + e_n) = \alpha I + \beta = \frac{\beta}{1 - \alpha}
 \end{aligned}$$

where

$$\alpha = \sum_{n=1}^N a_n d_n \quad \text{and} \quad \beta = \sum_{n=1}^N a_n \int_{x_0}^{x_N} (c_n x + f_n) dx$$

We need to show that under the standard assumptions, $|\alpha| < 1$, we have

$$|\alpha| = \left| \sum_{n=1}^N a_n d_n \right| \leq \sum_{n=1}^N |a_n d_n|$$

and

$$\sum_{n=1}^N a_n = 1 \quad \text{and} \quad |a_n| < 1$$

The x_n are ordered, consequently $a_n x_0 + e_n < a_n x_N + e_n$ which implies $a_n > 0$, hence $|a_n d_n| = a_n |d_n|$, and therefore

$$|\alpha| \leq \sum_{n=1}^N a_n |d_n|$$

which implies $|\alpha| < 1$. To show that

$$\int_{x_0}^{x_N} f_0(x) dx = \beta$$

we merely note that β is independent of d_n , and the result follows from above.

VI.2.17 We want the integral

$$I_1 = \int_{x_0}^{x_N} x f(x) dx$$

under the same assumptions as in exercise VI.2.16. Once again since $Tf = f$, we have

$$I_1 = \int_{x_0}^{x_N} x(Tf)(x) dx = \sum_{n=1}^N \int_{x_{n-1}}^{x_n} x(Tf)(x) dx$$

so we transform variables.

$$(Tf)(a_n x + e_n) = c_n x + d_n f(x) + f_n$$

and we replace x with $\frac{x-e_n}{a_n}$ to write the same integral as

$$\begin{aligned} \sum_{n=1}^N \int_{x_0}^{x_N} \frac{(x - e_n)}{a_n} (c_n x + d_n f(x) + f_n) d(a_n x + e_n) \\ = \sum_{n=1}^N \int_{x_0}^{x_N} \frac{x}{a_n} (c_n x + d_n f(x) + f_n) d(a_n x + e_n) \\ - \sum_{n=1}^N \int_{x_0}^{x_N} \frac{e_n}{a_n} (c_n x + d_n f(x) + f_n) d(a_n x + e_n) \end{aligned}$$

The first integral can be written as

$$\begin{aligned} \int_{x_0}^{x_N} \frac{x}{a_n} (c_n x + f_n) d(a_n x + e_n) + \int_{x_0}^{x_N} \frac{d_n}{a_n} x f(x) d(a_n x + e_n) \\ = \int_{x_0}^{x_N} (c_n x^2 + f_n x) dx + d_n \int_{x_0}^{x_N} x f(x) dx \\ = \int_{x_0}^{x_N} (c_n x^2 + f_n x) dx + d_n I_1 \end{aligned}$$

The other half looks like

$$e_n \int_{x_0}^{x_N} (c_n x + f_n) dx + a_n e_n \int_{x_0}^{x_N} f(x) dx$$

which using the notation I_0 for the integral I in exercise VI.2.16 yields

$$e_n \int_{x_0}^{x_N} (c_n x + f_n) dx + d_n e_n I_0$$

We now let

$$\gamma = \sum_{n=1}^N \int_{x_0}^{x_N} (c_n x^2 + f_n x) dx$$

$$\delta = \sum_{n=1}^N e_n \int_{x_0}^{x_N} (c_n x + f_n) dx$$

$$\epsilon = \sum_{n=1}^N d_n$$

$$\phi = \sum_{n=1}^N d_n e_n$$

Then we have

$$\begin{aligned} I_1 &= \gamma + \epsilon I_1 - (\phi I_0 + \delta) \\ &= \frac{\gamma - (\phi I_0 + \delta)}{1 - \epsilon} \\ &= \frac{\gamma - \delta - (\phi \beta / (1 - \alpha))}{1 - \epsilon} \end{aligned}$$

where we have used the notation of VI.2.16 for the integral I_0 . We therefore must restrict the d_n such that $\epsilon \neq 1$. Once again, γ, δ are independent of d_n , and we arrive at the integral $I_1 = \gamma - \delta$ in the case that $d_n = 0$ for all n .

Answers to Chapter VI, section 3

The exercises in Section 3 are all experimental in nature.

Answers to Chapter VI, section 4

VI.4.1 Define the metric on $\mathbb{R} \times Y$ by

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + \theta d_Y(y_1, y_2)$$

where $a = \max\{a_i : i = 1, \dots, N\}$.

$$d_Y(M_n(a, y), M_n(b, y)) \leq c|a - b|$$

$$d_Y(M_n(x, a), M_n(b, y)) \leq s d_Y(a, b)$$

and $\theta = \frac{1-a}{2c}$. $c, s > 0$, and under the usual assumptions about the ordering of x_i , $1 > a > 0$. We wish to show that under these assumptions, the IFS is hyperbolic with respect to the metric d . Under each map, $(x, y) \mapsto (L_n(x), M_n(x, y))$ so that

$$\begin{aligned} d(w_n(x_1, y_1), w_n(x_2, y_2)) &= d((L_n(x_1), M_n(x_1, y_1)), (L_n(x_2), M_n(x_2, y_2))) \\ &= |L_n(x_1) - L_n(x_2)| + \theta d_Y(M_n(x_1, y_1), M_n(x_2, y_2)) \\ &= |a_n x_1 - a_n x_2| + \theta d_Y(M_n(x_1, y_1), M_n(x_2, y_2)) \\ &= |a_n| |x_1 - x_2| + \theta d_Y(M_n(x_1, y_1), M_n(x_2, y_2)) \\ &\leq a |x_1 - x_2| + \theta d_Y(M_n(x_1, y_1), M_n(x_2, y_2)) \end{aligned}$$

The second half of this expression can be decomposed using the triangle inequality to read

$$\begin{aligned} d_Y(M_n(x_1, y_1), M_n(x_2, y_2)) &\leq d_Y(M_n(x_1, y_1), M_n(x_2, y_1)) + d_Y(M_n(x_2, y_1), M_n(x_2, y_2)) \\ &\leq c|x_1 - x_2| + s d_Y(y_1, y_2) \end{aligned}$$

Putting this together with the statements above, we have

$$\begin{aligned} d(w_n(x_1, y_1), w_n(x_2, y_2)) &\leq a|x_1 - x_2| + \theta c|x_1 - x_2| + s\theta d_Y(y_1, y_2) \\ &\leq \frac{2a+1-a}{2}|x_1 - x_2| + s\theta d_Y(y_1, y_2) \\ &\leq \frac{1+a}{2}|x_1 - x_2| + s\theta d_Y(y_1, y_2) \end{aligned}$$

Since $0 < a < 1$, we have $1 + a < 2$ so let $s' = \max(\frac{1+a}{2}, s) < 1$ then

$$d(w_n(x_1, y_1), w_n(x_2, y_2)) \leq s'|x_1 - x_2| + s'\theta d_Y(y_1, y_2) = s'd((x_1, y_1), (x_2, y_2))$$

so the IFS is hyperbolic as desired.

VI.4.2 By Theorem 4.1 in Chapter VI, the attractor exists. We devise the operator $T : \mathcal{F} \rightarrow \mathcal{F}$ as in Theorem 2.2 in Chapter VI to be

$$(Tf)(x) = M_n(f(L_n^{-1}(x))) \quad x \in [x_{n-1}, x_n]$$

where $\mathcal{F}$ is the set of continuous functions $f : [x_0, x_N] \rightarrow Y$. L_n is $a_n x + e_n$ which is continuous with continuous inverse, f is assumed continuous, and M_n is again linear, so that Tf is a continuous function on each of the $[x_{n-1}, x_n]$. We must assure that T takes $\mathcal{F}$ into itself. We have

$$\begin{aligned} L_1(x_0) &= a_1 x_0 + e_1 = \frac{x_1 - x_0}{x_N - x_0} x_0 + \frac{x_N x_0 - x_0 x_1}{x_N - x_0} = x_0 \\ L_N(x_N) &= a_N x_N + e_N = \frac{x_N - x_{N-1}}{x_N - x_0} x_N + \frac{x_N x_{N-1} - x_N x_0}{x_N - x_0} = x_N \\ L_n(x_{n-1}) &= a_n x_{n-1} + e_n = \frac{x_n - x_{n-1}}{x_N - x_0} x_{n-1} + \frac{x_N x_{n-1} - x_0 x_{n-1}}{x_N - x_0} = x_{n-1} \\ L_n(x_n) &= a_n x_n + e_n = \frac{x_n - x_{n-1}}{x_N - x_0} x_n + \frac{x_N x_{n-1} - x_n x_0}{x_N - x_0} = x_n \end{aligned}$$

so the endpoints of each segment conincide with the next and last, and x_0, x_N map to themselves. Hence these points are fixed under L_n and therefore under L_n^{-1} . By definition,

$$\begin{aligned} (Tf)(x_0) &= M_n(x_0, F_0) = (x_{n-1}, F_{n-1}) \\ (Tf)(x_N) &= M_n(x_N, F_N) = (x_n, F_n) \end{aligned}$$

hence $T : \mathcal{F} \rightarrow \mathcal{F}$ as desired. We now show that T is a contraction mapping. Let $f, g \in \mathcal{F}$. Then we have

$$\begin{aligned} (Tf)(x) &= M_n(L_n^{-1}, f(L_n^{-1}(x))) \\ (Tg)(x) &= M_n(L_n^{-1}, g(L_n^{-1}(x))) \end{aligned}$$

With the given definition of a metric, and the restrictions imposed on M_n we have

$$\begin{aligned} d(Tf, Tg) &= \max_{x \in [x_{n-1}, x_n]} d_Y(M_n(L_n^{-1}(x), f(L_n^{-1}(x))), M_n(L_n^{-1}(x), g(L_n^{-1}(x)))) \\ &\leq \max_{x \in [x_{n-1}, x_n]} s d_Y(f(L_n^{-1}(x)), g(L_n^{-1}(x))) \\ &\leq d(f, g) \end{aligned}$$

as desired.

VI.4.8 We want to show we can generate polynomial graphs with affine hidden variable interpolation. Using the proof given in exercise VI.4.2, we have $(Tf)(x) = M_n(L_n^{-1}(x), f(L_n^{-1}(x)))$. Consequently, the attractor of the IFS is given by

$$f(x) = M_n(L_n^{-1}(x), f(L_n^{-1}(x)))$$

where we let $f(x) = (y(x), z(x))$. We are interested only in fitting the first component, the second component may be chosen for ease of manipulation of the equations. We use the notation of the book, namely

$$\begin{aligned} L_n(x) &= a_n x + e_n \\ M_n \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= A_n \begin{pmatrix} y \\ z \end{pmatrix} + \begin{pmatrix} c_n x + f_n \\ k_n x + g_n \end{pmatrix} \\ A_n &= \begin{pmatrix} d_n & h_n \\ l_n & m_n \end{pmatrix} \end{aligned}$$

From this we get

$$\begin{pmatrix} y(x) \\ z(x) \end{pmatrix} = A_n \begin{pmatrix} y(L_n^{-1}(x)) \\ z(L_n^{-1}(x)) \end{pmatrix} + \begin{pmatrix} c_n(L_n^{-1}(x)) + f_n \\ k_n(L_n^{-1}(x)) + g_n \end{pmatrix}$$

from which we extract the equation for $y(x)$ to get

$$y(x) = d_n y(L_n^{-1}(x)) + h_n z(L_n^{-1}(x)) + f_n + c_n L_n^{-1}(x)$$

Newton proved that it requires $N + 1$ interpolation points to generate an N^{th} degree polynomial, so that we assume that we have been given these points, and that they fit $y(x)$, and likewise assume

$$y(x) = \alpha_N x^N + \alpha_{N-1} x^{N-1} + \cdots + \alpha_0$$

Since L_n is linear, so is L_n^{-1} , so the degree of $y(L_n^{-1}(x))$ is also N , and we let (for notational convenience)

$$y(L_n^{-1}(x)) = \beta_{Nn} x^N + \beta_{N-1n} x^{N-1} + \cdots + \beta_{0n}$$

Then we have

$$\begin{aligned} \alpha_N x^N + \cdots + \alpha_0 &= d_n y(L_n^{-1}(x)) + h_n x(L_n^{-1}(x)) + f_n + c_n L_n^{-1}(x) \\ &= d_n \beta_{Nn} x^N + \cdots + d_n \beta_{0n} + h_n z(L_n^{-1}(x)) + f_n + c_n L_n^{-1}(x) \end{aligned}$$

which gives

$$(\alpha_N - d_n \beta_{Nn}) x^N + \cdots + (\alpha_0 - d_n \beta_{0n}) = h_n z(L_n^{-1}(x)) + f_n + c_n L_n^{-1}(x).$$

Now d_n , as in the non-hidden variable case, is independent of the fitting of the points (x_i, F_i) , hence on each subinterval, we may choose

$$d_n = \frac{\beta_{Nn}}{\alpha_N}$$

Then the left hand side is a polynomial of degree $N - 1$ so we may choose h_n , and let

$$z(x) = \gamma_{N-1} x^{N-1} + \cdots + \gamma_0$$

in such a way so that if

$$\delta_{N-1n} x^{N-1} + \cdots + \delta_{0n} = z(L_n^{-1}(x))$$

then we have

$$h_n \delta_{in} = \alpha_i - d_n \beta_{in}$$

for $i \geq 2$ and

$$h_n \delta_{1n} x + h_n \delta_{0n} + c_n L_n^{-1}(x) + f_n = (\alpha_1 - d_n \beta_{1n}) x + (\alpha_0 - d_n \beta_{0n})$$

by equation of coefficients. Thus we may generate a hidden variable fractal interpolation function for a polynomial of degree N .

Answers to Chapter VI, section 5

VI.5.1 From Theorem 5.1 in Chapter VI, using three maps, we may choose

$$\begin{aligned}w_1 \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & a_1 & b_1 \\ 0 & c_1 & d_1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} 0 \\ e_1 \\ f_1 \end{pmatrix} \\w_2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & a_2 & b_2 \\ 0 & c_2 & d_2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ e_2 \\ f_2 \end{pmatrix} \\w_3 \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & a_3 & b_3 \\ 0 & c_3 & d_3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} \frac{2}{3} \\ e_3 \\ f_3 \end{pmatrix}\end{aligned}$$

For the Sierpinski triangle, the appropriate values are given by $a_i = d_i = \frac{1}{2}$, $b_i = c_i = 0$, and $e_1 = f_1 = 0$, $e_2 = \frac{1}{2}$, $f_2 = 0$ and $e_3 = 0$, $f_3 = \frac{1}{2}$.

Answers to Chapter VII, section 1

VII.1.1 Let $\{\mathbb{R}^2, f\}$ be given by

$$f(x, y) = \begin{cases} (2x, 2y - 1) & \text{if } y > 1/2, \\ (2x - 1, 2y) & \text{if } x > 1/2, \\ (2x, 2y) & \text{otherwise.} \end{cases}$$

If $(x, y) > 0$ and $(x, y) \notin S$, then eventually $y > 1/2$ since y is doubled if $x > 1/2$ or if neither $x, y > 1/2$. Once this happens we have

$$f^n(x, y) = (2^n x, 2^n y - 2^{n-1})$$

Given $y > 1/2$, we have

$$\lim_{n \rightarrow \infty} 2^n y - 2^{n-1} = \lim_{n \rightarrow \infty} (2y - 1)2^{n-1} = \infty$$

and $x \rightarrow \infty$, as well. Hence these points escape to infinity. If $y < 0$, $x > 0$, then eventually $2^n x > 1/2$, if $x < 1/2$. Once this happens we have

$$\lim_{n \rightarrow \infty} 2^n x - 2^{n-1} = \lim_{n \rightarrow \infty} (2x - 1)2^{n-1} = \infty$$

for any point whose x coordinate is larger than $1/2$. The y coordinate remains negative, and $2^n y \rightarrow -\infty$. Hence these points also escape, in this case in the direction of $(\infty, -\infty)$. For points $y < 0$, $x < 0$, each coordinate remains negative indefinitely and these points diverge as $2^n x \rightarrow -\infty$ and $2^n y \rightarrow -\infty$, so these points escape to $(-\infty, -\infty)$.

VII.1.3 The condition $x^2 + y^2 > R$ determines the color based on the iteration number. This is passage past a circle of radius $\sqrt{R}$. Since if $f(x, y) = (2x, 2y)$ we have $f^n(x, y) = (2^n x, 2^n y)$, and

$$(2^n x)^2 + (2^n y)^2 = 2^{2n}(x^2 + y^2) > R \Rightarrow x^2 + y^2 > \frac{R}{2^{2n}}$$

we have concentric circles of radius $\frac{\sqrt{R}}{2^n}$.

VII.1.6 These two functions,

$$w_1(z) = r e^{i\theta} z - 1 \quad w_2(z) = r e^{i\theta} z + 1$$

take a disk of radius R , to a disk of radius rR , rotate it by θ , and move its center to either ± 1 . Consequently, if we can find a disk which is mapped into itself by both maps, we know that the attractor of the IFS is interior to the disk (see exercise VII.1.8). We thus look for R as a function of r such that this is the case, and restrict it such that its two images do not intersect. This will force the attractor of the IFS to be totally disconnected.

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We make one further simplification: If the rotation were 0, we could examine the transformation on the real line, since these transformations take circles to circles, this will correctly predict the values of R . This reduces the problem to looking at $w_i(x) = -rx \pm 1$. We look for a cross section of the ball $[a, b]$ such that the images

$$w_1([a, b]) = [-rb - 1, -ra - 1] \quad w_2([a, b]) = [-rb + 1, -ra + 1]$$

both lie within $[a, b]$. These two intervals are of identical length and orientation, differing only by a shift of two along the real line. Consequently, our conditions are

$$a \leq -rb - 1 \quad b \geq -ra + 1$$

Solving at equality yields

$$\begin{aligned} a &= -rb - 1 \\ b &= -ra + 1 \\ a &= r^2a - r - 1 \\ (1 - r^2)a &= -r - 1 \\ a &= -\frac{1+r}{1-r^2} = -\frac{1}{1-r} \end{aligned}$$

Symmetry now dictates that $b = \frac{1}{1-r}$, so that a disk of radius $R = 1/(1-r)$ maps inside itself under each map of the original IFS. We now want the condition on r such that these images of the disk $B_R(\mathcal{O})$ do not intersect. This happens when a disk of radius rR , centered about 1 does not intersect the y -axis, namely that

$$\begin{aligned} r \cdot \frac{1}{1-r} &< 1 \\ r &< 1-r \\ 2r &< 1 \\ r &< 1/2 \end{aligned}$$

Consequently, for the original maps, as long as $r < 1/2$, the maps

$$w_1(A) \cap w_2(A) \subset w_1(B_R(\mathcal{O})) \cap w_2(B_R(\mathcal{O})) = \emptyset$$

and the IFS is totally disconnected. We must now define $f(z)$ such that on the attractor, we do this with three maps (letting $z = z_1 + iz_2$),

$$f(z) = \begin{cases} w_1 & \text{if } z_1 \leq -\frac{r}{1-r} \\ w_2 & \text{if } z_1 \geq \frac{r}{1-r} \\ re^{i\theta}z + \lambda & \lambda = \frac{z_1(1-r)}{r} \end{cases} \quad \text{otherwise}$$

VII.1.7 This is essentially the general case of the preceding problem. Let $\{A; f\}$ be the associated shift dynamical system of a totally disconnected hyperbolic IFS on $\mathbb{R}^2$, where A is the attractor of the IFS. Then we must have

$$f(a) = w_i^{-1}(a) \quad \text{for } a \in w_i(A).$$

Since the IFS is totally disconnected, we have

$$w_i(A) \cap w_j(A) = \emptyset \quad \text{if } i \neq j$$

Furthermore, since the IFS is hyperbolic, $w_i(A)$ is compact for each i . Consequently, in $\mathbb{R}^2$ this means $d(w_i(A), w_j(A)) > 0$ if $i \neq j$. Then we may extend $\{A; f\}$ using any function $g(x)$ which smoothly connects the maps w_i^{-1} in the intervening distance. For example, since there are finitely many maps, we may define

$$\epsilon = \min_{i \neq j} d(w_i(A), w_j(A)) > 0.$$

Now construct functions which are equal to w_i^{-1} on $w_i(A)$, and decrease to zero at a distance less than $\epsilon/2$ from these sets, by taking functions

$$\tilde{w}_i(x) = \begin{cases} (1 - \frac{d(x, w_i(A))}{k\epsilon})w_i^{-1}(x) & \text{if } d(x, w_i(A)) \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

with $k \geq 2$. Finally, sum these together to get

$$g_k(x) = \sum_{i=1}^N \tilde{w}_i(x)$$

Because the distance ϵ was chosen to be less than the distance between images of the attractor under w_i , these functions are continuous, agree with f on the attractor, and since k can be taken at any value greater than 2, there are infinitely many of them.

VII.1.9 We get the iteration formula by generating the real and imaginary parts of $f(z) = z^3 - \lambda$. Letting $z = x + iy$, we have

$$\begin{aligned} z^3 &= (x + iy)^3 = x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3 \\ &= x^3 + i3x^2y - 3xy^2 - iy^3 \\ &= (x^3 - 3xy^2) + i(3x^2y - y^3) \end{aligned}$$

so that the new formulas, letting $\lambda = \lambda_1 + i\lambda_2$ are

$$\text{newx} = x^3 - 3xy^2 - \lambda_1$$

$$\text{newy} = 3x^2y - y^3 - \lambda_2$$

The choice of R can be derived as follows: Writing $z = re^{i\theta}$, we have $z^3 = r^3e^{i3\theta}$, which has a distance from the origin of r^3 compared to the distance r of z . Subtracting λ will have maximal effect on this distance when z^3 and λ lie on the same ray from the origin, namely when $\lambda = \rho e^{i\phi}$ and $\phi = 3\theta$. Consequently the problem reduces to this maximal case, namely that

$$r^3 - |\lambda| \geq r$$

or

$$r^3 - r - |\lambda| \geq 0$$

The minimum value for $|\lambda|$ is zero, and the inequality achieves equality for this λ value when $r = 1 + |\lambda|$. Choosing this as a candidate value for R , we find that

$$\begin{aligned} (1 + |\lambda|)^3 - (1 + |\lambda|) - |\lambda| &= |\lambda|^3 + 3|\lambda|^2 + 3|\lambda| + 1 - 1 - 2|\lambda| \\ &= |\lambda|^3 + 3|\lambda|^2 + |\lambda| \geq 0 \end{aligned}$$

so that for any $\lambda \in \mathbb{C}$, choosing $R > |\lambda| + 1$ will work in the escape time algorithm.

Answers to Chapter VII, section 2

VII.2.1 In this case $X = \hat{\mathbb{C}}$, we must show the presence of sets such that $W(A) = A$, where $W(A) = w_1(A) \cup w_2(A)$. Choose $A = \{\mathcal{O}\}$, then $w_1(A) = w_2(A) = \mathcal{O}$. Likewise, choose $A = \{\infty\}$, then $w_1(A) = w_2(A) = \{\infty\}$. Consequently, the unrestricted IFS in this case does not contain a unique invariant set. In point of fact, neither of these sets above are attractors of the IFS. The system is not contractive around these points, which are repellers. The point $z = 1$ is also a fixed point of the system, it is a repeller, and is part of $J(f)$.

VII.2.2 Let $\lambda \in (3/4, 5/4)$ and $f(z) = z^2 - \lambda$. We want to look at solutions to

$$z^2 + z + (1 - \lambda) = 0.$$

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Using the quadratic equation we obtain

$$z = \frac{-1 \pm \sqrt{1-4+4\lambda}}{2} = \frac{-1 \pm \sqrt{4\lambda-3}}{2}$$

We let

$$z_1 = \frac{-1 + \sqrt{4\lambda-3}}{2}$$

$$z_2 = \frac{-1 - \sqrt{4\lambda-3}}{2}$$

The easiest way to examine cycles of period 2 here is to check to see whether $f(z_1) = z_2$ and $f(z_2) = z_1$ (different approaches are used for different problems, this avoids quadratic equations here). We have

$$z_1^2 = \frac{1}{4} - \frac{\sqrt{4\lambda-3}}{2} + \frac{4\lambda-3}{4} = (\lambda - \frac{1}{2}) - \frac{\sqrt{4\lambda-3}}{2}$$

$$z_2^2 = \frac{1}{4} + \frac{\sqrt{4\lambda-3}}{2} + \frac{4\lambda-3}{4} = (\lambda - \frac{1}{2}) + \frac{\sqrt{4\lambda-3}}{2}$$

Subtracting λ from each of these yields the desired values $f(z_1) = z_2$, and $f(z_2) = z_1$, hence this is a cycle of period 2.

To check whether the cycle is attractive, we have

$$f^{2'}(z) = 4z^3 - 4\lambda z$$

We can simplify this for each z_i by making use of the period 2 properties:

$$z_1^3 = (z_2 + \lambda)z_1 = z_1z_2 + \lambda z_1$$

$$z_2^3 = (z_1 + \lambda)z_2 = z_1z_2 + \lambda z_2$$

giving us

$$f^{2'}(z_1) = f^{2'}(z_2) = 4z_1z_2$$

and

$$4z_1z_2 = 4 \frac{1 - (4\lambda - 3)}{4} = \frac{4 - 4\lambda}{4} = 1 - \lambda$$

For $\lambda \in (3/4, 5/4)$ we have $(1 - \lambda) \in (-1/4, 1/4)$ so that

$$|f^{2'}(z_i)| \leq \frac{1}{4} < 1$$

Hence this is an attractive cycle of period 2.

VII.2.3 Let λ be specified and suppose $a \cdot i \in J_\lambda$. Then $-a^2 \in J_\lambda$ and $a^4 \in J_\lambda$ are real line points. The set J_λ is connected, therefore there is an extreme point of this form which maps to the left most extremity of J_λ , on both sides of the real axis. The pair of IFS cut J_λ at the negative extremity and map this image to the left and right side, depending on the definition of the square root in the two IFS equations.

VII.2.5 For the dynamical system $\{\hat{C}; f(z) = z^2 - \lambda : \lambda > 2\}$, we wish to show that the orbit of the critical point $\{\mathcal{O}\}$ converges to the Point at Infinity. We know that the basin of attraction for ∞ contains at least the points for which $r > 1/2 + \sqrt{1/4 + \lambda}$ so that if $\lambda > 2$, we have $r > 1/2 + 3/2 = 2$. $f^2(\mathcal{O}) = f(-\lambda)$ which is 2 when $\lambda = 2$ and increases for increasing λ . Hence $\lim_{n \rightarrow \infty} f^n(\mathcal{O}) = \infty$. By Theorem 2.1 in Chapter VII, the IFS $\{X = \hat{C} \setminus B(\infty, \epsilon); +\sqrt{z + \lambda}, -\sqrt{z + \lambda}\}$ possesses a unique attractor.

Answers to Chapter VII, section 3

VII.3.1 The Newton's transformation associated with $F(z)$ is given by

$$f(z) = z - \frac{F(z)}{F'(z)} = z - \frac{z^4 - 1}{4z^3}$$

To see that 1 is indeed fixed, we compute

$$f(1) = 1 - 0 = 1$$

Taking the derivative with respect to z evaluated at 1, we have

$$\begin{aligned} f'(z) &= 1 - \frac{(4z^3)^2 - 12z^2(z^4 - 1)}{(4z^3)^2} = 1 - \frac{16z^6 - 12z^6 + 12z^2}{16z^6} \\ &= \frac{12z^6 - 12z^2}{16z^6} = \frac{3}{4} - \frac{3}{4z^4} \end{aligned}$$

evaluated at 1 we have $f'(z) = 3/4 - 3/4 = 0$ so that 1 is an attractive (sometimes called super-attractive because $f'(z) = 0$) fixed point of the Newton's transformation.

VII.3.2 The Newton's transformation associated with $F(z) = z^2 + 1$ is

$$\begin{aligned} f(z) &= z - \frac{z^2 + 1}{2z} = z - \frac{1}{2}\left(z + \frac{1}{z}\right) \\ &= \frac{1}{2}z - \frac{1}{2z} = 1/2\left(z - \frac{1}{z}\right) \end{aligned}$$

so that the corresponding IFS is got by taking inverses of $f(z)$ to yield

$$\begin{aligned} w &= 1/2\left(z - \frac{1}{z}\right) \\ 2wz &= z^2 - 1 \\ z^2 - 2wz - 1 &= 0 \\ z &= \frac{2w \pm \sqrt{4w^2 + 4}}{2} = w \pm \sqrt{w^2 + 1} \end{aligned}$$

Consequently, the appropriate IFS is

$$w_1(z) = z + \sqrt{z^2 + 1} \quad \text{and} \quad w_2(z) = z - \sqrt{z^2 + 1}$$

For any $z \in \mathbb{R}$, $z^2 + 1 > 0$, hence $w_i(z) \in \mathbb{R}$. For $z = \infty$, we have $w_1(z) \rightarrow \infty$, and $w_2(z) \rightarrow 0$, consequently $\mathbb{R} \cup \{\infty\} \mapsto \mathbb{R} \cup \{\infty\}$, and the map is onto, by virtue of the behavior of $w_2(z)$ for large z . Then if $A = \mathbb{R} \cup \{\infty\}$, $W(A) = A$. We must make sure we can exclude the fixed points of the transformation. These are given by

$$\frac{1}{2}\left(z - \frac{1}{z}\right) = z$$

or $z_f = \pm i$. The derivative of $f(z)$ at these points is $1/2(1 + 1/(z^2)) = 0$ so these are attractive. Hence removal of a small ϵ -ball around $\pm i$ results in a space $X \subset \hat{\mathbb{C}}$ on which A is a unique invariant set. Therefore A is the attractor of the IFS.

VII.3.5 The Newton's transformation associated with $F(z) = e^z - 1$ is given by

$$f(z) = z - \frac{e^z - 1}{e^z} = (z - 1) + e^{-z}$$

It has two fixed points in $\hat{\mathbb{C}}$, for we have

$$z = (z - 1) + e^{-z} \rightarrow e^{-z} = 1 \rightarrow z = 0$$

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indicating that 0 is a fixed point. We also have

$$\lim_{z \rightarrow \infty} (z - 1) + e^{-z} = \infty$$

so that ∞ is also fixed. The derivative of $f(z)$ is given by

$$f'(z) = 1 - e^{-z}$$

which is zero at $z = 0$, and approaches 1 at infinity. Hence the attractive fixed point is 0, ∞ is a neutral fixed point.

VII.3.7 We wish to find a Möbius transformation $h(z)$ which establishes a conjugacy between $f(z)$ and $g(z)$. This means finding $h(z)$ such that

$$f \circ h(z) = h \circ g(z)$$

where

$$f(z) = \frac{1}{2}\left(z - \frac{1}{z}\right) \quad g(z) = z^2 \quad h(z) = \frac{az+b}{cz+d}$$

We mention at the outset that such a transformation is indeterminate with respect to scaling factors, so there will be choices among a, b, c , and d . Writing out the two terms of the conjugacy we have

$$h \circ g(z) = \frac{az^2 + b}{cz^2 + d}$$

and

$$\begin{aligned} f \circ h(z) &= \frac{1}{2} \left(\frac{az+b}{cz+d} - \frac{cz+d}{az+b} \right) = \frac{1}{2} \left(\frac{(az+b)^2 - (cz+d)^2}{(cz+d)(az+b)} \right) \\ &= \frac{1}{2} \left(\frac{a^2z^2 + 2abz + b^2 - c^2z^2 - 2cd - d^2}{acz^2 + bcz + adz + bd} \right) \\ &= \frac{1}{2} \left(\frac{(a^2 - c^2)z^2 + 2(ab - cd)z + (b^2 - d^2)}{acz^2 + (ad + bc)z + bd} \right) \end{aligned}$$

so that

$$\frac{2az^2 + 2b}{cz + d} = \frac{(a^2 - c^2)z^2 + 2(ab - cd)z + (b^2 - d^2)}{acz^2 + (ad + bc)z + bd}$$

We may eliminate the linear terms in z by setting

$$ad + bc = 0 \quad ab - cd = 0$$

which give us

$$\begin{aligned} a &= -\frac{bc}{d} \Rightarrow -\frac{b^2c}{d} - cd = 0 \Rightarrow d^2 = -b^2 \\ d &= -\frac{bc}{a} \Rightarrow ab + \frac{c^2b}{a} = 0 \Rightarrow a^2 = -c^2 \end{aligned}$$

With these equalities assumed, we have

$$\frac{2az^2 + 2b}{cz^2 + d} = \frac{(a^2 - c^2)z^2 + (b^2 + d^2)}{acz^2 + bd}$$

Since, as mentioned, we have choices of scale, we set $a = 1$. This gives us

$$\frac{2z^2 + 2}{cz^2 + d} = \frac{(1 - c^2)z^2 + (b^2 + d^2)}{cz^2 + bd}$$

and comparing denominators leads to $b = 1$. Then we have

$$\begin{aligned} 2 &= 1 - c^2 & \text{and} & & 2 &= 1 - d^2 \\ c &= \pm i & \text{and} & & d &= \pm i \end{aligned}$$

and we again have a choice (this time the sign of the scale factor). We choose $c = i$. Then $ab - cd = 0$ or $1 - id = 0$, and $d = -i$. Consequently we have

$$h(z) = \frac{z+1}{iz-i}$$

Checking, we have

$$\begin{aligned} f(h(z)) &= \frac{1}{2} \left(\frac{z+1}{iz-i} - \frac{iz-i}{z+1} \right) = \frac{1}{2} \frac{z^2 + 2z + 1 - (-z^2 + 2z - 1)}{iz^2 - iz + iz - i} \\ &= \frac{1}{2} \frac{2z^2 + 2}{iz^2 - i} = \frac{z^2 + 1}{iz^2 - i} = h(g(z)) \end{aligned}$$

as desired.

Answers to Chapter VII, section 4

VII.4.1 We analyze this map by first noting that this map is equivalent to the map

$$f(z) = \begin{cases} z^2 - 1 & \text{for } z_1 > 0 \\ z^2 - 1 + \lambda z_1 & \text{for } z_1 \leq 0 \end{cases}$$

where z_1 is the real part of $z = z_1 + iz_2$. Consequently, for $\lambda = 0$, and for any open set lying entirely in the right half plane, the mapping is open. For $\lambda \in [-1, 0]$, a look at a small ϵ -ball about the origin shows why the map is still continuous, namely, the action of the term λx is to shift the points within the ball over towards the right. This produces a closed oval, with points to the interior mapping to the interior and the boundary mapping to the boundary. Consequently an interior point is still surrounded by an open ball of the image of interior points, and the map remains open. Specifically, given (x, y) , if we have a point (x_1, y) to the left of (x, y) , then since near the origin, $x^2 \ll x$, the λx term dominates, and the disk is reversed, $f(x_1, y)$ is to the right of $f(x, y)$, and points with the same x value map to the same x value, creating an oval around the center (x, y) . Then an open set maps to an open set.

The mapping is not open for $\lambda = 1$ however. To see this, choose first $\epsilon_0 < 1/2$, so that for points in an ϵ_0 -ball about the origin, $2x^2 < x$. Then for points on the boundary of the left half ball $x^2 + y^2 = \epsilon_0^2$ and the intersection with the y -axis, the portion on the axis maps to an interval on the real axis $[-\epsilon_0^2 - 1, -1]$. The rest of the boundary meets this interval at $-\epsilon_0^2 - 1$, but lies entirely to the left of it:

$$\begin{aligned} x^2 + y^2 &= \epsilon_0^2 \\ f(x, y) &= (x^2 - y^2 - 1 + x, 2xy) \\ &= x^2 - (\epsilon_0^2 - x^2) - 1 + x, 2xy \\ &= (-\epsilon_0^2 - 1) + 2x^2 + x, 2xy \end{aligned}$$

Since we have chosen ϵ_0 such that the x term dominates the $2x^2$ term this indeed lies to the left. If we look at this figure from $\epsilon = \epsilon_0$ as $\epsilon \rightarrow 0$, then the envelope of the images of the boundary form the boundary of the image of the ball $B_{\epsilon_0}(\mathcal{O})$. In particular, there are points interior to this ball which map to the boundary of the image. For any such point, a smaller ball around that point folds interior to this envelope, exposing it, hence there is no open set around such a point which maps to an open set, and the map is not open.

Finally, by choosing a ball around the origin sufficiently large, for any fixed λ , the x^2 and y^2 terms will dominate the λx term so that $f(B_R(\mathcal{O})) \supset B_R(\mathcal{O})$. In fact, as $(x, y) \rightarrow \infty$, the map approaches the analytic map $f(z) = z^2 - 1$.

Answers to Chapter VIII, section 1

VIII.1.3 We must first show that we satisfy the Lipschitz continuity requirement of Theorem 11.1 in Chapter III, namely for each w_i there is a k_i , independent of p, x , such that

$$d(w_{ip}(x), w_{iq}(x)) \leq k_i \cdot d_P(p, q)$$

For $w_1(x) = \lambda_1 x$, we have

$$d(\lambda_{1p}x, \lambda_{1q}x) = |\lambda_{1p}x - \lambda_{1q}x| = |x||\lambda_{1p} - \lambda_{1q}| = |x|d_P(p, q)$$

which is not independent of x . Similarly, for $w_2(x) = \lambda_2 x + 1 - \lambda_2$, we have

$$d(\lambda_{2p}x + 1 - \lambda_{2p}, \lambda_{2q}x + 1 - \lambda_{2q}) = |\lambda_{2p}x + 1 - \lambda_{2p} - \lambda_{2q}x - 1 + \lambda_{2q}| = |x - 1|d_P(p, q)$$

also not independent of x . We must restrict the domain as in exercise VIII.1.2. We have $-.9 \leq \lambda_1, \lambda_2 \leq .9$, and we look for an interval $[a, b]$ such that $W([a, b]) \subset [a, b]$ for all $p \in P$.

For $\lambda_1, \lambda_2 > 0$, $[0, 1]$ contains all the attractors. To see this, we have

$$w_1([0, 1]) = [0, \lambda_1] \quad w_2([0, 1]) = [\lambda_2, 1] \quad w_3([0, 1]) = [1/2, 1]$$

When $\lambda_1 < 0$, we have $w_1([0, 1]) = [\lambda_1, 0] \not\subset [0, 1]$, and likewise $w_2([0, 1]) = [1, 1 - \lambda_2] \not\subset [0, 1]$. One pushes the value for the necessary lower bound for the interval down, the other pushes the upper bound up. These values reach their minimum and maximum when $\lambda_1 = \lambda_2 = -.9$. Consequently, we will have found the proper interval $[a, b]$ when

$$W_{(-.9, -.9)}([a, b]) \subset [a, b].$$

Writing this out explicitly, we have

$$\begin{aligned} w_1([a, b]) &= [-.9b, -.9a] \\ w_2([a, b]) &= [-.9b + 1.9, -.9a + 1.9] \\ w_3([a, b]) &= [.5a + .5, .5b + .5] \end{aligned}$$

which means that a, b must satisfy

$$\begin{aligned} a &\leq \min(-.9b, -.9b + 1.9, .5a + .5) \\ b &\geq \max(-.9a, -.9a + 1.9, .5b + .5) \end{aligned}$$

Now in order to include the interval $[0, 1]$ needed by the positive values for λ_1, λ_2 , $a \leq 0$, and $b \geq 1$, which means that

$$\begin{aligned} \min(-.9b, -.9b + 1.9, .5a + .5) &= -.9b \\ \max(-.9a, -.9a + 1.9, .5b + .5) &= -.9a + 1.9 \end{aligned}$$

Setting the inequalities above to equality to solve we have

$$a = -.9b \quad b = -.9a + 1.9 \quad b = .18b + 1.9$$

which yields

$$b = \frac{1.9}{.82} \approx 2.3 \quad a \approx -2.1$$

so that, for convenience, the interval $[-3, 3]$ contains all the attractors, and we may prove the continuity on the IFS

$$\{[-3, 3]; w_1, w_2, w_3\}$$

to get it for all of $\mathbb{R}$. Looking back to our original x dependencies we now see that by substituting the endpoints of this interval for x in the distance expressions, we have the following Lipschitz conditions:

$$\begin{aligned} d(w_{1p}(x), w_{1q}(x)) &\leq 3 \cdot d(p, q) \\ d(w_{2p}(x), w_{2q}(x)) &\leq 4 \cdot d(p, q) \end{aligned}$$

Consequently, the attractor varies continuously with $\lambda \in P$.

Answers to Chapter VIII, section 2

VIII.2.2 We first prove the hint. Let N be the set of all points $B \in \mathcal{H}(X)$ such that B is disconnected. Suppose $B \in N$. Then by definition, there are sets $B_1, B_2 \subset B$, such that

$$B_1 \cup B_2 = B; B_1 \cap B_2 = \emptyset.$$

Furthermore, since these sets are closed in B , they are compact. Then if $\delta = h(B_1, B_2)$, $\delta > 0$. Suppose $B' \in \mathcal{H}(X)$ such that $h(B, B') < \delta/3$. Then for any $x \in B'$, we have $d(x, B) < \delta/3$, and for any $y \in B$, we have $d(y, B') < \delta/3$. Let

$$B'_1 = \{b \in B' : d(b, B_1) < \delta/3\}$$

$$B'_2 = \{b \in B' : d(b, B_2) < \delta/3\}$$

Since each point in B' must be close to a point in B , we must have

$$B'_1 \cup B'_2 = B'.$$

Now suppose that $x \in B'_1 \cap B'_2 \neq \emptyset$. Then

$$d(x, B_1) + d(x, B_2) \leq \frac{2\delta}{3} < \delta$$

and by the triangle inequality, we would have, for some $b_1 \in B_1$ and $b_2 \in B_2$,

$$d(b_1, b_2) \leq d(x, b_1) + d(x, b_2) < \delta$$

which is a contradiction. Hence

$$B'_1 \cap B'_2 = \emptyset$$

and B' is also disconnected. But B' was *any* set within $\epsilon = \delta/3$ of B , so there is an open ball $B_\epsilon(B) \subset N$, and N is open in $\mathcal{H}(X)$.

Now let p be a parameter value such that $A(p) \in N$, that is, $A(p)$ is disconnected. Then there is an $\epsilon > 0$ such that $B_\epsilon(A(p)) \subset N$. This ball is open, and by the assumptions of the problem, the map $A : P \rightarrow \mathcal{H}(X)$ is continuous, so $A^{-1}(B_\epsilon(A(p)))$ is an open set in P containing p such that every element corresponds to a disconnected attractor. Hence the set of points in P for which $A(p)$ is disconnected is open in P . The complement is $\mathcal{M}$, which must therefore be closed.

VIII.2.4 For any value of λ in the parameter space, the attractor A can be written as the set of points of the form

$$z = \sum_{k=0}^{\infty} \pm \lambda^k$$

Since if $z \in A$, then the point

$$-z = \sum_{k=0}^{\infty} \mp \lambda^k$$

is also in A , we have

$$A = -A = \{z : -z \in A\}$$

(in other words the attractor of an element of this family of IFS is itself radially symmetric). Defining

$$\lambda A = \{\lambda z : z \in A\}$$

to make the notation simple, we have

$$A = (\lambda A + 1) \cup (\lambda A - 1).$$

And

$$-\lambda A + 1 = \lambda(-A) + 1 = \lambda A + 1$$

$$-\lambda A - 1 = \lambda(-A) - 1 = \lambda A - 1$$

in other words, $A(\lambda) = A(-\lambda)$. It then follows that if the attractor of λ is connected, so is the attractor of $-\lambda$ since they are identical, so $\mathcal{M}$ is symmetric about the origin.

VIII.2.5 We have the IFS

$$\{\mathbb{C}; \lambda z + 1, \lambda z - 1\}$$

where $\lambda = (0, 1/\sqrt{2})$. We may rewrite this as the IFS

$$\{\mathbb{R}^2; w_1(x, y), w_2(x, y)\}$$

where

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 0 & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1/\sqrt{2} \begin{pmatrix} -y \\ x \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 0 & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1/\sqrt{2} \begin{pmatrix} -y \\ x \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{aligned}$$

Suppose we look at the action of these two maps on a rectangle centered at the origin. Each map flips the rectangle along the corners, reduces each side by $1/\sqrt{2}$, and then translates it either one unit to the left or to the right. We may solve for the position of a corner of such a rectangle, symmetric about the origin, such that it went to half the original rectangle. To do this, we must have (in absolute value)

$$\begin{aligned} x &= 1/\sqrt{2}(y) + 1 \\ y &= 1/\sqrt{2}(x) \end{aligned}$$

which has the solution $x = 2$, and $y = \sqrt{2}$.

Consequently, if we have the rectangle with corners $(\pm 2, \pm \sqrt{2})$, Then each map takes this rectangle to one of the horizontal halves of the original rectangle, just-touching along $x = 0$. The attractor is thus the rectangle, which tiles the plane.

VIII.2.7 Suppose that $w_1(A) \cap w_2(A) = \{z_0\}$, a single point. We wish to argue that if I is a path which intersects itself, then I is a point. Certainly such a path which is not trivial must be contained in either $w_1(A)$ or $w_2(A)$ or, since it must pass through z_0 this is a point of intersection. Then we may argue the same about the points $w_1(z_0)$ and $w_2(z_0)$, so these are points of self intersection. By induction, such a non-trivial non-self-intersecting path can not exist.

VIII.2.8 Let $e = e_1 e_2 \cdots e_n \cdots$ be a point in Σ a code space on the symbols $\{-1, 1\}$. For $\lambda \in \mathbb{C}$, define

$$f(\lambda) = e_1 + e_2 \lambda + e_3 \lambda^2 + \cdots$$

We want the radius of convergence of $f(\lambda)$. $f(\lambda)$ is bounded above by $1 + \lambda + \lambda^2 + \cdots = \sum \lambda^n$ (corresponding to $e = 1\bar{1}$). It is bounded below by $-1 - \lambda - \lambda^2 \cdots = -\sum \lambda^n$, corresponding to $e = -1\bar{1}$. These two series have a radius of convergence of $|\lambda| < 1$ and diverge at the endpoints. Consequently $f(\lambda)$ has a radius of convergence of 1. We show that the attractor for each λ is the set of points

$$\pm 1 \pm \lambda \pm \lambda^2 \pm \cdots$$

and that the map $f(\lambda)(e) = \phi(e)$. The attractor is the limit of the sequence in $\mathcal{H}(\mathbb{C})$ given by $\{W^n(\{0\})\}$. This sequence yields

$$0 \mapsto \pm 1 \mapsto \pm \lambda \pm 1 \mapsto \pm \lambda^2 \pm \lambda \pm 1 \mapsto \cdots$$

By induction, $A(\lambda)$ is the set of points

$$\pm 1 \pm \lambda \pm \lambda^2 \cdots$$

which is the image of Σ under both $f(\lambda)$ and $\phi : \Sigma \rightarrow A(\lambda)$.

Answers to Chapter VIII, section 3

VIII.3.3 The attractive orbit here is not a cycle, but rather aperiodic. There are other bubbles with periods equal to powers of 2 multiplied by the various odd numbers out past this point however. Try experimenting with the numbers around $\lambda = 1.75$ for instance.

Answers to Chapter VIII, section 4

All exercises in this section are programming experiments.

Answers to Chapter IX, section 1

IX.1.3 $p_1 \cdot p_3 = 0.03$ since the probabilities are independent.

IX.1.4 The attractor is a Sierpinski triangle with vertices at $0, i, 1$. The ball B_4 is located at $2 + i$ and has radius $\sqrt{2}$ and therefore intersects the attractor only at the point 1. The probability of getting only one string in code space is 0.

IX.1.5 $p_1 = .275$, $p_2 = .125$, and $p_3 = .5$. We want to know about the ball B_1 . B_1 includes only $w_1(A)$, hence one expects a probability of .275.

IX.1.6 There is a set of random strings (of measure 0 in code space) for which this number will not occur, e.g. B_2 in the above problem with the string $1\bar{1}$ has incidence 0.

Answers to Chapter IX, section 2

IX.2.2 X is the set of leaves, which is finite. $\mathcal{F} = 2^X$ the collection of all subsets of X . $\mathcal{G}$ is the collection of sets consisting of one leaf. Any set in $\mathcal{F}$ contains a finite number of leaves, therefore it is the finite union of sets in $\mathcal{G}$ so by definition, $\mathcal{G}$ generates $\mathcal{F}$.

IX.2.3 In order to prove that S is not in the field $\mathcal{F}$ we must show that it can not be written as a finite union of sets in $\mathcal{F}$. Suppose that $S = \bigcup_{n=1}^N A_n$ for some collection of $A_n \in \mathcal{F}$ where each A_n is the union of finitely many subintervals. Then at least one of the A_n must contain the disjoint union of an infinite number of intervals. But this contradicts the hypothesis about A_n . Hence $S \notin \mathcal{F}$.

IX.2.4 Let $X = [0, 1] \times [0, 1] \subset \mathbb{R}^2$, and $\mathcal{G}$ be the set of rectangles of the form $[p/q, r/s] \times [t/u, v/w]$ that is subrectangles with rational corners. Let $\mathcal{F}$ denote the field generated by $\mathcal{G}$. Let $S \in \mathcal{F}$. We need to show that S has rational area. i) $\bigcup_{n=1}^N A_n$ has rational area, where the A_n are subrectangles since the area of the union is given by $\sum_{n=1}^N \text{area}(A_n) - \sum_{i \neq j} \text{area}(A_i \cap A_j)$. ii) Similarly for any set of A_n as above, the complement $X \setminus \bigcup_{n=1}^N A_n = \bigcap_{n=1}^N X \setminus A_n$ has rational area, because $X \setminus A_n$ can be written as the finite union of open subrectangles. Consequently any set $S \in \mathcal{F}$ has rational area. The area of $B(\mathcal{O}, 1)$ is π which is irrational, consequently $B(\mathcal{O}, 1)$ is not in the field.

IX.2.5 Call the sets in the figure

a	G_1	G_2	b
c	$\emptyset$	e	f
g	h	i	j
l	m	X	o

Then we have the following list of equivalences, where those generated earlier in the list are used to generate the rest (once they have been established to be in the field).

$$\begin{aligned}
a &= X \setminus G_1 \\
X &= G_1 \cup a \\
\emptyset &= X \setminus X \\
c &= G_1 \setminus G_2 \\
e &= c \setminus G_1 \\
j &= X \setminus e \\
i &= G_1 \setminus e \\
f &= c \cup i \\
l &= X \setminus f \\
m &= i \cup l \\
h &= m \setminus e \\
o &= X \setminus h \\
g &= G_1 \setminus o \\
b &= X \setminus g
\end{aligned}$$

IX.2.6 Let e be the image of the binary number $a = .a_1a_2\cdots = \sum \frac{a_i}{2^i}$ under the map π which takes $a_i + 1 = e_i$, in other words $\pi(a) = e$. The complement of $e = \pi(a)$, a irrational, in Σ is not the union of a finite number of cylinder sets, which under the inverse of this map map to intervals of rational width around rational points in $[0, 1]$.

IX.2.7 X is a space, let $\mathcal{F} = 2^X$, the power set of X . Let $A \in \mathcal{F}$. Then since $X \setminus A \subset X$, we have $X \setminus A \in \mathcal{F}$. Since $X, \emptyset \subset X, \emptyset \in \mathcal{F}$. Compliments of subsets of X , and finite unions of subsets of X are subsets of X and are therefore elements of $\mathcal{F}$. Hence $\mathcal{F}$ is a field (in fact the biggest field over X).

IX.2.8 $\mathcal{B}$ is the σ -field generated by rectangles with rational corners on the unit square. These rectangles form a countable base in the metric topology induced by the Manhattan metric, as evidenced by the fact that the σ -field contains all balls of radius ϵ for all ϵ in the statement, and by the equivalence of metrics. To show this is the Borel field, the proof of Theorem 2.3 shows that the σ -field generated by a countable base for the topology of X is the Borel field.

IX.2.9 Σ is the codespace on the symbols $\{0, 1\}$. The cylinder sets are defined as all the sets, for all n , of the form

$$C(e_1 \cdots e_n) = \{e_\sigma \in \Sigma : e_{\sigma_1} = e_1, \dots, e_{\sigma_n} = e_n\}$$

Let e be a point in Σ . Then the cylinder set $C_n(e)$ is precisely the set of elements $x \in \Sigma$ such that $x_i = e_i$ for $i \leq n$. Consequently,

$$d(e, x) = \sum_{i=1}^{\infty} \frac{|e_i - x_i|}{3^i} = \sum_{i=n+1}^{\infty} \frac{|e_i - x_i|}{3^i} \leq 1/3^n$$

By choosing $\tilde{e}$ such that $\tilde{e}_i = e_i$ for $i \leq n$ and $\tilde{e}_i \neq e_i$ for $i > n$, we have that $C_n(e) = B_{\frac{1}{3^n}}(\tilde{e})$. The balls of radius 3^{-n} form a countable base for the metric topology on Σ , hence, by the proof of theorem 3, they generate the Borel field $\mathcal{B}$ on Σ .

IX.2.10 Let $\Delta \subset \mathbb{R}^2$ be the Sierpinski triangle, and $\mathcal{G}$ be the set of connected components of $\mathbb{R}^2 \setminus \Delta$. Notice that since by definition of component, this generating set is disjoint. Consequently, one cannot get a subset of a component (other than the component itself or the empty set) by union or compliment of the elements in $\mathcal{G}$. If $\mathcal{F}$ is the σ -field generated by $\mathcal{G}$ then $\mathcal{F}$ likewise contains no nonempty proper subsets of any component of $\mathbb{R}^2 \setminus \Delta$. Let $\epsilon > 0$ be chosen so small that if x is the center of the largest triangle inside the Sierpinski set, $B_\epsilon(x)$ fits entirely inside this component. Then $B_\epsilon(x)$ is not in $\mathcal{F}$, but it is in $\mathcal{B}$. Hence $\mathcal{F} \neq \mathcal{B}$. $\mathcal{F} \subset \mathcal{B}$ however, since any half plane is in $\mathcal{B}$ and any triangle is the union of the compliments of 3 half-planes.

IX.2.11 Let X be a metric space, $\mathcal{G}$ a generating set for the field $\mathcal{F}_1$, and for the σ -field $\mathcal{F}_2$. Let $\mathcal{F}_3$ be the σ -field generated by $\mathcal{F}_1$. Then

$$\mathcal{F}_3 = \mathcal{F}_1 \cup \left\{ \bigcup_{n=1}^{\infty} A_n : A_n \in \mathcal{F}_1 \right\}$$

We must show that this is the smallest σ -field containing $\mathcal{G}$. Suppose not. Then there is an $A \in \mathcal{F}_3$ such that $A \notin \mathcal{F}_2$. But $\mathcal{F}_1$ is the smallest σ -field containing $\mathcal{G}$ so the extra entry must be of the form

$$\bigcup_{n=1}^{\infty} A_n; \{A_n\} \subset \mathcal{F}_1.$$

But the smallest σ -field containing $\mathcal{G}$ must contain all of these sets, since it contains $\mathcal{F}_1$. Hence $\mathcal{F}_2 = \mathcal{F}_3$.

IX.2.12 Let $\mathcal{F}$ be any field of subsets of X . Choose $A \in \mathcal{F}$. Then $X \setminus A \in \mathcal{F}$ and the union of these two is in $\mathcal{F}$. The union is $A \cup X \setminus A = X$.

Answers to Chapter IX, section 3

IX.3.3 Let $X = [0, 1] \subset \mathbb{R}$. $\mathcal{F}$ is generated by subintervals, and for $a \leq b$ we define $\mu((a, b)) = \mu([a, b]) = b - a$, and $\mu(A)$ = sum of lengths of subintervals of A .

(i) $\mu : \mathcal{F} \rightarrow [0, \infty)$. By definition $\mu(A) \geq 0$. Since the sum of lengths of disjoint intervals is less than or equal to $\mu([0, 1]) = 1$ it follows that $\mu(A) < \infty$.

(ii) Let $[a, b]$ be an interval, and let $b = \lim_{n \rightarrow \infty} a_n$ for some sequence $\{a_n\}$ such that $a_1 \in [a, b]$. Then the sum of the lengths of intervals in the union

$$[a, a_1] \cup [a_1, a_2] \cup \cdots = [a, b]$$

is

$$\mu([a, a_1] + \sum_{n=1}^{\infty} (a_{n+1} - a_n)$$

This is a telescoping sum and is therefore equal to

$$\mu([a, a_1] + \mu([a_1, \lim_{n \rightarrow \infty} a_n]) = [a, b]$$

Thus since any element in the field which can be written as the union of countably many elements must be unions of this form together with finitely many other elements, μ is countably additive.

(iii) Let S be the set of rationals in $[0, 1]$. Then S is not in the field $\mathcal{F}$. To see this, we note that neither S nor its complement can be written as the union of finitely many intervals, since both contain countably many components, thus any finite union contains an element with countably many components.

(iv) Let $\hat{\mathcal{F}}$ be the σ -field generated by $\mathcal{F}$. For any field or σ -field, since

$$A \cap B = X \setminus (X \setminus A \cup X \setminus B)$$

the finite intersection of elements of a field are also in the field. Thus for any $x = p/q$ where p, q are integers, the set $\{x\} = [a, x] \cap [x, b] \in \mathcal{F}$ and we have $\mu(\{x\}) = 0$. The rationals (and therefore S) are a countable set, consequently we can write $S = \bigcup_{n=1}^{\infty} A_n$ where $\{A_n\}$ are the elements of S . Consequently, $S \in \mathcal{F}$. We have

$$\hat{\mu}(S) = \hat{\mu}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \hat{\mu}(A_n) = \sum_{x=p/q} \hat{\mu}(\{x\}) = 0$$

IX.3.4 Let Σ be the code space on symbols $\{0, 1\}$, and define $0 \leq p_1 \leq 1$, $p_2 = 1 - p_1$. We define μ on cylinder sets by

$$\mu(C(e_1 \cdots e_n)) = p_{e_1} \cdots p_{e_n}$$

To define μ on other sets in the field $\mathcal{F}$ there is really only one way to be consistent with this definition, and satisfy countable additivity. We describe this:

$$\mu(X) = \mu(C(1) \cup C(2)) = p_1 + p_2 = 1$$

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$$

where $A \cap B$ is in the field, since it is equal to $X \setminus ((X \setminus A) \cup (X \setminus B))$ by de Morgan's rules. The easiest way to see how this defines a countably additive measure on the field $\mathcal{F}$ is to note that for any finite set of codes $\{e_1, \dots, e_n\}$ the measure defined on the cylinder sets of these codes *taken in any order* is the same. Notice also that two cylinder sets are disjoint if they disagree on any specified symbol. Taken together, the measure of a union of these sets is the product of the codes on which they agree, if they are the same length, and this product times the product of the 'tail' of the longer string if they are different lengths. There are no elements of the field $\mathcal{F}$ which can be formed which specify values for an infinite number of places, since this is the disjoint union of an infinite number of components. Consequently, $S = \{x : x \text{ is odd}\}$ is not in $\mathcal{F}$, and $\mu(\{x : x_7 = 1\}) = p_1$. The set $S \in \hat{\mathcal{F}}$ since it can be written as a countable intersection (and therefore a countable union) of finite unions of cylinder sets. Its measure, as above, is the product of the probabilities on which the elements agree, namely all odd places, and is therefore $\lim_{n \rightarrow \infty} p_1^n = 0$. Notice that since the cylinder sets form a countable basis for the metric topology on Σ , $\hat{\mathcal{F}}$ is the Borel field on Σ .

IX.3.5 Since multiplication is commutative, it is easier to define the indices in reverse order. Thus we want the intervals

$$w_{e_n} \circ \dots \circ w_{e_1}([0, 1])$$

as a generator for the field $\mathcal{F}$. Notice that the interval $w_1([0, 1])$ has measure p_1 and the interval $w_2([0, 1])$ has measure p_2 . Define $\mu(X) = 1$. Then we must have

$$\mu(w_1 \circ w_{e_n} \circ \dots \circ w_{e_1}([0, 1]) \cup w_2 \circ w_{e_n} \circ \dots \circ w_{e_1}([0, 1])) = p_1 p_{e_n} \dots p_{e_1} + p_2 p_{e_n} \dots p_{e_1}$$

in order to have even finite additivity, but

$$p_1 p_{e_n} \dots p_{e_1} + p_2 p_{e_n} \dots p_{e_1} = (p_1 + 1 - p_1) p_{e_n} \dots p_{e_1} = p_{e_n} \dots p_{e_1}$$

so that, since these are the subintervals of $w_{e_n} \circ \dots \circ w_{e_1}([0, 1])$ the 'middle third' has measure 0. Hence we have $\mu(B) = 0$ for any $B \in \mathcal{F}$ such that $B \subset X \setminus A$. This implies countable additivity, since every element of $\mathcal{F}$ must be the union of closed or open intervals beginning or ending on the boundary of a subinterval of the above form, and each subinterval is the countable union of subintervals, and satisfies countable additivity. Hence μ is a measure on $\mathcal{F}$. This provides $\mu(A) = 1$, $\mu(X \setminus A) = 0$, $\mu([1/3, 2/3]) = 0$.

IX.3.6 We define $\mathcal{F}$ on $[0, 1]$ by using the IFS

$$\{[0, 1], w_1 = \frac{1}{2}x, w_2 = \frac{1}{2}x + \frac{1}{2}, p_1, p_2\}$$

as in the previous problem. Here the subintervals touch, but the definition from problem 4 of $\mu(X) = 1$ yields that the intersection of two intervals at their boundaries has measure zero. In this case the field is the set of finite unions of subintervals which have boundaries of the form $k/2^n$. By expressing the numbers on the interval in binary, since any element of the interval is the countable sum $\sum_{n=1}^{\infty} (e_n - 1)/2^n$ for $e_i \in \{1, 2\}$ we generate all the subintervals of the interval, hence the extension of $\mathcal{F}$ is $\mathcal{B}([0, 1])$. In order to generate the measure which takes the measure of an interval as the length of an interval, it is necessary and sufficient to have translation independence. This requires $p_1 = 1/2$.

IX.3.7 Let (X, d) be a compact metric space, μ a Borel measure on X such that $\mu(X) \neq 0$. Let S be the support of μ on X . Then $\mu(S) = \mu(X) \neq 0$, so that $S \neq \emptyset$. S is closed which implies that S is compact. Hence $S \in \mathcal{H}(X)$.

IX.3.8 Let μ be a measure on σ -field $\mathcal{F}$, and let $\bar{\mathcal{F}}$ be the class of sets of the form $A \cup B$ where $A \in \mathcal{F}$ and $B \subset C \in \mathcal{F}$ such that $\mu(C) = 0$. $\bar{\mathcal{F}}$ is a σ -field since clearly $X \setminus B = X \setminus C \cup C \setminus B$ and $C \setminus B$ is a subset of an element of $\mathcal{F}$ of measure zero. The same implies with unions of this form. Define $\bar{\mu}(A \cup B) = \bar{\mu}(A) = \mu(A)$. Then if we have a countable union of sets of this form, we have

$$\bar{\mu}\left(\bigcup_{n=1}^{\infty} (A_n \cup B_n)\right) = \bar{\mu}\left(\left(\bigcup_{n=1}^{\infty} A_n\right) \cup \left(\bigcup_{n=1}^{\infty} B_n\right)\right)$$

which is the union of an element of $\mathcal{F}$ with a subset of an element of $\mathcal{F}$ of measure zero. Hence

$$\bar{\mu}(\bigcup_{n=1}^{\infty} (A_n \cup B_n)) = \bar{\mu}(\bigcup_{n=1}^{\infty} A_n)$$

and this latter obeys countable additivity by assumption. Hence $\bar{\mathcal{F}}$ is a σ -field, and $\bar{\mu}$ is a measure on it.

Answers to Chapter IX, section 4

IX.4.1 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a piecewise constant function with finitely many discontinuities, located at $\{x_1, \dots, x_n \subset \mathbb{R}\}$. Let $x_0 = 0, x_{n+1} = 1$. Then f has a single value on each of the intervals $I_0 = [x_0, x_1], \dots, I_n = [x_n, x_{n+1}]$, and we let $y_i = f(x)$, $x \in [x_i, x_{i+1}]$. Then if $\chi_i = \chi_{[x_i, x_{i+1}]}$ is the characteristic function on the interval I_i , we may rewrite

$$f(x) = \sum_{i=0}^n y_i \chi_i$$

If μ is the Borel measure on $[0, 1]$ given by $\mu([a, b]) = b - a$, we have, by definition

$$\int_{[0,1]} f d\mu = \sum_{n=0}^n y_i \mu(I_i) = \sum_{n=0}^n y_i (x_{i+1} - x_i).$$

For the Riemann integral of f on $[0, 1]$ we have

$$\begin{aligned} \int_0^1 f(x) dx &= \int_{x_0}^{x_1} f(x) dx + \dots + \int_{x_n}^{x_{n+1}} f(x) dx \\ &= \int_{x_0}^{x_1} y_0 dx + \dots + \int_{x_n}^{x_{n+1}} y_n dx \\ &= y_0(x_1 - x_0) + \dots + y_n(x_{n+1} - x_n) \\ &= \int_{[0,1]} f d\mu \end{aligned}$$

IX.4.2 We want a measure on $(\mathbb{R}, \text{Euclidean})$, defined on the field $\mathcal{G} = \{A : A \text{ is a rectangle}\}$, which corresponds to area. This means (assuming $a < b, c < d$), if $A = [a, b] \times [c, d]$ or the equivalent expression using open or half open intervals, we define $\mu(A) = (b - a)(d - c)$, and for unions of rectangles define $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$. Let $A \in \mathcal{F}$ such that $A = \bigcup_{n=1}^{\infty} A_n$, $A_n \in \mathcal{F}$ with the A_n disjoint. Then for the union to be in $\mathcal{F}$ we must be able to express it as a finite union of rectangles or the compliments of rectangles. Notice that the compliment of a finite union of rectangles is expressible on X as a finite union of rectangles. The measure of this region is then finitely additive by hypothesis, therefore μ is countably additive. Suppose there were another measure μ' on $\mathcal{F}$ which yielded the area of rectangles in X . Then since both measures need to be countably additive, and since they agree on each subrectangle of X , the two measures agree on $\mathcal{F}$. Hence $\mu' = \mu$. The extension of μ to $\mathcal{B}$ is also a measure which yields the area of sets $A \in \mathcal{B}$, because $\mathcal{B}$ contains all open sets in X , the infimum represented in the Caratheodory extension theorem can be replaced by an infimum on open sets, and is therefore the area of the closure of A .

To show that Δ is in $\mathcal{B}$ it suffices to note that it is closed. This follows since the compliment is then open, and can be covered by open sets in $\mathcal{F}$. This cover has a finite subcover, which implies that $X \setminus \Delta$ is a countable union of sets in $\mathcal{F}$, hence $\Delta \in \mathcal{B}$.

IX.4.3 The integral is defined on the intersection of two strips with Δ . To calculate it, we first determine the measure of an arbitrary strip $[a, b] \subset [0, 1]$ intersected with Δ . The measure is countably additive, so note that given $A \subset \Delta$, with the measure of A being $\mu(A)$, that we have

$$\mu(A) = \mu(w_1(A) \cup w_2(A)) + \mu(w_3(A)) = (p_1 + p_2)\mu(A) + p_3\mu(A) = .5\mu(A) + .5\mu(A)$$

Suppose that for $n < N$ $\mu([k/2^n, (k+1)/2^n] \times \mathbb{R} \cap \Delta) = 1/2^n$ for $0 \leq k \leq 2^n$. That is, the measure of Δ falling within a

strip of length $1/2^n$ positioned on a boundary of $[0, 1]$ subdivided into equal $1/2^n$ intervals is its length. We note that each such strip intersected with Δ is a vertical stack of disjoint Sierpinski triangles. We now subdivide the strip into two parts. The left-hand part contains a stack of twice as many triangles, half the size, while the right-hand is a stack of as many triangles, half the size. Each triangle in the original strip is produced by transforming Δ by some sequence $(w_i \circ \cdots \circ w_k)$ of transformations, those on the stack in the left-hand portion are either of the form $w_i \circ \cdots \circ w_k \circ w_1$ or $w_i \circ \cdots \circ w_k \circ w_2$ while those on the right have an included transformation by w_3 . Consequently the measure of the $1/2^n$ width strip is divided equally on its two halves. By induction, and finite additivity, the measure of any strip beginning and ending on boundaries of the form $k/2^n$ is its length. By countable additivity, the measure concentrated on any strip $[a, b] \times \mathbb{R} \cap \Delta$ is its length.

Having demonstrated this, the integral is $1 \cdot 2/3 + (-1) \cdot 1/3 = 1/3$. We will use the above result in further exercises.

IX.4.4 Let $\alpha, \beta \in \mathbb{R}$, and f, g simple functions. Then we have

$$\begin{aligned}\alpha f &= \alpha \sum y_i \chi_{I_i}(x) = \sum (\alpha y_i) \chi_{I_i}(x) \\ \beta g &= \beta \sum z_j \chi_{J_j}(x) = \sum (\beta z_j) \chi_{J_j}(x)\end{aligned}$$

are simple functions. By definition, $\alpha f + \beta g$ is αf where $g = 0$, and βg where $f = 0$ and the sum elsewhere. We need to write the collection of sets $\bigcup I_i \cup \bigcup J_j$ as the union of disjoint sets. This is done by writing

$$K_k = I_k \setminus \bigcup J_j, K_{m+j} = J_j \setminus \bigcup I_i, K_{m+n+(i+j)} = I_i \cap J_j, K_k = \emptyset \text{ elsewhere.}$$

Define $w_i = \alpha y_i, w_j = \beta z_j, w_k = \alpha y_i + \beta z_j$ on the sets defined above. Then $w(x) = \sum w_k \chi_{K_k}$ is a simple function, and equal to $\alpha f + \beta g$. The integral of the sum is the sum of the integrals by the above definition of $w(x)$.

IX.4.6 We need to rewrite the function, as in the above problem, as a simple function with disjoint sets. Writing $C = A \setminus B, D = B \setminus A, E = A \cap B$, we have

$$f(x) = \chi_C + (2.3)\chi_D + (1 + 2.3)\chi_E$$

To evaluate the integral, we need the measures on each of these sets. We define them in terms of strings: C is the set of strings beginning with 1, and not having a 2 in the second place, i.e., $C = C(1, 1)$ the cylinder set beginning in two 1's. D is the set of strings not beginning with 1 which have a 2 in the second place, or, $D = C(2, 2)$. E is the set beginning with 1 with a 2 in the second place, or, $E = C(1, 2)$. By the definition of μ we have $\mu(C) = p_1 p_1 = (.4)^2$, $\mu(D) = p_2 p_2 = (.6)^2$, $\mu(E) = p_1 p_2 = (.4)(.6)$. The integral is given by

$$\int_{\Sigma} f d\mu = (.4)^2 + (2.3)(.6)^2 + (1 + 2.3)(.4)(.6) = 1.78$$

IX.4.7 Define the Borel measure δ_a by $\mu(B) = 1$ if $a \in B$ and 0 otherwise. Let f be continuous $f : (X, d) \rightarrow \mathbb{R}$. For each n , denote by $\{B_i\}_{i=1}^{k_n}$ a Borel partition of X into disjoint sets B_i where $\text{diam}(B_i) \leq 1/n$. Define $\{f_n\}$ to be a sequence of simple functions defined on each partition by $f_n = \sum y_i \chi_{B_i}$ with $y_i = f(x_i)$ on the sets which do not include a and $y_{i_a} = a$ on the set in each partition $a \in B_{i_a}$. Then since the measure of all the sets is zero with the exception of B_{i_a} on which it is 1, we have

$$\int_X f_n d\delta_a = f(a) \forall n.$$

Hence the limit as $n \rightarrow \infty$ is $f(a)$. Our partition is independent of the choice of x_i with the exception of the choice of y_{i_a} , but the function is continuous, so that as $n \rightarrow \infty$ another choice of x_{i_a} approaches a as the diameter of B_{i_a} goes to zero.

IX.4.8 We proceed by first deducing that since $f(x, y)$ is continuous, and the measure on $\blacksquare$ is defined on rectangles by their area, the value of the integral is the Riemann integral of $f(x, y)$. By definition, the Riemann integral is done by taking a sequence of rectangular partitions of $\blacksquare$, and constructing two sequences of simple functions, choosing y_i, y'_i by

$$y_i = \inf\{f(x, y) : (x, y) \in B_i\}; y'_i = \sup\{f(x, y) : (x, y) \in B_i\}$$

By Theorem 4.1, since $f(x, y)$ is continuous, the two sequences converge to the same limit, namely $\int_{\blacksquare} f d\mu$, since they converge and converge to the same limit, the Riemann integral is defined and is equal to this limit. The value of the integral, by taking the Riemann integral, is $1/2$.

IX.4.10 Referring to exercise 3.4, the measure of any strip intersecting Δ is simply its width. The function $f(x, y) = x^2$ is independent of y so we partition the space into strips, their intersection with Δ produces decreasing diameters. $f(x)$ is continuous, hence this partition and the choice of x_i are immaterial. Choose x_i to be the left-hand boundary of each strip, intersecting some point on Δ . Then the value of the integral is just $\int_0^1 x^2 dx = 1/3$. Let $\mathbf{X}$ be the set of pixels on a computer display. For simplicity, define the distance between the pixel (i, j) on the i^{th} row and in the j^{th} column and the pixel (k, ℓ) as $|i - k| + |j - \ell|$, which is a Manhattan metric. Then $\mathbf{X}$ may be related to the subset of $\mathbb{R}^2$ consisting of integer coordinate points in the first quadrant of the same size. This set is closed and totally bounded. Hence $\mathbf{X}$ is compact. Since there are finitely many pixels, we may define the generator for the Borel subsets to be all the subsets consisting of one pixel each. This defines the Borel field on $\mathbf{X}$ to be the power set of $\mathbf{X}$. Thus any set of pixels is a Borel subset. Suppose there are J rows and K columns so that the total number of pixels is $N = J \times K$. Define a Borel measure by $\mu(A) = (\text{number of pixels in } A)/N$. Any function from $\mathbf{X} \rightarrow \mathbb{R}$ is continuous because the pixels themselves are open sets, hence the inverse image of an open set in $\mathbb{R}$ is always open.

We will choose the function $f(i, j) = i + j$. Any function on the finite set of pixels can be described by its value on each pixel, hence its value at a pixel times the characteristic function at that pixel, hence the sum of such terms makes any function simple. Then the integral of f over $\mathbf{X}$ is

$$\int_{\mathbf{X}} f d\mu = \sum_{\mathbf{X}} \frac{i+j}{N} = \frac{1}{N} \sum_{\mathbf{X}} (i+j)$$

where the summation notation indicates summing over all pixels in $\mathbf{X}$. Since there are K pixels with first value i , and J pixels with second value j , we have

$$\begin{aligned} \frac{1}{N} \sum_{\mathbf{X}} (i+j) &= \frac{1}{N} \left(\sum_{n=1}^K n + \sum_{n=1}^J n \right) \\ &= \frac{1}{N} \left(\frac{K(J+1)}{2} + \frac{J(K+1)}{2} \right) \\ &= \frac{1}{N} \frac{N(J+1) + N(K+1)}{2} \end{aligned}$$

so the integral is

$$\int_{\mathbf{X}} f d\mu = \frac{J+K+2}{2}$$

Answers to Chapter IX, section 5

IX.5.1 Let

$$\mathbf{X} = \{(i, j) : i, j \in \{1, 2, 3, \dots, K\}\}$$

and define a metric $d : \mathbf{X} \rightarrow \mathbb{R}$ by

$$d((i_1, j_1), (i_2, j_2)) = |i_1 - i_2| + |j_1 - j_2|$$

Then $(\mathbf{X}, d)$ is a compact metric space. Define μ to be an atomic measure with

$$\mu((i, j)) = \frac{i+j}{K^3 + K^2}$$

and let ν be another atomic measure with

$$\nu((i, j)) = \frac{1}{K^2}$$

for every $(i, j) \in \mathbf{X}$.

We wish to calculate the distance in the Hutchinson metric between μ and ν , given by

$$d_H(\mu, \nu) = \sup \left\{ \int_X f d\mu - \int_X f d\nu : f \in C(\mathbf{X}, \mathbb{R}), |f(x) - f(y)| \leq d(x, y) \forall x, y \in \mathbf{X} \right\}$$

where $C(\mathbf{X}, \mathbb{R})$ is the set of continuous functions from $\mathbf{X}$ to $\mathbb{R}$. Now $\mathbf{X}$ is a discrete space under a metric which is the restriction of the Manhattan metric on $\mathbb{R}^2$, consequently $f : \mathbf{X} \rightarrow \mathbb{R}$ is always continuous. As well, the integrals are the integrals of simple functions. Hence for every f , we the integrals can be expressed

$$\begin{aligned} \int_X f d\mu &= \sum_X f(i, j) \frac{i+j}{K^3 + K^2} \\ \int_X f d\nu &= \sum_X f(i, j) \frac{1}{K^2} \end{aligned}$$

so the difference is, for any function f ,

$$\sum_X f(i, j) \left(\frac{i+j}{K^3 + K^2} - \frac{1}{K^2} \right) = \frac{1}{K^2} \sum_X f(i, j) \left(\frac{i+j}{K+1} - 1 \right)$$

Intuitively, one is tempted to pick the function $f(i, j) = i + j$, which among the set of valid functions is the largest of the form

$$f(i, j) = \alpha i + \beta j$$

However, the range of the term $(i+j)/(K+1) - 1$ is from $2/(K+1) - 1 = -(K-1)/(K+1)$ to $2K/(K+1) - 1 = (K-1)/(K+1)$ which gives a negative coefficient to $f(i, j)$ over almost half of $\mathbf{X}$ detracting from the value of the sum. We must therefore endeavor to pick $f(i, j)$ as large as we can in absolute value, negative when the measure subtraction term is negative, and positive when it is positive. That is, we maximize by choosing the largest value for $|f|$ allowable under the distance constraint where $|(i+j)/(K+1) - 1| = |C(i, j)|$ is largest. The interested reader can confirm that this exceeds the intuitive result. The value of $C(i, j)$ is constant along each diagonal from $(1, n)$ to $(n, 1)$ and is equal to zero along the major diagonal from $(1, K)$ to $(K, 1)$. It is anti-symmetric with respect to this diagonal, negative above it and positive below.

We thus place the largest values for $|f|$ at $(1, 1)$ and (K, K) , and since the distance between these is $2K - 2$, we set them to $\pm(K-1)$. We then proceed outward from these points in Manhattan metric circles. The distance between such circles (which are the diagonals where they intersect $\mathbf{X}$) is always the horizontal distance between the extensions of the diagonals, thus we end up satisfying the distance constraint by putting f to

$$\begin{array}{ccccc} -(K-1) & -(K-2) & \cdots & -1 & 0 \\ -(K-2) & -(K-3) & \cdots & 0 & 1 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ -1 & 0 & \cdots & (K-3) & (K-2) \\ 0 & 1 & \cdots & (K-2) & (K-1) \end{array}$$

This may be summed by taking twice the sum above the diagonal, and noticing that there are n entries where $(K-n)$ is the absolute value of the entry, yielding

$$\begin{aligned}
& \frac{2}{K^2(K+1)} \sum_{n=1}^K n(K-n)^2 \\
&= \frac{2}{K^2(K+1)} \sum_{n=1}^K (nK^2 - 2n^2K + n^3) \\
&= \frac{2K^2}{K^2(K+1)} \sum_{n=1}^K n - \frac{4K}{K^2(K+1)} \sum_{n=1}^K n^2 + \frac{2}{K^2(K+1)} \sum_{n=1}^K n^3 \\
&= \frac{2K^2K(K+1)}{2K^2(K+1)} - \frac{4KK(K+1)(2K+1)}{K^2(K+1)} + \frac{2K^2(K+1)^2}{4K^2(K+1)} \\
&= \frac{K-1}{6} = d_H(\mu, \nu)
\end{aligned}$$

IX.5.2 Since $\mu \in \mathcal{P}(\mathbf{X})$ the measure of the entire space is $\mu(\mathbf{X}) = 1$. This value is identical with the value on the support of μ which we will call S , that is, this implies $\mu(S) = 1$. In particular it is non-zero, and since, by Theorem 3.1 in Chapter IX statement (3) we have $\mu(\emptyset) = 0$, $S \neq \emptyset$. Then by Theorem 3.4 in Chapter IX, the support of a Borel measure on a metric space is closed. Since $\mathbf{X}$ is compact, a closed subset of $\mathbf{X}$ is also compact, hence S is compact. Since S is compact and non-empty, S , the support of μ is an element of $\mathcal{H}(\mathbf{X})$.

Answers to Chapter IX, section 6

IX.6.1 We want to show that the Markov operator is indeed a transformation on $\mathcal{P}(\mathbf{X})$, that is, $M(\nu) : \mathcal{P}(\mathbf{X}) \rightarrow \mathcal{P}(\mathbf{X})$. It suffices to show that

$$\nu \circ w_i^{-1} : \mathcal{P}(\mathbf{X}) \rightarrow \mathcal{P}(\mathbf{X})$$

since a sum of these operators is a Borel measure if they are, and $\sum_i p_i = 1$ indicates that they are normalized if each entry is.

(i) $w_i^{-1} : \mathbf{X} \rightarrow \mathbf{X}$, since for hyperbolic IFS these are invertible maps. In addition, if we define $w_i^{-1}(\{x\}) = \emptyset$ whenever $x \notin w_i(\mathbf{X})$ then $w_i^{-1}(\mathbf{X}) = \mathbf{X}$. Hence $\nu \circ w_i^{-1}(\mathbf{X}) = 1$, and this is a normalized Borel measure. Hence $M(\nu)(\mathbf{X}) = 1$.

(ii) $\nu \circ w_i^{-1}$ is a map $\mathbf{X} \rightarrow [0, \infty)$ (actually $\mathbf{X} \rightarrow [0, 1]$). Furthermore, by definition, if $\{A_j\}_{j=1}^\infty$ are mutually disjoint,

$$\begin{aligned}
\nu \circ w_i^{-1} \left(\bigcup_{j=1}^\infty A_j \right) &= \nu \left(\bigcup_{j=1}^\infty w_i^{-1}(A_j) \right) \\
&= \sum_{i=1}^\infty \nu(w_i^{-1}(A_i)) = \sum_{j=1}^\infty (\nu \circ w_i^{-1})(A_j)
\end{aligned}$$

so that $\nu \circ w_i^{-1}$ is countably additive. Since w_i^{-1} maps open sets to open sets (because w_i is a contraction, it is continuous), and since $w_i^{-1}(\mathbf{X} \setminus A) = w_i^{-1}(\mathbf{X}) \setminus w_i^{-1}(A)$, w_i^{-1} takes the generators for the Borel subsets into themselves, and preserves the operation of set complementation. Hence $w_i^{-1} : \mathcal{B} \rightarrow \mathcal{B}$, and the expression $\nu \circ w_i^{-1}$ is well defined. Hence we have $\nu \circ w_i^{-1} : \mathcal{P}(\mathbf{X}) \rightarrow \mathcal{P}(\mathbf{X})$ and therefore $M(\nu) : \mathcal{P}(\mathbf{X}) \rightarrow \mathcal{P}(\mathbf{X})$, as desired.

IX.6.2 We define L by

$$L = \{f \in C^0(\mathbf{X}, \mathbb{R}) : |f(x) - f(y)| < d(x, y) \forall x, y \in \mathbf{X}\}$$

and let $f \in L$. Define $\tilde{f} = s^{-1} \sum_i p_i f_i \circ w_i$, and we wish to show that $\tilde{f} \in L$. Since $\tilde{f}$ is a scalar times the sum of compositions of continuous functions, $\tilde{f}$ is clearly in $C^0(\mathbf{X}, \mathbb{R})$. We must therefore only verify the distance condition. For all $x, y \in \mathbf{X}$ we have

$$\begin{aligned}
|\tilde{f}(x) - \tilde{f}(y)| &= |s^{-1} \sum_i |f \circ w_i(x) - f \circ w_i(y)| \\
&\leq s^{-1} \sum_i p_i |f \circ w_i(x) - f \circ w_i(y)| \\
&\leq s^{-1} \sum_i p_i |w_i(x) - w_i(y)| \text{ since } f \in L \\
&\leq s^{-1} \sum_i p_i s |x - y| \text{ since } s = \max_i s_i \\
&\leq s^{-1} \cdot s |x - y| \cdot \sum_i p_i \\
&\leq |x - y| = d(x, y) \text{ since } \sum_i p_i = 1
\end{aligned}$$

Hence $\tilde{f} \in L$.

IX.6.7 Let X consist of three points: $X = \{A, B, C\}$, and let an IFS be defined by

$$\{X; w_1, w_2; p_1 = .6, p_2 = .4\}$$

where

$$\begin{aligned}
w_1(A) &= B & w_1(B) &= B & w_1(C) &= B \\
w_2(A) &= C & w_2(B) &= A & w_2(C) &= C
\end{aligned}$$

Furthermore, let $\mu_0(A) = \mu_0(B) = \mu_0(C) = 1/3$. We now let $\mu_n = M^n(\mu_0)$ and find constant entries for the *Markov transition matrix* $\tilde{M}$, $a, b, \dots, i$, such that

$$\begin{pmatrix} \mu_n(A) \\ \mu_n(B) \\ \mu_n(C) \end{pmatrix} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} \mu_{n-1}(A) \\ \mu_{n-1}(B) \\ \mu_{n-1}(C) \end{pmatrix}$$

By definition,

$$M(\mu) = \sum_i p_i \mu \circ w_i^{-1} = .6(\mu \circ w_1^{-1}) + .4(\mu \circ w_2^{-1})$$

so that

$$\mu_n = .6(\mu_{n-1} \circ w_1^{-1}) + .4(\mu_{n-1} \circ w_2^{-1})$$

and we have

$$\begin{aligned}
\mu_n(A) &= .6(\mu_{n-1} \circ w_1^{-1}(A)) + .4(\mu_{n-1} \circ w_2^{-1}(A)) = 0 + .4\mu_{n-1}(B) \\
\mu_n(B) &= .6(\mu_{n-1}(A) + \mu_{n-1}(B) + \mu_{n-1}(C)) + 0 \\
\mu_n(C) &= 0 + .4(\mu_{n-1}(A) + \mu_{n-1}(C))
\end{aligned}$$

giving us

$$\tilde{M} = \begin{pmatrix} 0 & .4 & 0 \\ .6 & .6 & .6 \\ 0 & .4 & .4 \end{pmatrix}$$

For a discrete space like this, $\tilde{M}$ completely determines $M: \mathcal{P}(X) \rightarrow \mathcal{P}(X)$, by defining μ_n at each point in the space. In the limit, $\tilde{M}^n \rightarrow \tilde{M}$, an operator which takes every point in $\mathcal{P}(X)$ to some measure μ_f , and which is therefore necessarily singular. By definition $\mu_f = \tilde{M}(\mu_f)$. This is an eigenvalue equation which means that μ_f is an eigenvector of the eigenvalue 1.

IX.6.8 We first look at the sequence of measures $\{\mu_n\}_{n=0}^\infty$. We have

$$\begin{aligned}\mu_1(A) &= \sum_i p_i \mu_0(w_i^{-1}(A)) \\ \mu_2(A) &= \sum_i p_i \mu_1(w_i^{-1}(A)) \\ &= \sum_i p_i \left(\sum_j p_j \mu_0(w_j^{-1}(w_i^{-1}(A))) \right) \\ &= \sum_i p_i \left(\sum_j p_j \mu_0(w_j^{-1} \circ w_i^{-1}(A)) \right)\end{aligned}$$

Since the sum after n such iterations is finite, the order is not important and we can write

$$\mu_n(A) = \sum_{i,j,\dots,k} p_i p_j \cdots p_k \mu_0(w_k^{-1} \circ \cdots \circ w_j^{-1} \circ w_i^{-1}(A))$$

where the sum indicates a sum over all the indices $i, j, \dots, k$.

μ_0 is concentrated on A since, by assumption, $\mu_0 \in \mathcal{P}(X)$, and $\mu_0(A) = 1$ so the support of μ_0 is a subset of A . If $x \notin A$, all the terms in the above sum are zero because of this, hence the support of μ_n is again a subset of A , and we must have $\mu_n(A) = 1$, because $M : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$. Now let μ be the unique invariant measure of the operator M , that is $M(\mu) = \mu$. From the formula just derived with respect to μ_0 for $M^n(\mu_0)$, and from the invariance of μ we must have, for any set B ,

$$\mu(B) = \sum_{i,j,\dots,k} p_i p_j \cdots p_k \mu(w_k^{-1} \circ \cdots \circ w_j^{-1} \circ w_i^{-1}(B))$$

Now choose $B = w_i \circ w_j \circ \cdots \circ w_k(A)$. Then we have

$$\mu(w_i \circ w_j \circ \cdots \circ w_k(A)) = \sum_{m,n,\dots,l} p_m p_n \cdots p_l (w_m^{-1} \circ w_n^{-1} \circ \cdots \circ w_l^{-1}(w_i \circ w_j \circ \cdots \circ w_k(A)))$$

where the sum is over the same number of indices as in the composition for B . Since each element of the sum is positive or zero, the sum is greater than or equal to each of its entries, and we have

$$\begin{aligned}\mu(w_i \circ w_j \circ \cdots \circ w_k(A)) &\geq p_i p_j \cdots p_k (w_k^{-1} \circ \cdots \circ w_j^{-1} \circ w_i^{-1}(w_i \circ w_j \circ \cdots \circ w_k(A))) \\ &\geq p_i p_j \cdots p_k \mu(A) = p_i p_j \cdots p_k\end{aligned}$$

Equality holds precisely when this is the only contributing term of the sum, that is

$$\mu(w_m^{-1} \circ w_n^{-1} \circ \cdots \circ w_l^{-1}(w_i \circ w_j \circ \cdots \circ w_k(A))) = 0$$

whenever the ordered set of indices $(m, n, \dots, l) \neq (i, j, \dots, k)$, which says for all of these ordered sets the set

$$w_m^{-1} \circ w_n^{-1} \circ \cdots \circ w_l^{-1}(w_i \circ w_j \circ \cdots \circ w_k(A))$$

is at least of measure zero, if not empty. It is empty for the totally disconnected case, hence equality holds when the IFS is totally disconnected.

IX.6.11 The measure μ has support on A , consequently it suffices to restrict to Borel subsets of A . We first let the IFS be totally disconnected. Then the sets of the form

$$W_i = w_{\sigma_n} \circ w_{\sigma_{n-1}} \circ \cdots \circ w_{\sigma_1}(A)$$

for all n form a countable basis on the space A . Hence since for the totally disconnected case, these are all both closed and open, for any Borel subset B of A we have

$$B = \bigcup_{i=1}^{\infty} W_i$$

for some disjoint choice of W_i . But

$$\phi^{-1}(B) = \phi^{-1}\left(\bigcup_{i=1}^{\infty} W_i\right) = \bigcup_{i=1}^{\infty} \phi^{-1}(W_i) = \bigcup_{i=1}^{\infty} C_i$$

where C_i are disjoint cylinder sets in Σ . The measure ρ on Σ by observation gives the measure on any cylinder set as

$$\rho(C(\sigma_1, \dots, \sigma_k)) = p_1 \cdots p_k$$

which is precisely the measure μ applied to the corresponding basis set W_i on A . Consequently, the measures agree, and we have

$$\rho(\phi^{-1}(B)) = \mu(B)$$

When the attractor is not totally disconnected, we have for each W_i finitely many cylinder sets in the image $\phi^{-1}(W_i)$, which are disjoint, and correspond to terms in the sum given in exercise IX.6.8. Consequently the measures agree on Borel subsets of A in this case as well. Since the Borel subsets of A are precisely the subsets $A \cap B$ where B is a Borel subset of $\mathbf{X}$, and since the support of the measure is A , the equality holds for any Borel subset of $\mathbf{X}$.

Answers to Chapter IX, section 7

IX.7.1 We want to prove that $\mu([x, x + \delta]) = \delta$, where μ is the invariant measure of the IFS

$$\{[0, 1]; w_1 = 1/2x, w_2 = 1/2x + 1/2; 1/2, 1/2\}$$

To do this it suffices to show that this expression holds on any closed interval of the form $[k/2^n, (k+1)/2^n] \subset [0, 1]$, since we may then build an approximation of any interval from these, good to any error, by the disjoint union of subsets of this form which cover the interval. Since μ is defined as the infimum of the measures of such covers, we let the error go to zero.

To show that $\mu([k/2^n, (k+1)/2^n]) = 1/2^n$ we note that for each iteration of $W(A)$ the image is 2 times the number of subintervals of A of half the length of those of the iteration before. Each of these subintervals is of the form

$$w_{\sigma_1} \circ \cdots \circ w_{\sigma_k}(A)$$

where there are k terms in the composition for $W^k(A)$. Consequently, these intervals of length $1/2^k$ have measure

$$\mu(w_{\sigma_1} \circ \cdots \circ w_{\sigma_k}(A)) = p_{\sigma_1} \cdots p_{\sigma_k}$$

Since each term in the product is $1/2$, the right hand side is $1/2^k$ as desired.

IX.7.2 We must prove the μ applied to any rectangle in $\blacksquare$ gives the area of the rectangle, where μ is the invariant measure of the IFS

$$\{\blacksquare; w_1, w_2, w_3, w_4; 1/4, 1/4, 1/4, 1/4\}$$

and the w_i are given by

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \\ w_3 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \\ w_4 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix} \end{aligned}$$

that the invariant measure μ is uniform, that is, that for any rectangle, we have $\mu([a, b], [c, d]) = (c - a)(d - b)$. We argue as in the previous problem: For any square with a side of length $\frac{1}{2^n}$ of the form

$$S = ([\frac{k}{2^n}, \frac{k+1}{2^n}], [\frac{\ell}{2^n}, \frac{\ell+1}{2^n}])$$

we have some sequence $i, j, \dots, k$ of digits, of length n such that this square is equal to

$$S = w_i \circ w_j \circ \dots \circ w_k(A)$$

and hence its measure is

$$\mu(S) = p_i p_j \dots p_k = (\frac{1}{2^n})^2$$

so that the measure has the uniform property on such squares. These squares form a countable base for the Borel subsets of A , consequently any rectangle may be written as

$$([a, b], [c, d]) = \bigcup_{i=1}^{\infty} S_i$$

where the S_i have been chosen such that $\mu(S_i \cap S_j) = 0$, that is, they meet only on the edges. Consequently, by countable additivity, we have $\mu([a, b], [c, d]) = (c - a)(d - b)$, for this rectangle, consequently the measure of any rectangle in A is its area.

IX.7.4 The distance between two points in $([0, 1]_p, d_p)$ can be written

$$d_p(a, b) = |d_p(a, 0) - d_p(b, 0)|$$

We start with the sets in each space corresponding to addresses in a cylinder set in code space, namely

$$\phi(C(\sigma_1, \sigma_2, \dots, \sigma_n)) \text{ and } \phi_p(C(\sigma_1, \sigma_2, \dots, \sigma_n)).$$

If the measures agree on the size of these sets, which form a base for the Borel field in each space then since we can rewrite f on these as

$$f(C(\sigma_1, \sigma_2, \dots, \sigma_n)) = \phi_p(\phi^{-1}(C(\sigma_1, \sigma_2, \dots, \sigma_n)))$$

we will have demonstrated that the measures are equal. On $[0, 1]$, almost by definition, the invariant measure ν applied to a cylinder set yields

$$\nu(C(\sigma_1, \sigma_2, \dots, \sigma_n)) = p_1 p_2 \dots p_n,$$

that is, the probability of landing in the cylinder set on n iterations.

On $[0, 1]_p$, the cylinder set corresponds to an interval with endpoints

$$.\sigma_1 \sigma_2 \dots \sigma_n \bar{0} \quad \text{and} \quad .\sigma_1 \sigma_2 \dots \sigma_n \overline{(N-1)} = .\sigma_1 \sigma_2 \dots (\sigma_n + 1) \bar{0}$$

How wide is this interval? Start with $n = 1$. The interval $(.\sigma_1, .(\sigma_1 + 1))$ can be read from the IFS we used to construct $[0, 1]_p$, to be of width p_{σ_1} . A contraction of the same relative subdivision of $[0, 1]$ is placed inside it to make our “tick marks” by the IFS, so that the portion of this subinterval which is also in the interval $(.\sigma_1 \sigma_2, .\sigma_1 (\sigma_2 + 1))$ is thus $p_{\sigma_1} p_{\sigma_2}$. If this proceeds in this fashion up to σ_k , then the subinterval which is again divided by the action of the IFS is of length $p_{\sigma_1} p_{\sigma_2} \dots p_{\sigma_k}$, and we take a fraction of this equal to $p_{\sigma_{k+1}}$ times its length. Consequently by induction, the length of the image of a cylinder set is

$$\mu(\phi_p(C(\sigma_1, \sigma_2, \dots, \sigma_n))) = d(.\sigma_1 \sigma_2 \dots \sigma_n, .\sigma_1 \sigma_2 \dots (\sigma_n + 1)) = p_{\sigma_1} p_{\sigma_2} \dots p_{\sigma_n}$$

which agrees with the value on $[0, 1]$ for the image of the same set under the invariant measure ν . Hence $\nu(A) = \mu(f(A))$, and f is measure preserving.

IX.7.5 The diagonal of the square forms right triangles with the vertical lines generated by the tick marks on the x -axis, and with the horizontal grid lines generated by tick marks on the y -axis. The slope of this line is 1, hence the distance between

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horizontal grid lines on the diagonal is $\sqrt{2}$ times the distance on the x -axis. Consequently, a point picked at random on the diagonal has a p_i chance of falling between the $i - 1^{\text{th}}$ grid mark and the i^{th} grid mark, and so forth. With probability 1 then, a point on the diagonal will have each digit occurring in proportion to the p_i spacing on the x -axis in its x coordinate, and the digits occur with equal frequency in its y coordinate.

Answers to Chapter IX, section 8

There are no mathematical exercises in section 8.